

Further Mathematics

Solutions

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Further Mathematics

Data Analysis

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Further Mathematics: Solutions

Chapter 1: Data Analysis

Exercise 1: Data Types & Representation

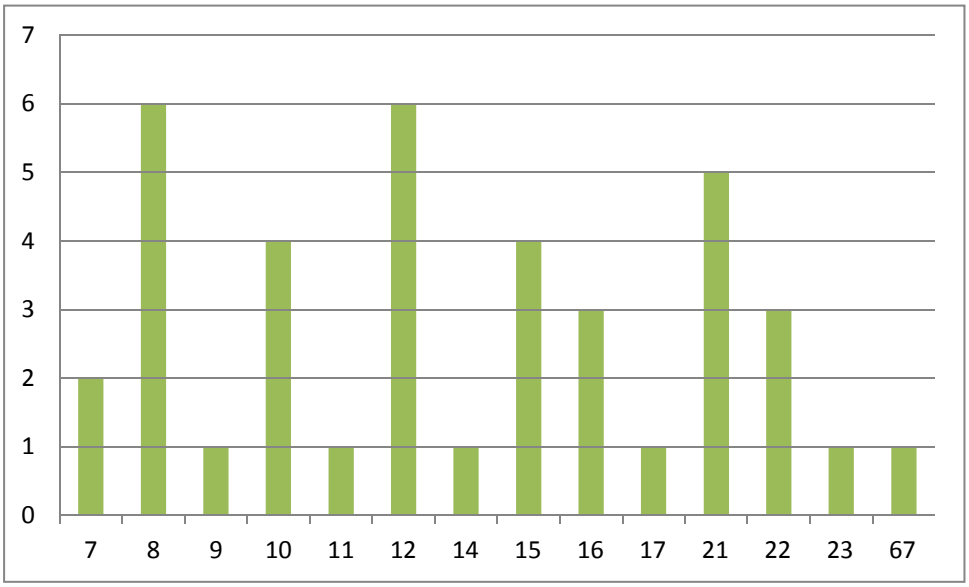
1)

a) Data
b) Sample
c) Data
d) Sample
e) Data

2)

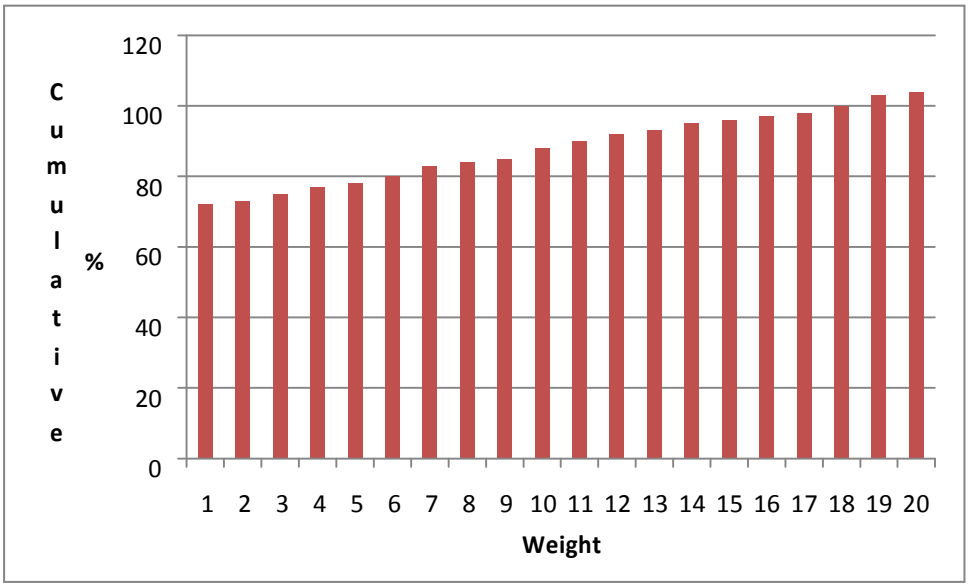
a) Quantitative, continuous
b) Quantitative, discrete
c) Categorical
d) Categorical
e) Categorical
f) Quantitative, discrete

3)



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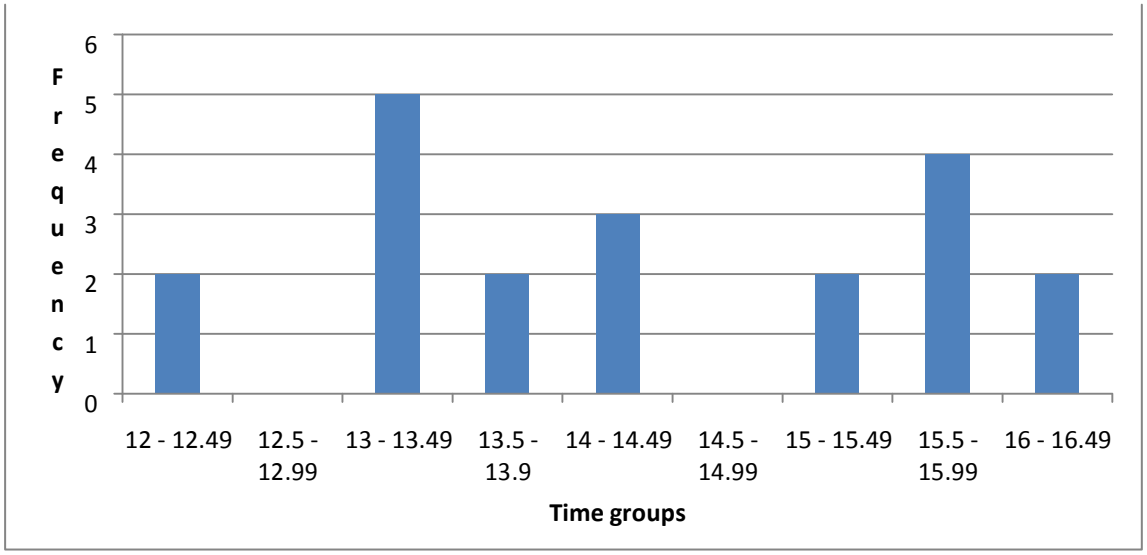
4)



5)

a) 12.0 to 12.49; 12.5 to 12.99, 13 to 13.49, 13.5 to 13.99, 14 to 14.49, 14.5 to 14.99, 15 to 15.49, 15.5 to 15.99, 16 to 16.49

b)



c) 14.4

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d) 13 – 13.49

6)

a)

Stem	Leaf
2	0 3 3 5 5 6 6 7
3	0 0 1 2 3 4 7
4	1 1 2 2 3
5	3

b)

Stem	Leaf
6	2 6 6 9
7	0 1 3 3 4 5 6 8
8	0 1 3 8
9	2 9 9

c)

Stem	Leaf
1	0 2
2	0 1 1 1 2 2 2 4 6 7
3	0 1 2 2 2 5 7
4	0 3 3 5 6 7 9
5	0 6
6	0

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7)

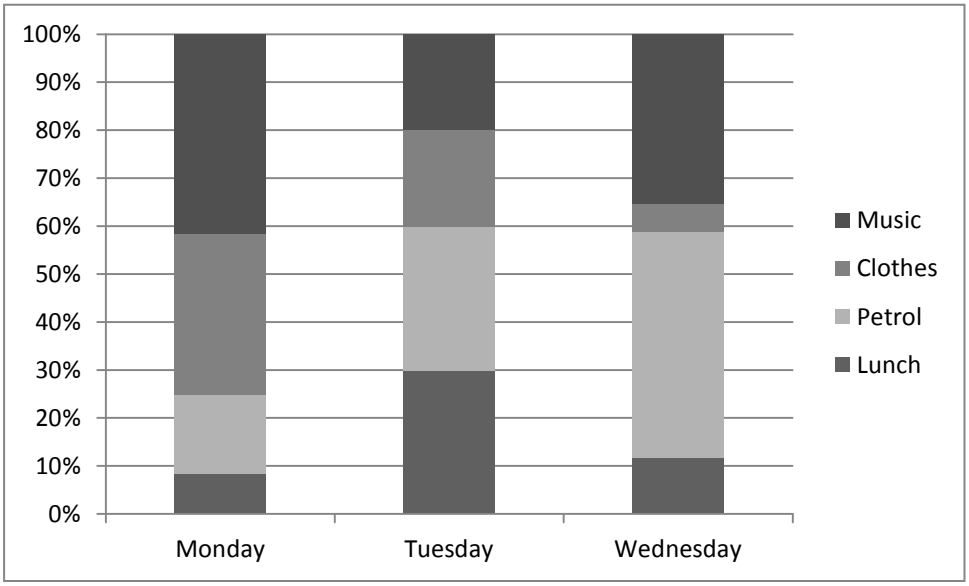
a) Skewed left

b) Skewed right

c) Skewed right

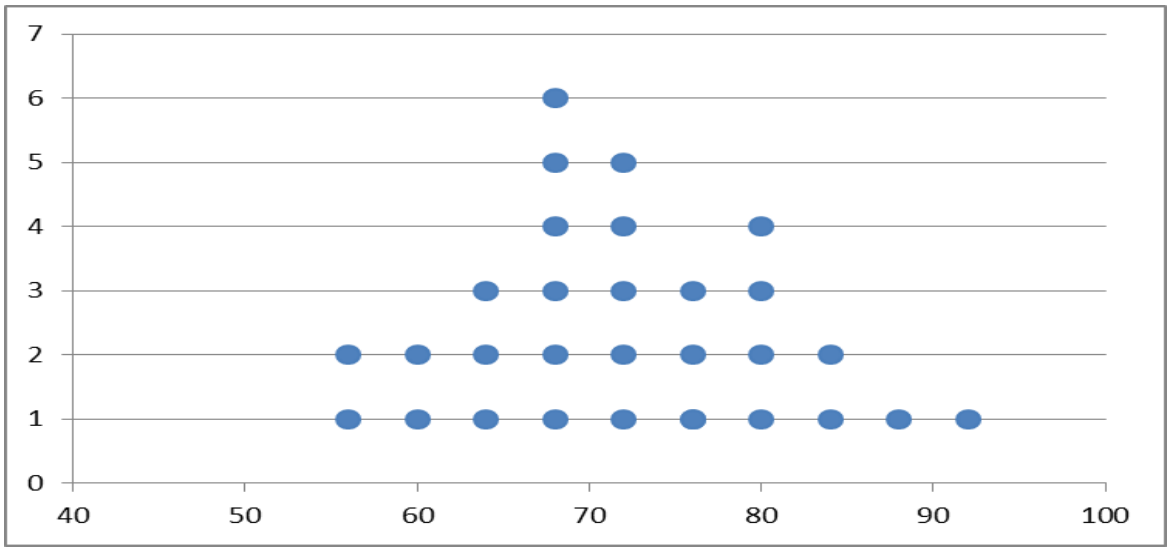
d) Normal

8)



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9)



10)

Score range	Cumulative frequency	Cumulative %
26-30	2	6.25
31-35	2	6,25
36-40	3	9.4
41-45	5	15.6
46-50	8	25
51-55	8	25
56-60	12	37.5
61-65	18	56.25
66-70	23	71.9
71-75	26	81.25
76-80	26	81.25
81-85	28	87.5
86-90	29	90.6
91-95	31	96.9
96-100	32	100

32 students sat the test, 24 passed

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11)

Stem	Leaf
3	4 8
4	2 3 5 8 9
5	1 3 7 8
6	0 1 3 7
7	1 4 7 9
8	5

Stem	Leaf
2	3 9
3	5 9
4	6 7 9
5	2 3 3 9
6	7
7	3 9
8	6
9	1 7
10	1
11	7
12	6

Team A median scores was 57.5

Team B median score was 56

Team A had the least range of scores and was more consistent

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Further Mathematics: Solutions

Chapter 1: Data Analysis

Exercise 2: Summary Statistics

1)

a) Mean = 8
Mode = 7
Median = 7

b) Mean = 16
Mode = 8, 13, 15 and 20
Median = 13

c) Mean = 12.6
Mode = 4 and 16
Median = 15

d) Mean = 8.6
Mode = 1
Median = 8

e) Mean = 9.7
Mode = 1, 3, 5 and 17
Median = 11

2)

a) Mean = 4.26
Mode = 7
Median = 5

b) Mean = 23.39
Mode = 25
Median = 23.5

c) Mean = 14
No mode
Median = 14

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d) Mean = 39.82
Mode = 7
Median = 5

3) An outlier biases the mean toward it, has no effect on the mode, and usually no effect on the median (depending on the distribution of scores around the original median)

4)

a)

Stem	Leaf
4	6 7 7 8
5	2 6 7 8
6	0 1 2
7	1 2 7 8
8	0 1 3 6 9

Mean = 65.55
Mode = 47
Median = 61.5
Range = 43

b)

Stem	Leaf
4	7 8 8 8 9 9
5	0 1 4 6 7
6	2 5 9
7	5 9
8	2 4 8 9

Mean = 62.2
Mode = 48
Median = 56.5
Range = 42

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c)

Stem	Leaf
4	7 8
5	0 1 2 3 6 6
6	1 1 9
7	4 6 7 9
8	1 4 8 9
9	0

Mean = 67.1
Mode = 56, 61
Median = 65
Range = 43

5) Mean = 23
Mode = 24
Median = 24

6) The mean will increase slightly, the mode will remain unchanged, and the median may change slightly depending on the distribution of scores around the median

7) 92

8) 75%

9) 76.67%

10) Due to the fact that in the second distribution there are more students contributing to the higher average, therefore the overall score for this group (girls) was higher than in the previous example

11) y = 16, x = 9

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12)

a) 33

b) 49

c) 88

d) 56

13)

a) 17

b) 25

c) 67

d) 33

14) A No, since the IQR measures the difference between points within a data set, and the range measures the difference between points at either end of the data set

15) Yes but only in the case where all values of the data set are equal

16)

a) 0

b) 1.58

c) 4.74

d) 37.35

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Further Mathematics: Solutions

Chapter 1: Data Analysis

Exercise 3: Normal Distribution

1)

a) The score is the mean

b) The score is one standard deviation above the mean

c) The score is two standard deviations below the mean

d) The score is more than two standard deviations above the mean

2) The score is one standard deviation above the mean

3) 9.5

4) 10

5) 17

6) 68%

7) Mean = 90, s.d. = 6.5

8) Three

9)

NAME	SCORE
James	C
Mark	E
Karen	B
Janine	C
Carol	C
June	C
Peter	D
Kevin	D
Brian	C
Alan	C

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10)

a) 96.4 to 103.6 kg

b) 34%

c) 8

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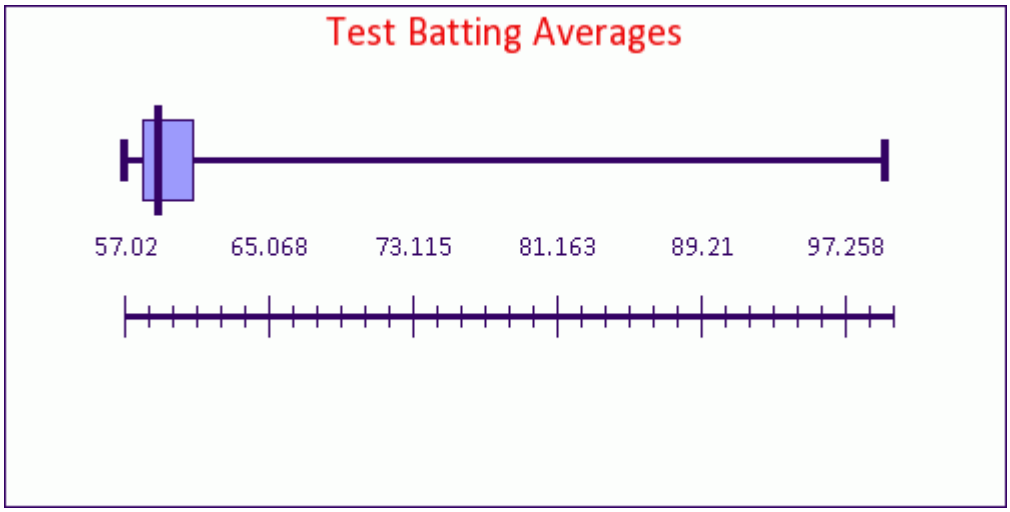
Further Mathematics: Solutions

Chapter 1: Data Analysis

Exercise 4: Box Plots

1)

a)



b) The mean will be higher since it is influenced by the outlier

c) Mean = 63.22, standard deviation = 12.97

d) The IQR, since the standard deviation is affected by the outlier

2)

a) 40

b) 75%

c) 86%

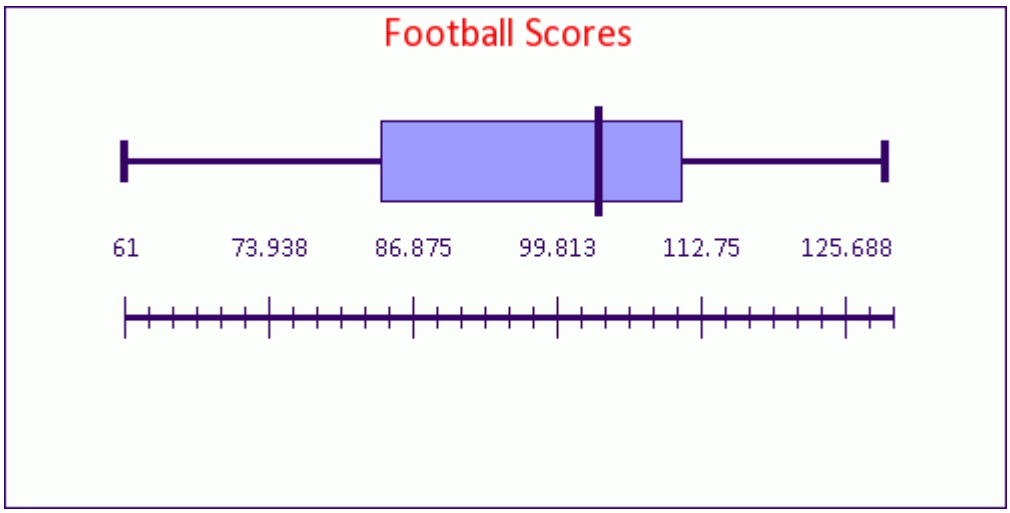
d) 25%

e) The test was easy, since the majority of the class scored well on it, the median was 86%, and three quarters of the class scored 70% or better

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3)

a)



b) 27

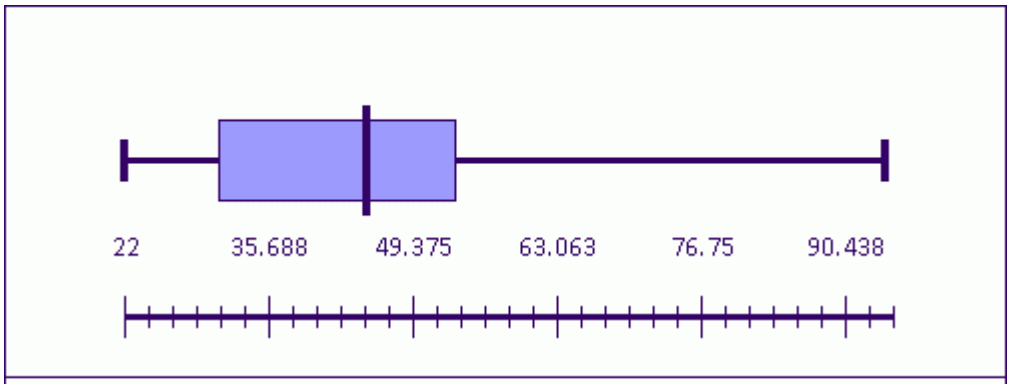
c) 103.5

d) There is a wide spread of scores, indicating that there are some evenly matched teams and some not so evenly matched. The spread may also be due to differing conditions in different parts of the country and over time

4) Hobart and Perth have similar distributions, but over a different set of data. Perth's range of temperatures is higher and slightly more widespread. Darwin has a small range of temperatures which are higher at all times than at any time in Hobart, and for most times in Perth.

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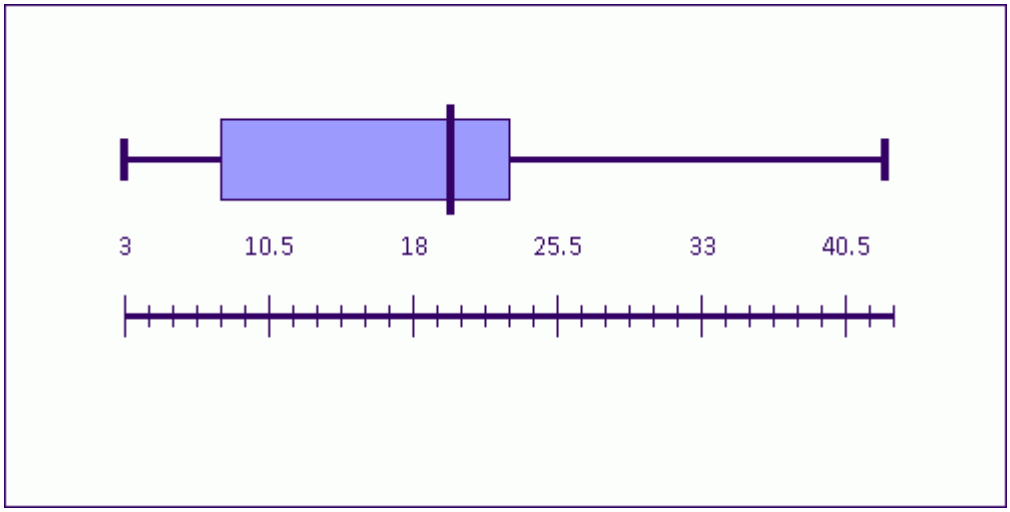
5)



22,24,27,27,29,30,31,31,31,33,33,35,35,45,45,45,47,48,49,5
51,52,52,55,63,63,65,66,67,95,

Quartile 1 = 31
Median = 45
Quartile 3 = 53.5
Mean = 44.9

6)



7) A

a) Water World (230)

b) Water World (0 and 400)

c) Water World

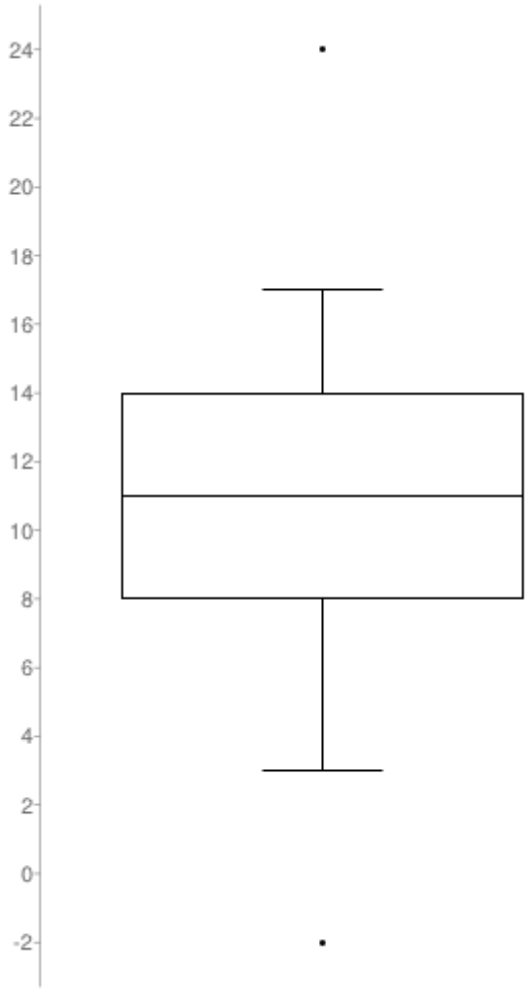
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d) Water World

e) Great Lake; its distribution of rainfall is less random

8)

a)



b) IQR = 6,

6 x 1.5 = 9

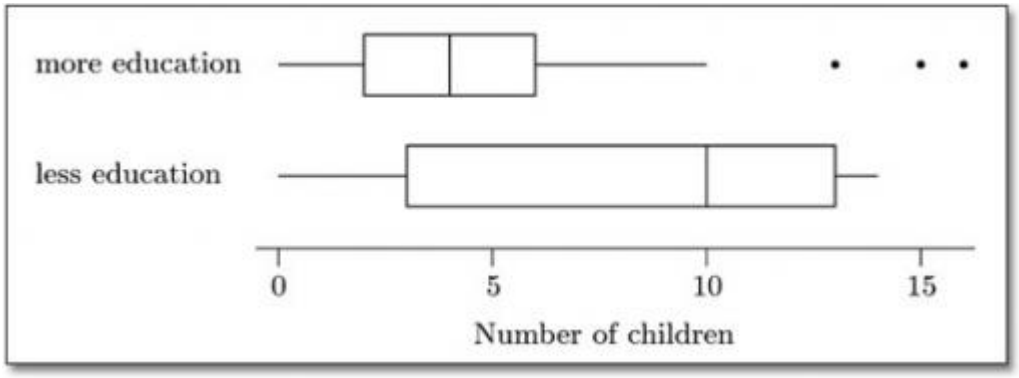
14 + 9 = 23

8-9 = -1

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9)

a)



b) The outliers are 16, 15, and 13 from the set of more education

c) The outliers increase the range of the "educated" set

d) With outliers, the range of the data for the "educated" group is higher. The median drops with the removal of the outliers, as does the IQR

e) Generally people with more education are likely to have fewer children. The reasons for this cannot be drawn from the graphs, however studies tied to educational outcomes (e.g. awareness of opportunities, busier lifestyle, social responsibility) could be conducted as a result of these findings

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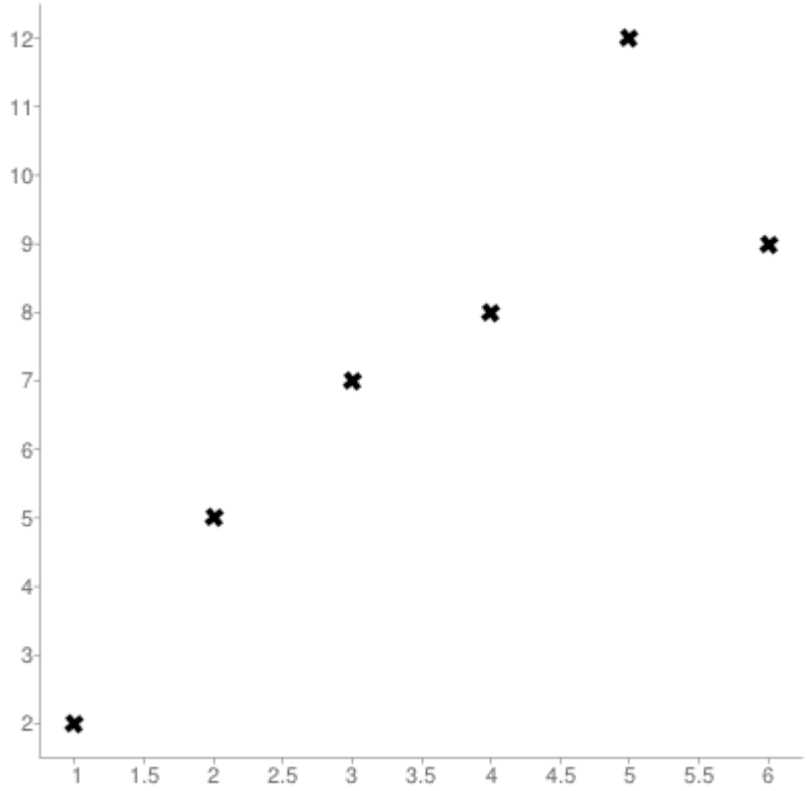
Further Mathematics: Solutions

Chapter 1: Data Analysis

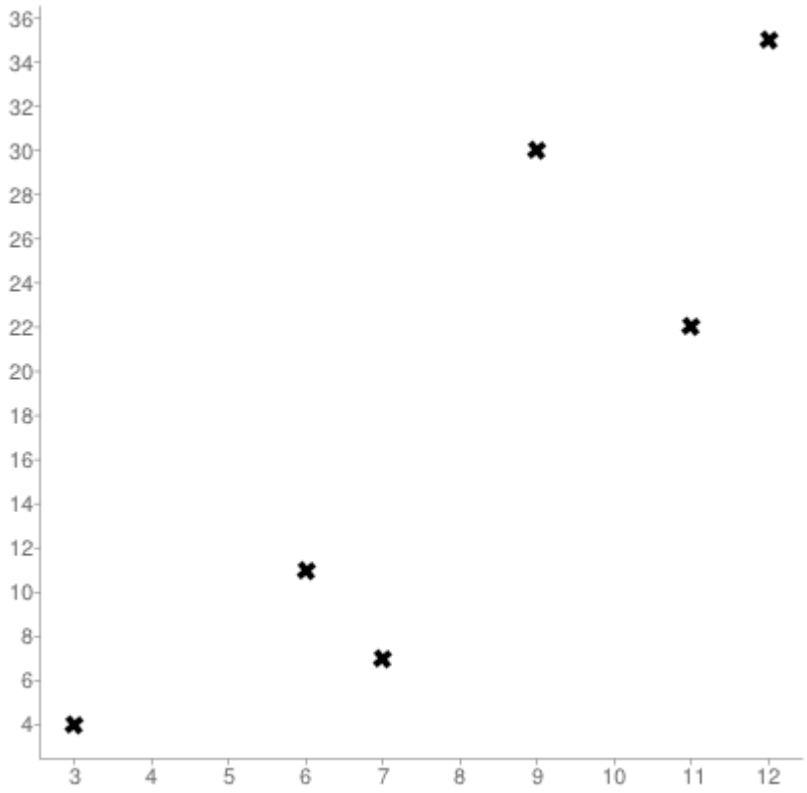
Exercise 5: Correlation

1)

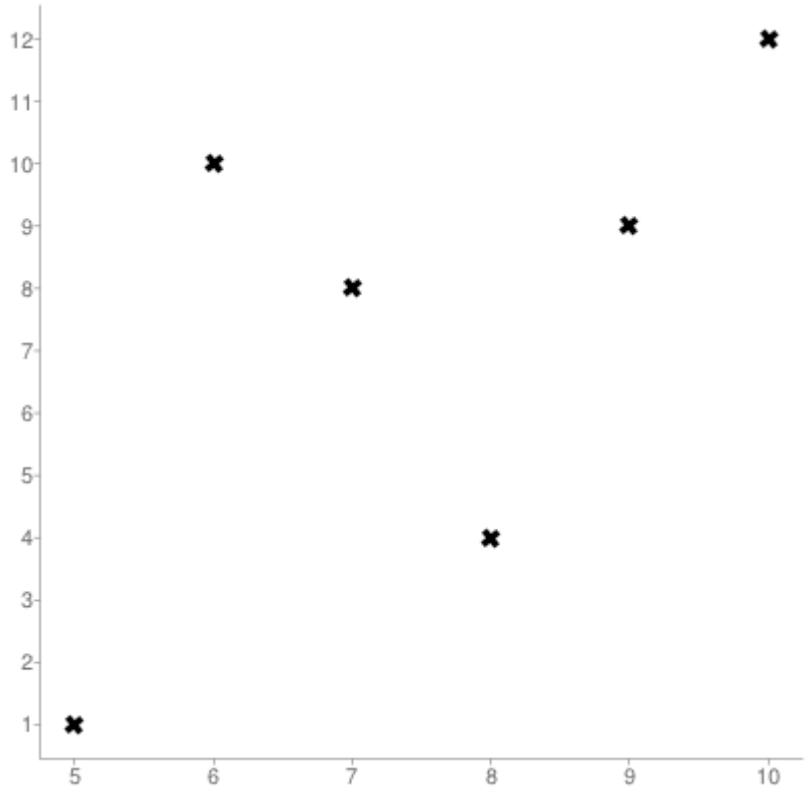
a)



b)

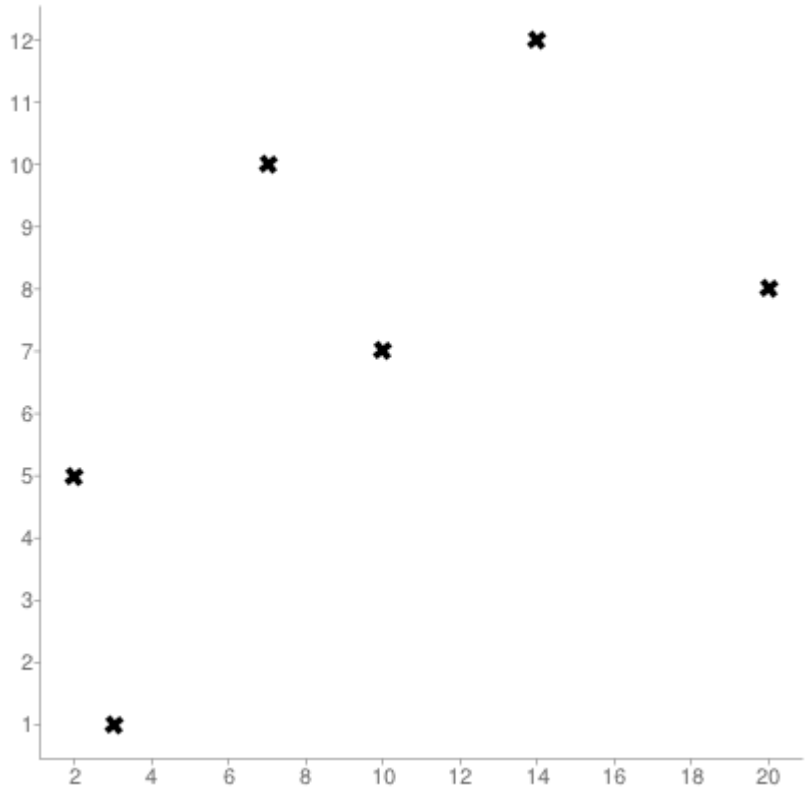


c)

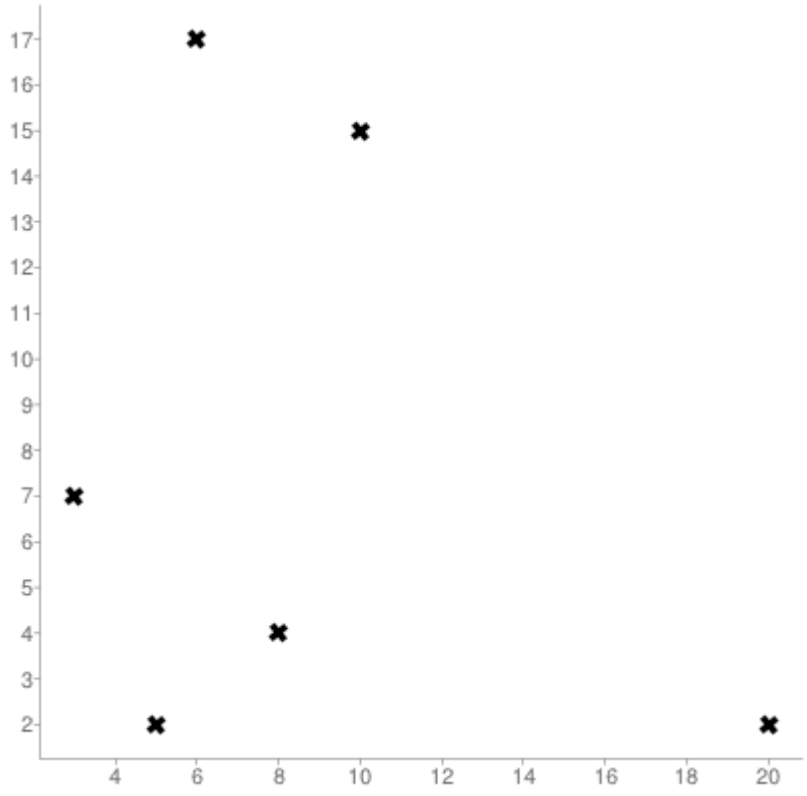


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d)

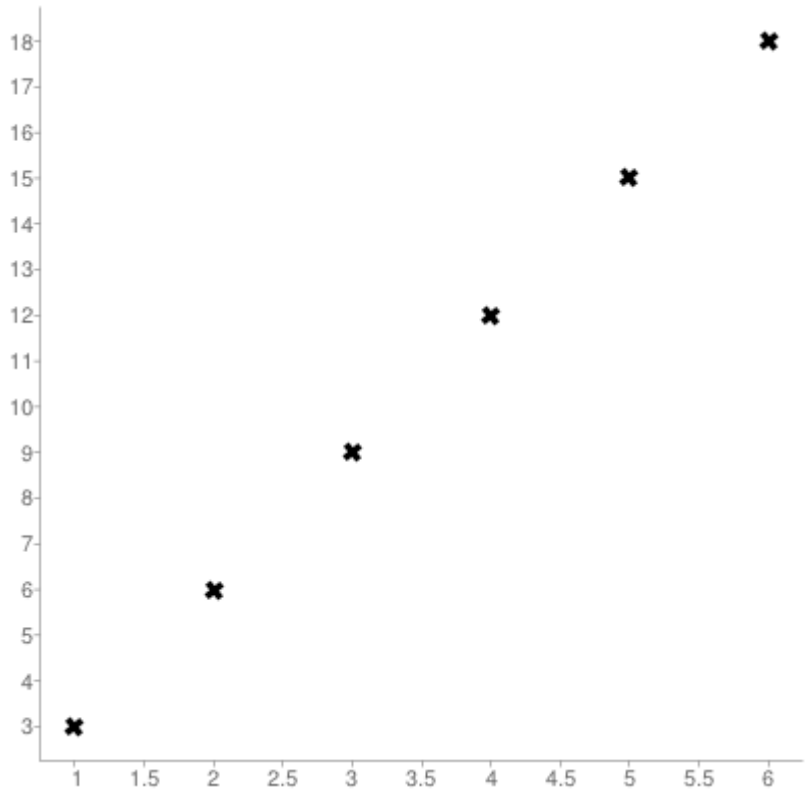


e)



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f)



2)

a) Strong positive

b) Strong positive

c) Medium positive

d) Medium positive

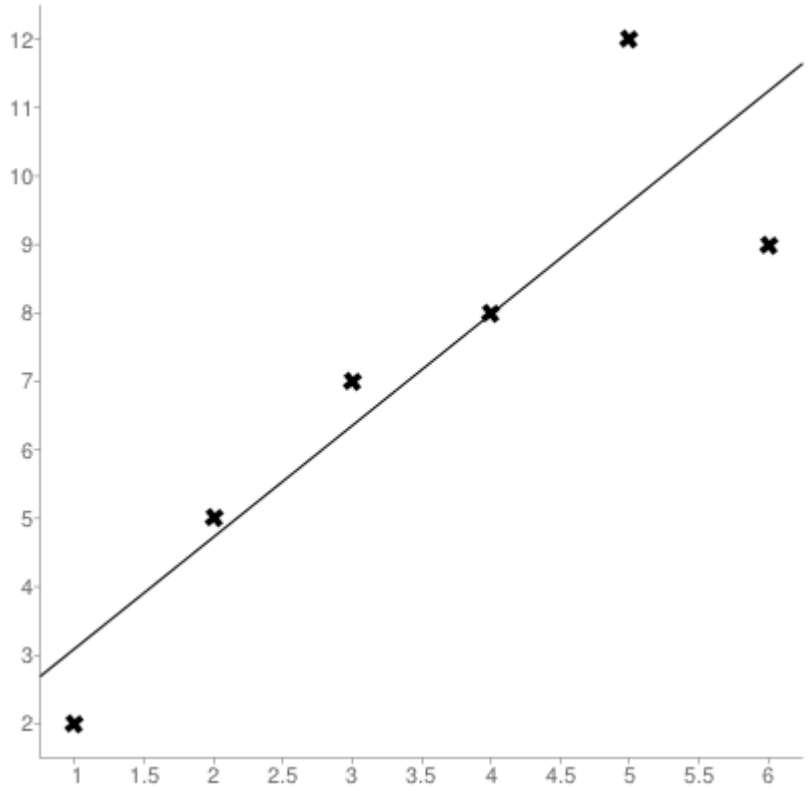
e) Weak negative

f) Perfect positive

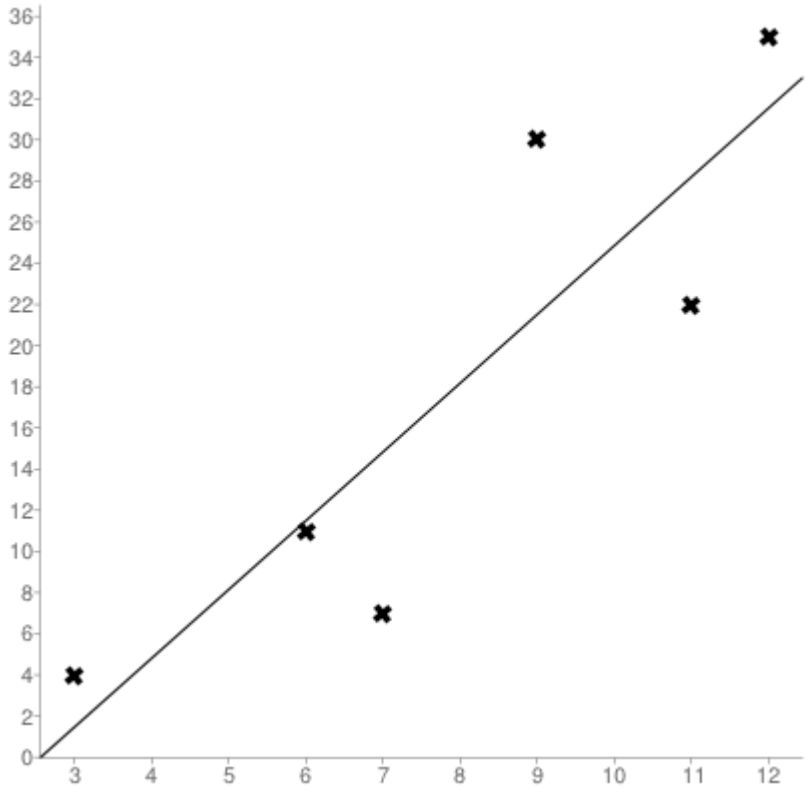
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3)

a)

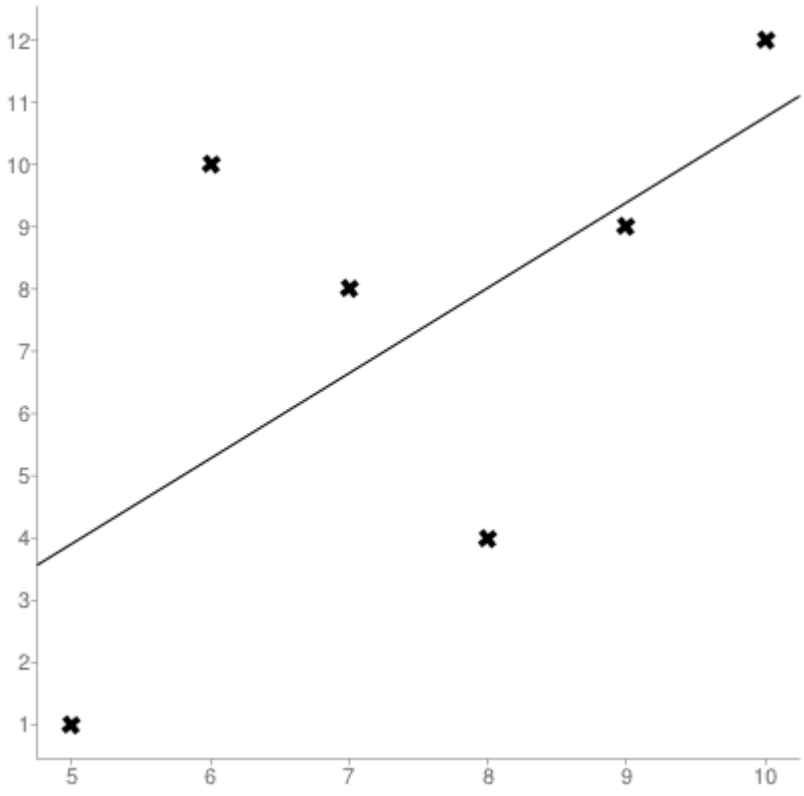


b)

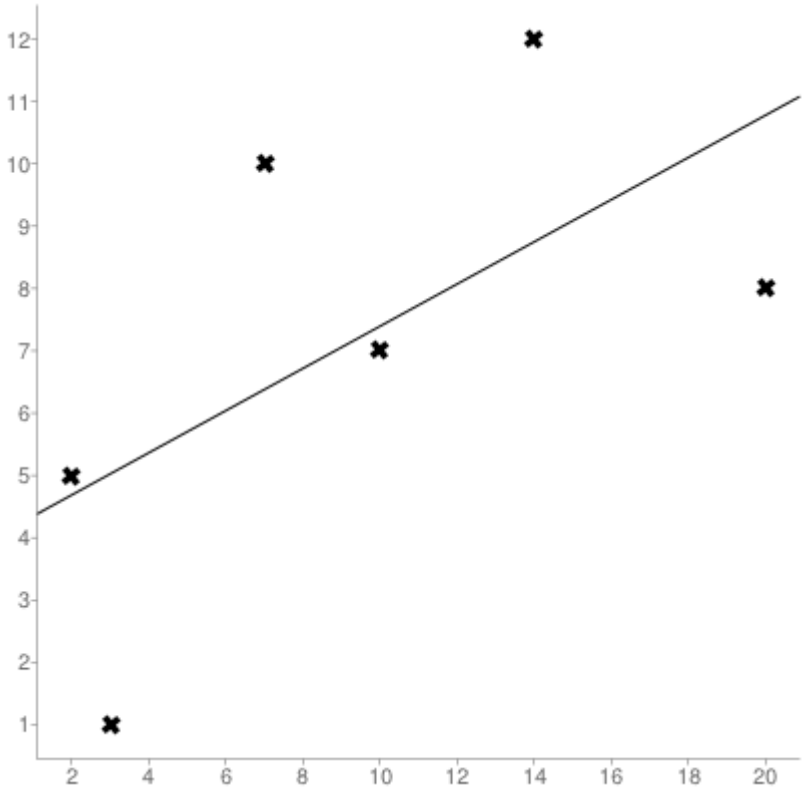


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c)

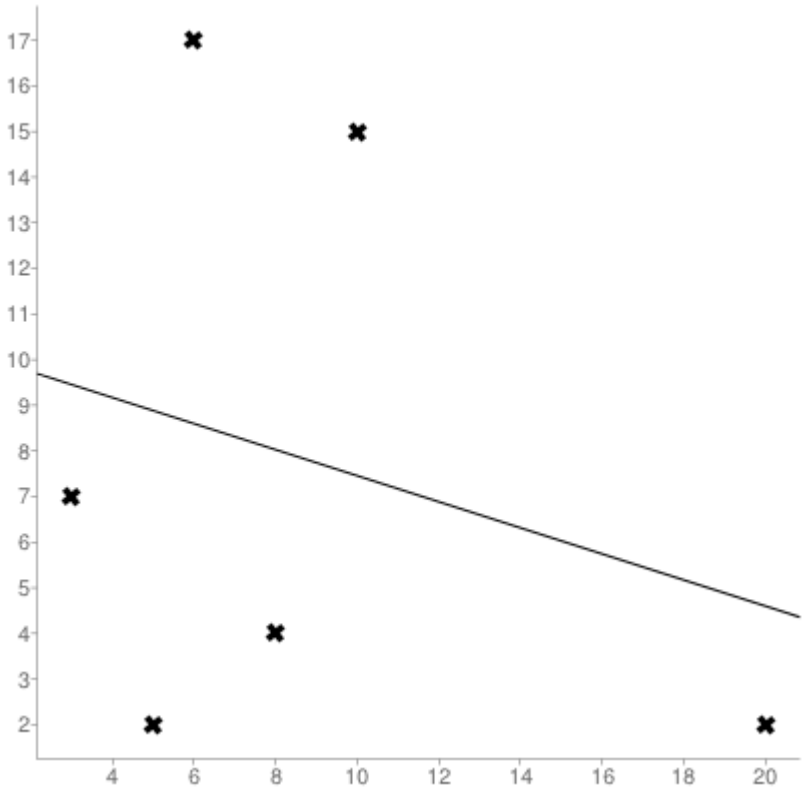


d)

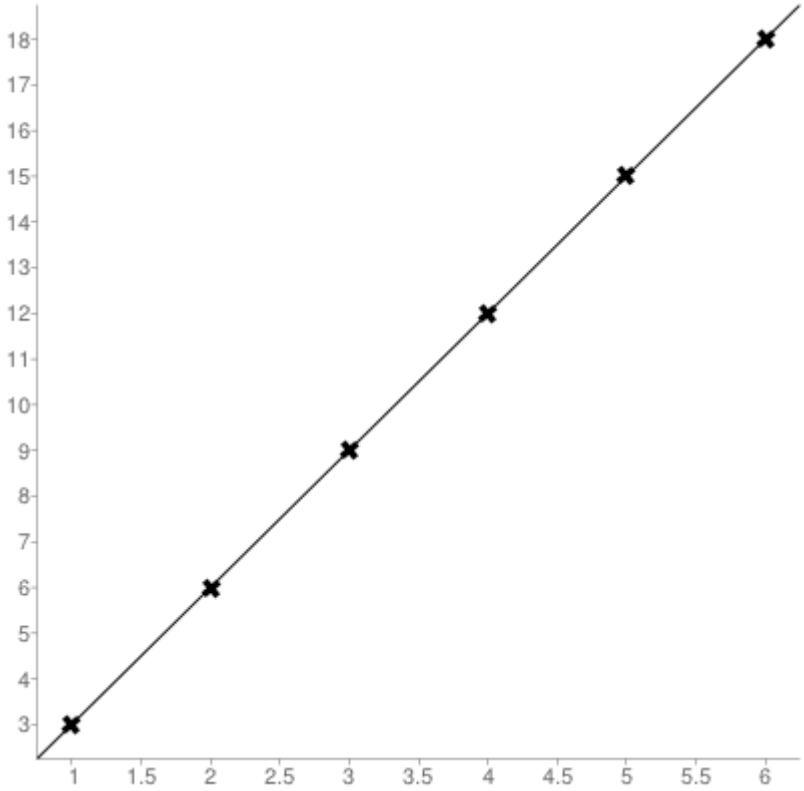


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e)



f)



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Equations of lines of best fit

1) $y = 1.63x + 1.47$

2) $y = 3.34x - 8.55$

3) $y = 1.37x - 2.95$

4) $y = 0.34x + 4$

5) $y = -0.29x + 10.3$

6) $y = 3x$

4)

a) 6.36

b) 1.47

c) 1.16

d) 5.02

e) 9.43

f) 9

5) The value of x is outside the range of data collected, to predict this point would require extrapolation, which is inaccurate

6)

a) Perfect positive

b) Strong positive

c) Weak negative

d) Medium positive

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e) Strong negative

f) No relationship

7) No; the high correlation is due to a common third factor, namely the seasons. When it is summer in Australia (the time of high air conditioner sales), it is winter in Canada (the time of high blanket sales). One event does not cause the other, they are linked by the common factor

8) The high correlation coefficient is due to a third factor; the relative wealth of the countries. A country whose population generally own one or more TV sets per household is likely to have a higher GDP, and the population more disposable income. With these comes such things as better nutrition, education and health care, all of which contribute to increased life expectancy

9) The data should show a high positive correlation, but only for a certain range of data. People generally get taller as they get older, but this usually stops at around 20 years old. From this point there would be no correlation since a 25 year old and a 50 year old would be around the same height. In fact as people get old they tend to shrink slightly, so for higher ages there may well be a negative correlation

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Further Mathematics

Number Patterns

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Further Mathematics: Solutions

Chapter 2: Number Patterns

Exercise 1: Arithmetic Sequences

1)

a) 2

b) 3

c) 6

d) -4

e) 12

2)

a) 2

b) 9

c) 8

d) 100

e) 10

3) 16

4) 174

5) 10

6) 29.2

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7)

a) $a = \frac{2x+2}{3}, d = \frac{x-2}{6}$

b) $\frac{10}{3}, \frac{11}{3}, 4$

c) $0, -\frac{1}{2}, -1$

8) 58

9) A: 13, 16, 19, 22

B: 13, 14, 15, 16

10) 11th term with a value of 20

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Further Mathematics: Solutions

Chapter 2: Number Patterns

Exercise 2: Geometric Sequences

1)

a) 2

b) 1.5

c) 0.5

d) 0.2

e) 1.5

f) $\frac{3}{10}$

2)

a) 2

b) $\frac{16}{9}$

c) 100

d) 1600

3) 162

4) 137438953472

5) 0.4

6) -2

7) $r = \frac{1}{4}, a = 384$

8) 3

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9) 243, 32

10) $\frac{81}{16}(x+3)$

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Further Mathematics: Solutions

Chapter 2: Number Patterns

Exercise 3: Sum to Infinity

1)

a) 16

b) $\frac{100}{9}$

c) $\frac{48}{3}$

d) 1.25

e) $\frac{51.2}{7}$

f) Cannot be calculated since r>1

2) 1

3) 6

4) 0.25

5) $0.\dot{9} = 0.9999\ldots = 0.9 + 0.09 + 0.009 + \cdots$

This is a geometric series with $a = 0.9 \; and \; r = 0.1$

Sum to infinity $= \frac{0.9}{1-0.1} = 1$

6) You would get the same amount ($40)

7) $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots . = 1$

8) Series is $10 + 8 + 6.4 + \cdots$

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$a = 10, r = 0.8$

Sum to infinity = 50

Therefore will not lose required weight

9) 12 kg

10) $6x$

Further Mathematics: Solutions

Chapter 2: Number Patterns

Exercise 4: Difference Equations

1)

a) 2, 5, 8, 11, 14

b) 4, 3, 2, 1, 0

c) 6, 9, 13.5, 19.75, 29.625

d) 2.5, 5, 10, 20, 40

e) 3, 7, 4, -3, -7

2)

a) $T_{n+1} = T_n + 2, T_1 = 1$

b) $T_{n+1} = T_n \div 2, T_1 = 100$

c) $T_{n+2} = T_{n+1} + T_n, T_1 = 1, T_2 = 2$

d) $T_{n+1} = T_n \div 4, T_1 = 100$

3) $T_{n+1} = T_n \times 0.25, T_1 = 1000$

4) $T_{n+1} = T_n \times 1.1, T_1 = 12$

19.3 degrees

5)

a) $T_{n+1} = T_n \times 1.02, T_1 = 20000$

b) $21224.16

c) $20000(1.02)^n$

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6) During the tenth hour

7)

a) $T_{n+1} = T_n \times 1.05, T_1 = 300$

b) 6 years

8)

a) $T_{n+1} = 0.9(T_n) + 25, T_1 = 25$

b) $250(1 - 0.9^n)$

c) 250

9)

a) Increase n by 1

Equation becomes

$F_{n+2} \times F_n - F_{n+1}^2$

Remember that $F_{n+2} = F_n + F_{n+1}$

Equation becomes $(F_n + F_{n+1})F_n - F_{n+1}(F_n + F_{n-1})$

$= F_n^2 - F_{n+1} \times F_{n-1}$

This is the opposite of the original equation

Therefore the equation has two solutions; itself or the opposite of itself, which depends on the value of n

b) Let $k = 2$

Then $F_{n+k} = F_{n+2} = F_{n+1} + F_n$

Let k=3

Then $F_{n+k} = F_{n+3} = F_{n+2} + F_{n+1} = F_{n+1} + F_n + F_{n+1} = 2F_{n+1} + F_n$

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Let $k = 4$

Then

$$F_{n+k} = F_{n+4} = F_{n+3} + F_{n+2} = 2F_{n+1} + F_n + F_{n+1} + F_n = 3F_{n+1} + 2F_n$$

Similarly it can be shown that when $k = 6$

$$F_{n+k} = F_{n+6} = 8F_{n+1} + 5F_n$$

Note that for any k, the coefficients of $F_{n+1}\ and\ F_n\ are\ F_k\ and\ F_{k-1}$

For example when $k = 6, F_6 = 8, F_5 = 5$

10) 12 days

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Further Mathematics

Geometry & Trigonometry

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Further Mathematics: Solutions

Chapter 3: Geometry & Trigonometry

Exercise 1: Pythagoras' Theorem

1)

a) 5 cm

b) 10 cm

c) 10.82 cm

d) 25.06 cm

e) 14.77 cm

f) 13.73 cm

2)

a) 5 cm

b) 24 cm

c) 22.45 cm

d) 14.14 cm

e) 8.49 cm

f) An equilateral triangle cannot form a right angled triangle

3) 9.43 km

4) 1.32 m

5) 1.9 m

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6) 8.27 m

7) 50 cm

8) 152.79 m

9) 107.7 cm

10) $\sqrt{50}$ cm

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Further Mathematics: Solutions

Chapter 3: Geometry & Trigonometry

Exercise 2: Similarity

1)

a) SSS

b) AA

c) SAS

d) Not similar

e) SAS

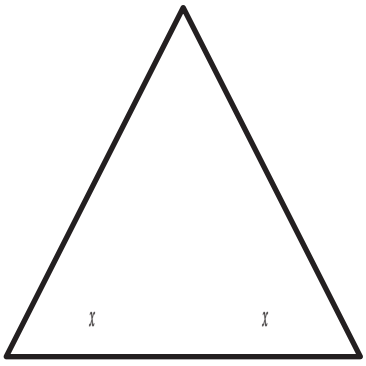
2) Their bases should be parallel, to ensure the corresponding angles are equal

3) A and C, since the ratio of their sides are the same

4) A and C by SAS

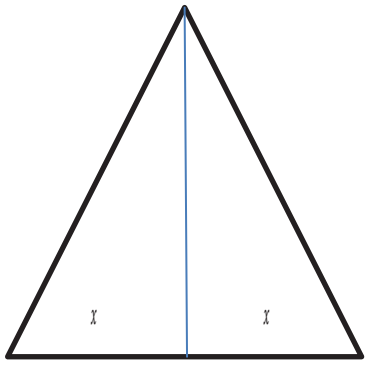
5) They share a common angle, and their base angles are equal since their bases are parallel

6)



Draw a perpendicular bisector from the apex to the base. The two triangles formed are similar (AA). Therefore the two corresponding sides are equal

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7)

a) AA

b) SAS

c) AA

d) SSS

e) SAS

f) AA

8) 5 metres

9) 16 metres

10) 3 metres

11) 1 cm

12) 5 cm

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Further Mathematics: Solutions

Chapter 3: Geometry & Trigonometry

Exercise 3: Volume & Surface Area

1)

a) $48\pi \text{ mm}^2$

b) $300\pi \text{ mm}^2$

c) $0.48\pi \text{ m}^2$

d) $18\pi \text{ m}^2$

2)

a) $400\pi \text{ cm}^2$

b) $0.1225\pi \text{ m}^2$

c) $192\pi \text{ m}^2$

d) $768\pi \text{ mm}^2$

3)

a) $24\pi \text{ cm}^2$

b) $65\pi \text{ cm}^2$

c) 39 cm^2

d) Approx 26 cm^2

4) $\frac{20\pi}{3} \text{ cm}^3$

5) $\frac{20}{3} \text{ cm}^3$

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6) 2.5 cm

7)

a) $\frac{32\pi}{3}$ mm^3

b) $\frac{4000\pi}{3}$ cm^3

c) $\frac{131.072\pi}{6}$ m^3

d) $\frac{16000\pi}{3}$ mm^3

8) 6.35 mm

9) 9 mm

10) 8 m

11)

a) 256π cm^2

b) 150π cm^2

c) 100π cm^2

d) 128π cm^2

12) $216 + 2\pi\sqrt{40} - 4\pi$ cm^2

(SA of cube + SA of cone – SA of hole)

13)

a) 13230 + 425.25π cm^3

b) 41472 + 5376 = 46848 cm^3

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c) $250\pi + 100\pi = 350\pi \text{ cm}^3$

14) 10 mm

—

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Further Mathematics: Solutions

Chapter 3: Geometry & Trigonometry

Exercise 4: Change of Scale

1) 22.5 cm

2)

a) It is tripled

b) It is tripled

3)

a) It is halved

b) It is quartered

4)

a) It is increased by a factor of 9

b) It is increased by a factor of 27

5)

a) It is increased by a factor of 4

b) It is increased by a factor of 8

6) It is increased by a factor of 16

7)

a) The SA of the bases increases by a factor of 4 and the SA of the body increases by a factor of 6

b) It is increased by a factor of 12

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8) 600 mm

9) Approximately 235π cm^2

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Further Mathematics: Solutions

Chapter 3: Geometry & Trigonometry

Exercise 5: Trigonometry (I)

1)

a) 2.5 cm

b) 7 cm

c) 2.5 cm

d) 12.44 cm

2)

a) 53

b) 31

c) 24

d) 30

3) 48 degrees

4) 31 degrees

5) 1225.69 cm^2

6) 19.2 km, 51.34 degrees

7) Angle of elevation is A, angle of depression is D. The angles are equal

8) 83.9 metres

9) 214.45 m

10) 14.32 metres

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11) 37.32 metres

12) 46.2 metres

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Further Mathematics: Solutions

Chapter 3: Geometry & Trigonometry

Exercise 6: Trigonometry (II)

1)

a) 7.88

b) 8.77

c) 7.78

d) 36.1 degrees

e) 50.4 degrees

f) 24.7 degrees

2)

a) 6.96

b) 15.33

c) 29.24

d) 28.2°

e) 41.4°

f) 65°

3)

a) 3.21

b) 67.5

c) 17.2

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d) 118.3

e) 107.1

f) 55.2

4)

a) 15.7 or 18.2 metres

b) 5.14 km

c) 54.7 m

d) 52.1 m

e) 166.23 m

f) 53.9 km

5)

a = 156.4 or 181.3

b = 15.4

d = 130.4

6) 154.4°

7) 1.81 km

8) 6.59 km

9) 381 km

10) 7 km and 80.6°

11) 22.95 km

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12) 9.54 km

13) 65 km

14) 48.4 and 61.7 km respectively

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Further Mathematics

Graphs & Relations

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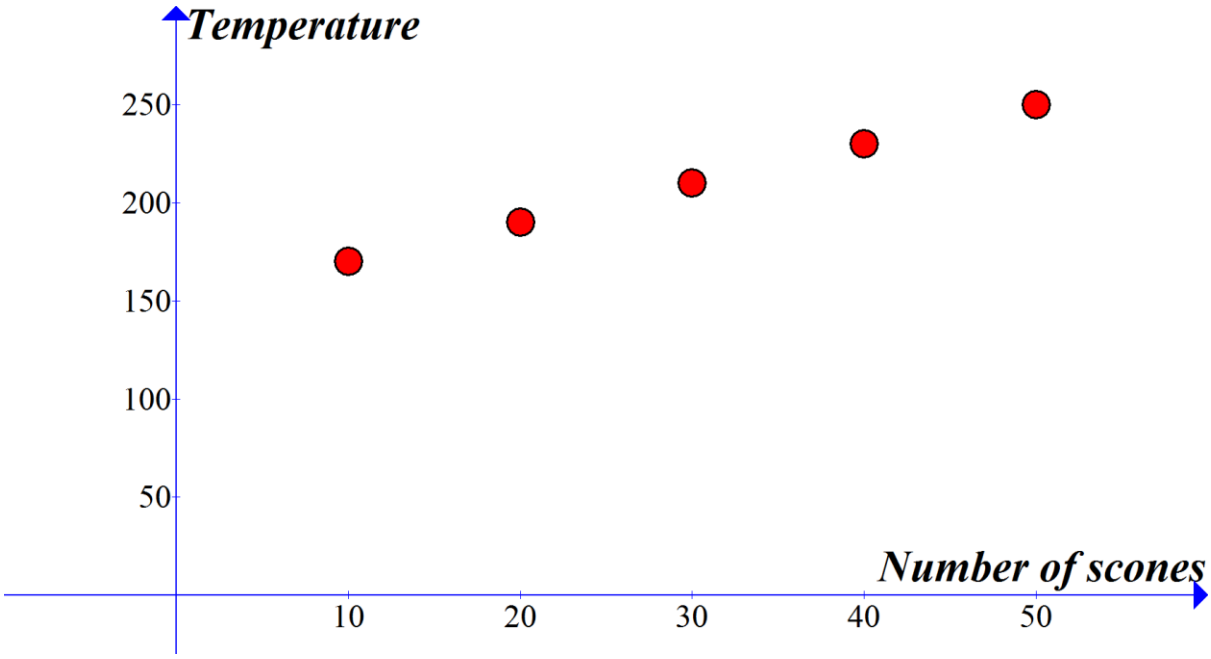
Further Mathematics: Solutions

Chapter 4: Graphs & Relations

Exercise 1: Linear Relationships

1)

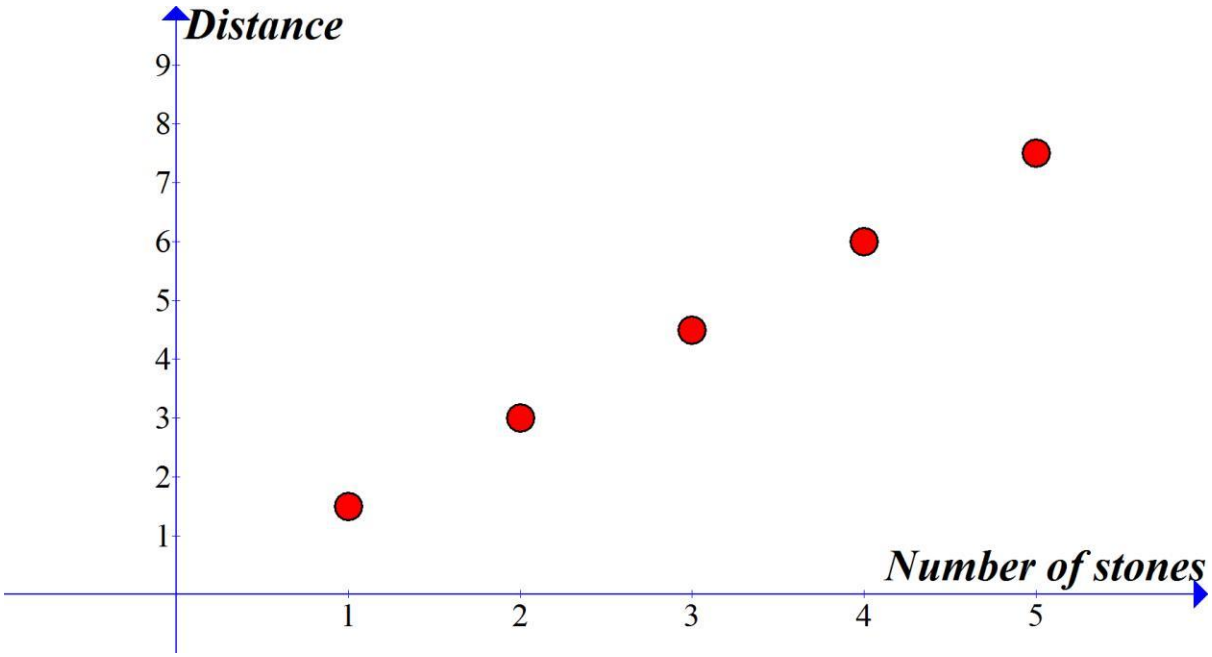
Number of scones	10	20	30	40	50
Temperature	170	190	210	230	250



Points cannot be joined because

- Relationship does not extend for all values of x, for example 500 scones
- There would be values of fractional scones on the x axis which does not make sense

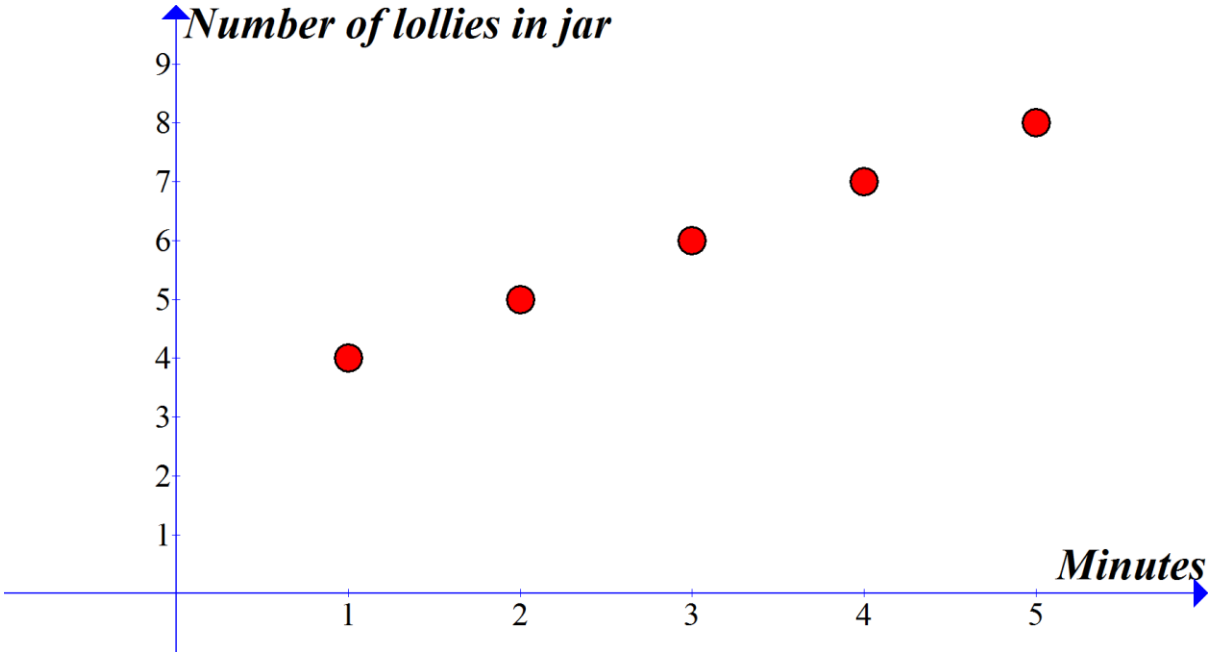
2)



Points cannot be joined since

- The river would not go on forever, therefore some points would be meaningless
- There would be fractional values of number of stones

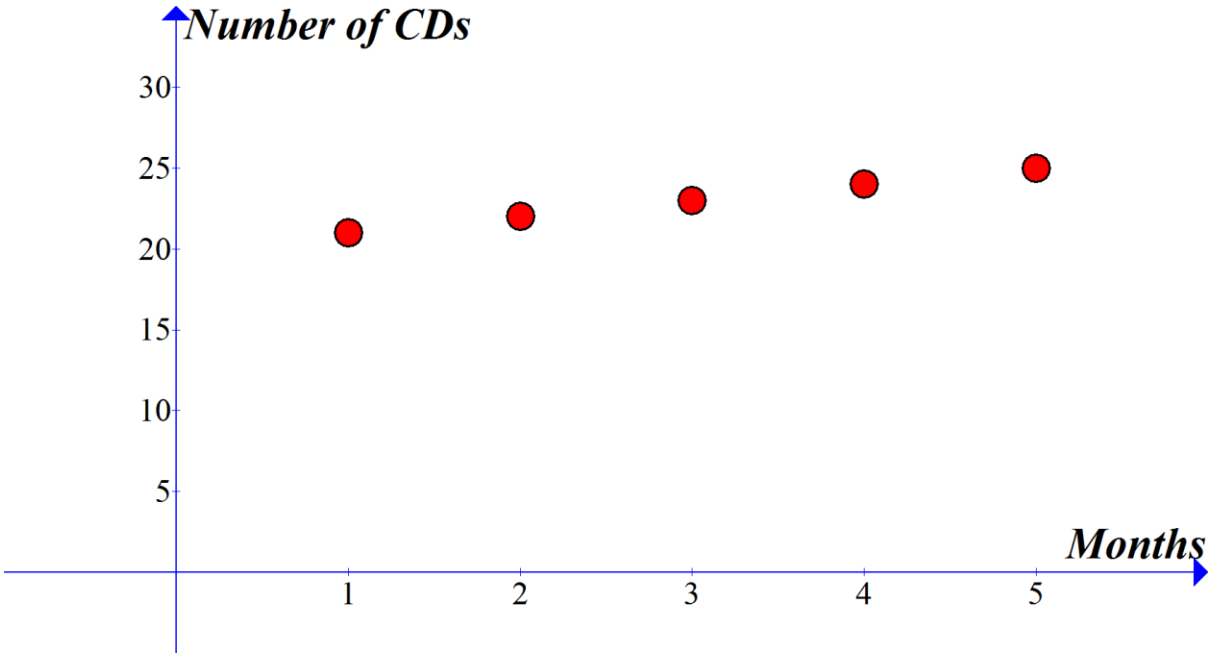
3)



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Points should not be joined since the jar must have a limit of lollies it can hold, and there would be fractional amount of lollies in the jar at certain times according to the graph if the points were joined

4)



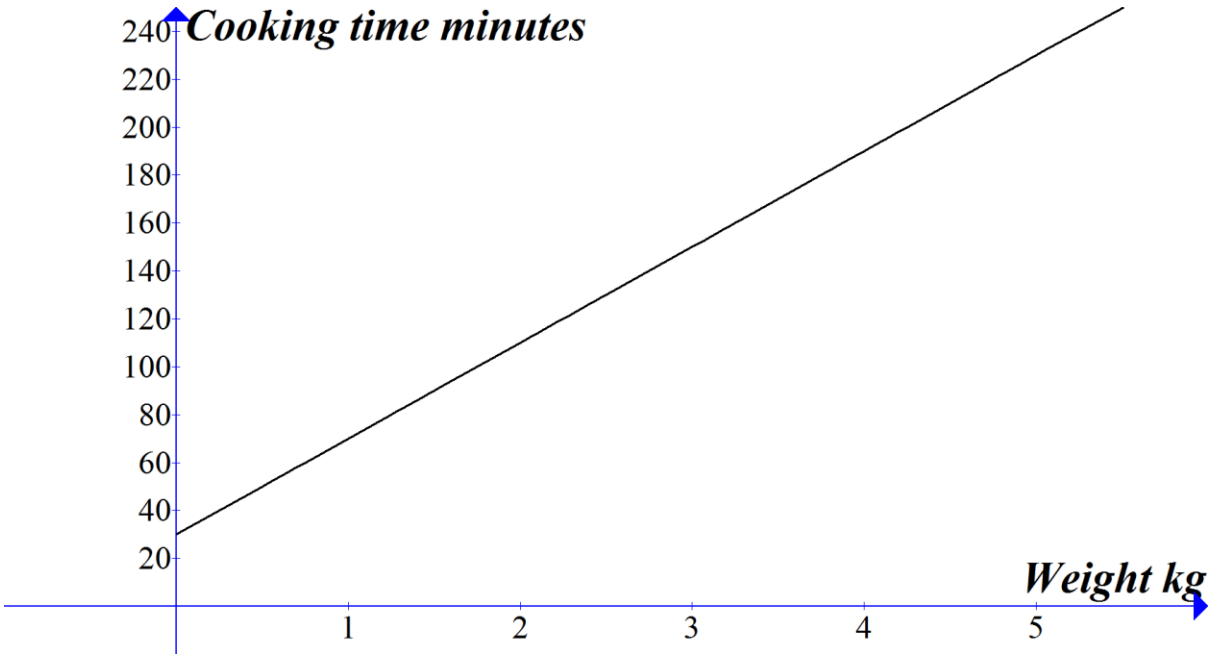
The points should not be joined since

- He buys a CD each month, therefore part months are meaningless
- There would be fractional numbers of CDs at various times

5)

A 1.5 kg roast should cook for 90 minutes

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Gradient is 40 which is the incremental time for each kg of meat

The y intercept is 30 which is the time added to the actual cooking time for a certain weight

No, there would be a limit to the weight of meat and the time for cooking

6)



He charges $65 for 2 hours and $95 for 3.5 hours

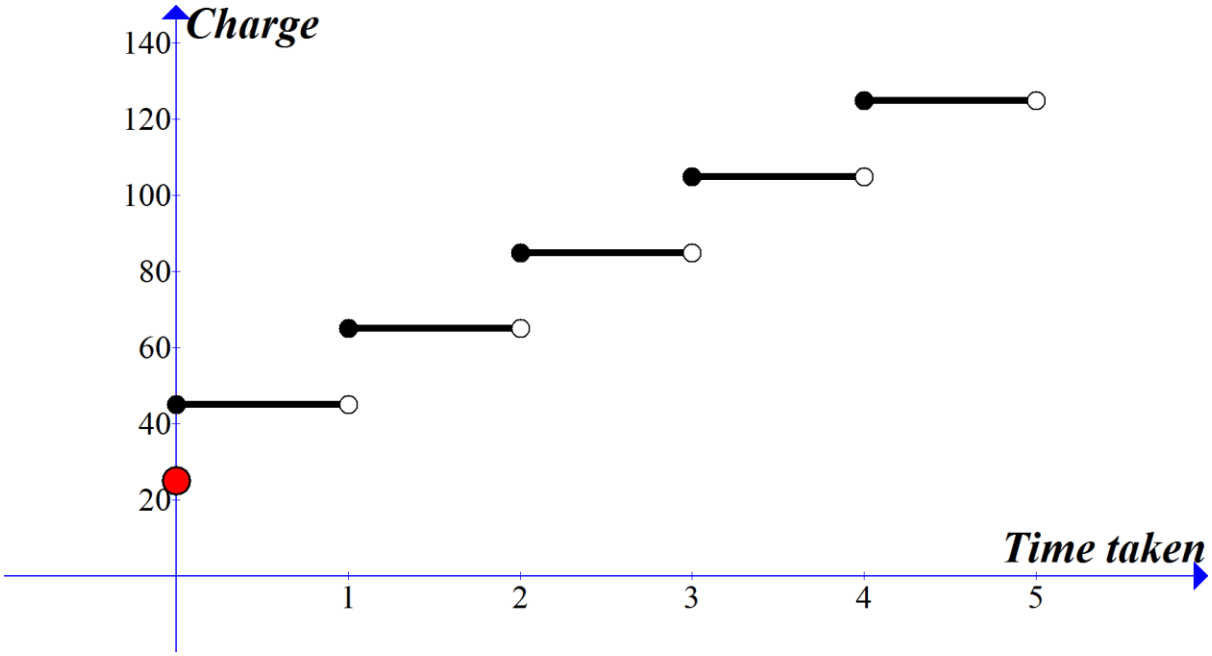
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Gradient is 20 which is the charge per hour after callout fee

The y intercept is 25 which is his callout fee before any work is commenced

The graph cannot be extended indefinitely, since there would be a limit to how much time could be spent on one job

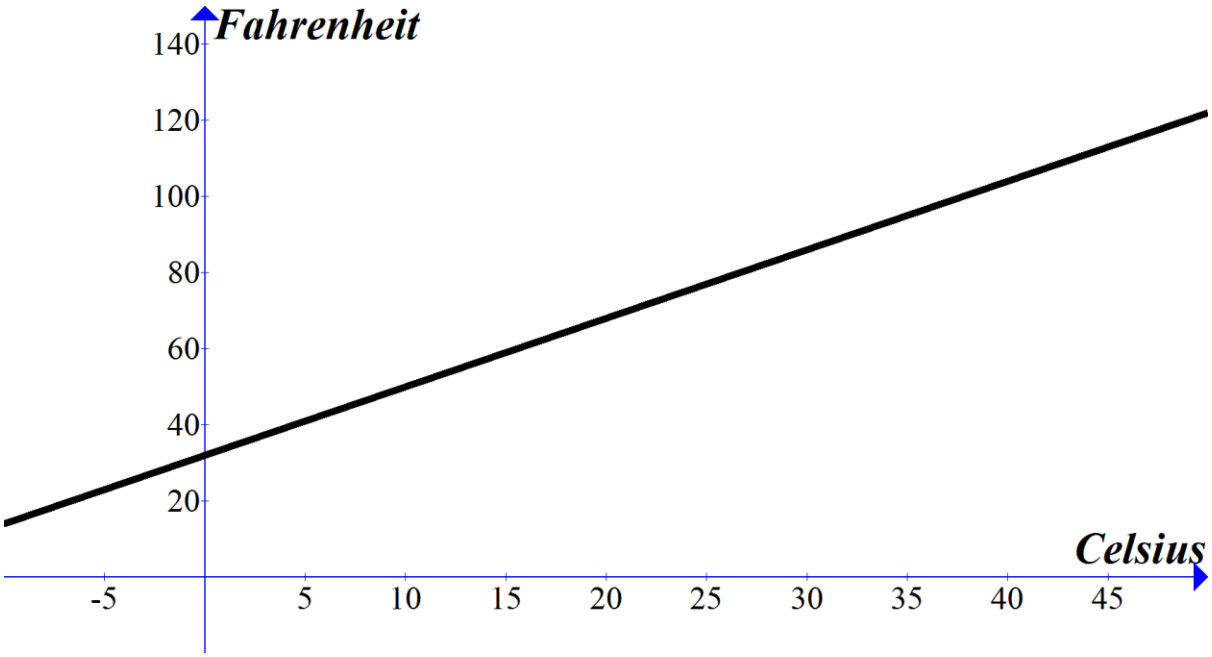
7)



He charges $65 for 2 hours, and $105 for 3.5 hours

The graph differs since It is a step function which reflects the different charging rates

8)



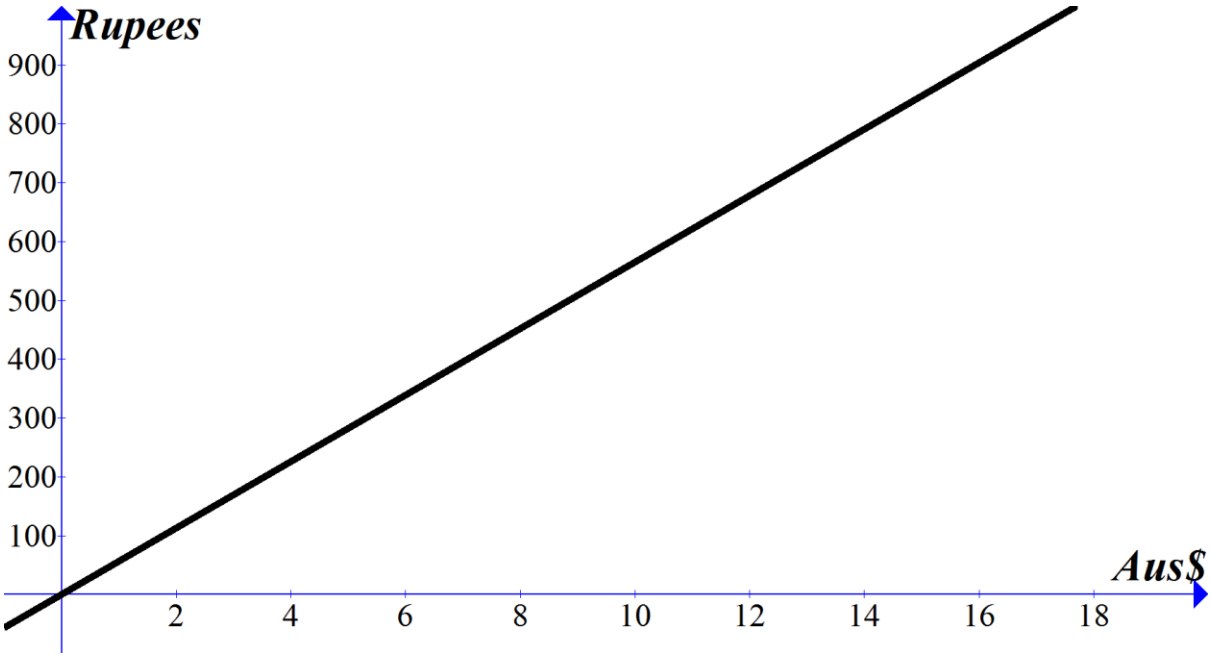
The gradient is 1.8 which is the change in degrees Fahrenheit for each change of 1 degree Celsius

The y intercept is 32 degrees which is the temperature in Fahrenheit for 0 degrees Celsius

Although there must be physical limits, to all intents and purposes the graph can be extended to both high low and fractional values of both variables

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9)



The gradient is 56.5 which is the number of extra rupees each additional dollar can buy

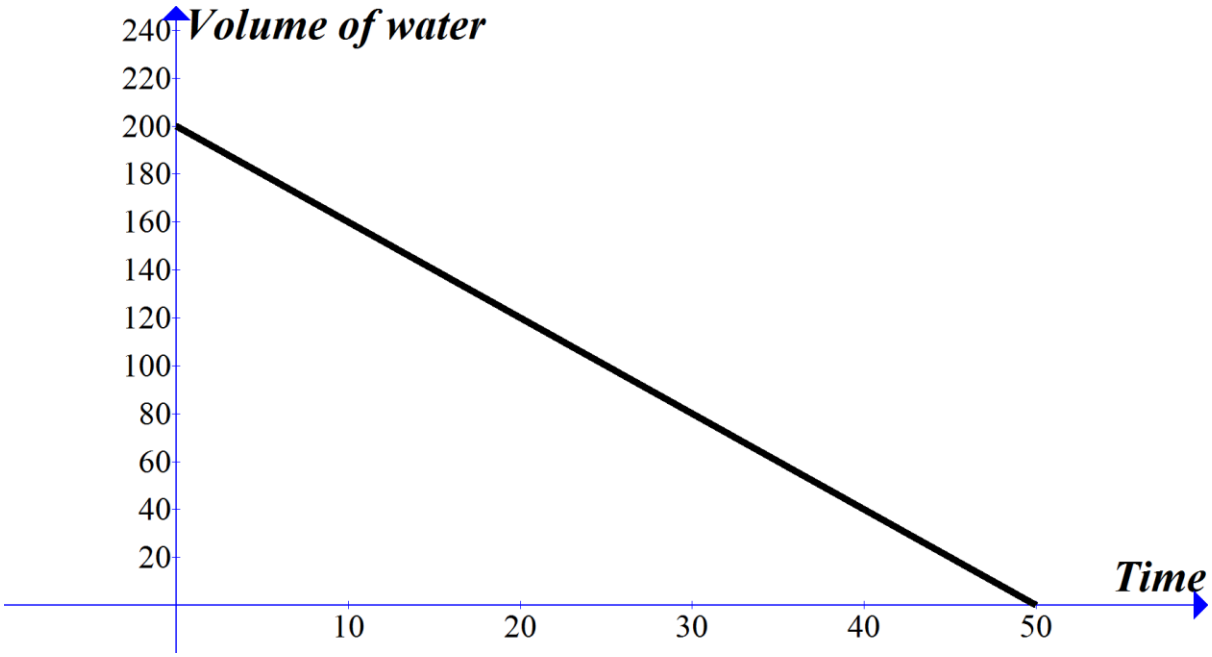
The y intercept is 0; no dollars will get you no rupees!

22,600 rupees

$30

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10)



The bath is empty after 50 seconds

The gradient is (-4) which is the volume of water leaving the bath for each minute of time passed, it is negative indicating a decreasing relationship

The y intercept is 200 which is the initial volume of water (t-0)

No, it is not valid for negative values of t and once the bath is empty (t=50) the relationship is invalid since it is not modelling a situation

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Further Mathematics: Solutions

Chapter 4: Graphs & Relations

Exercise 2: Simultaneous Equations

1)

a) (3, -2)

b) (6, 2)

c) (1, 1)

d) (2, 2)

e) (3, 1)

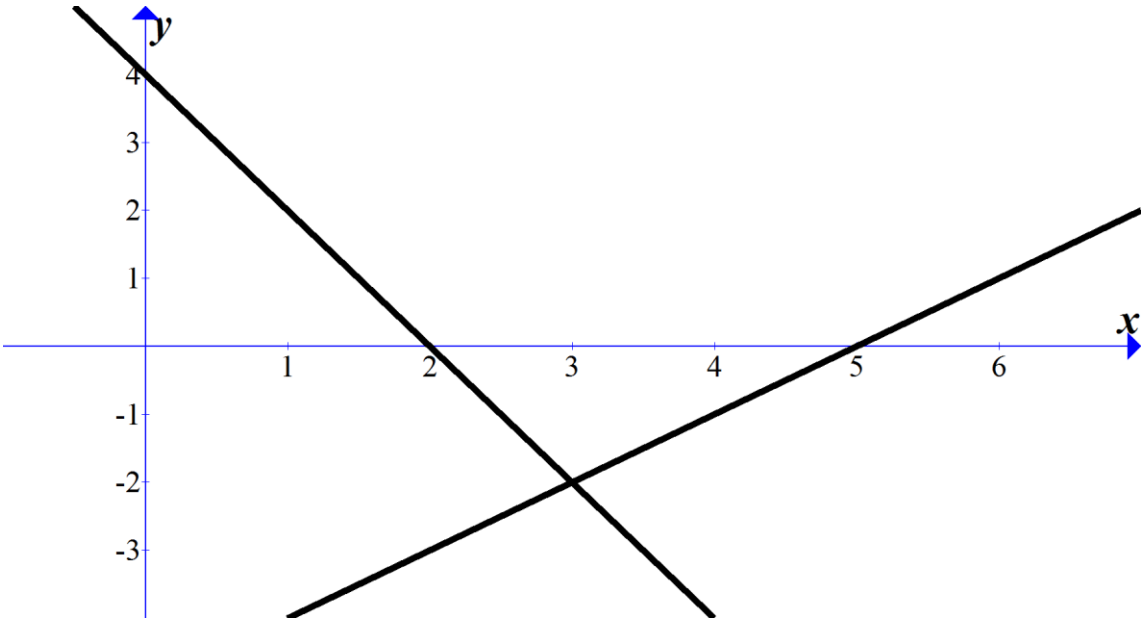
f) (4, 1)

g) (0, 1)

2) Tables should verify answers to q1

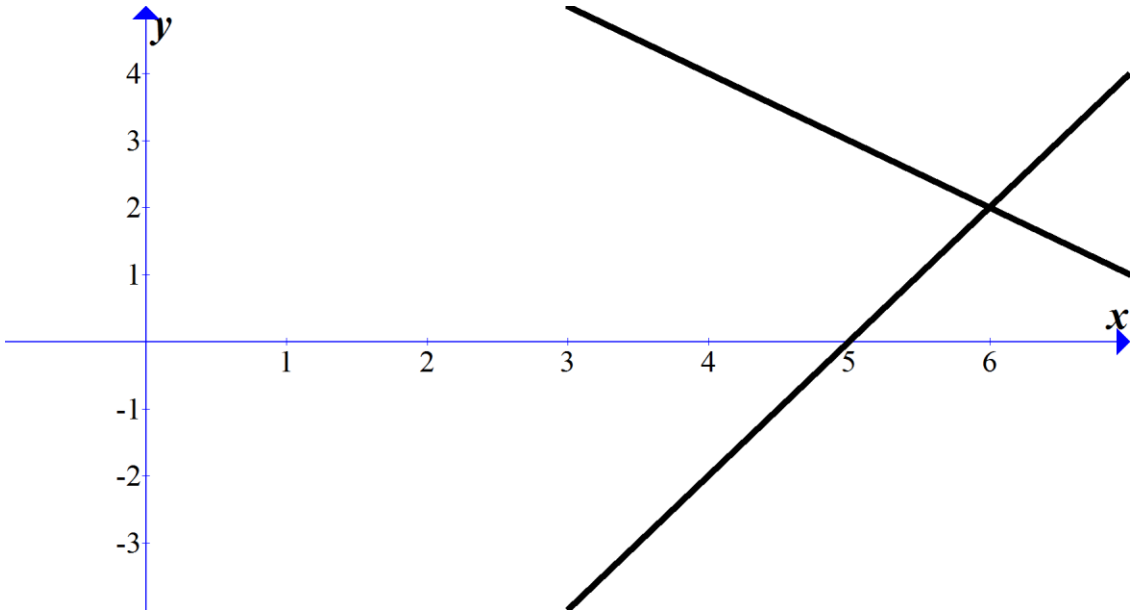
3)

a)

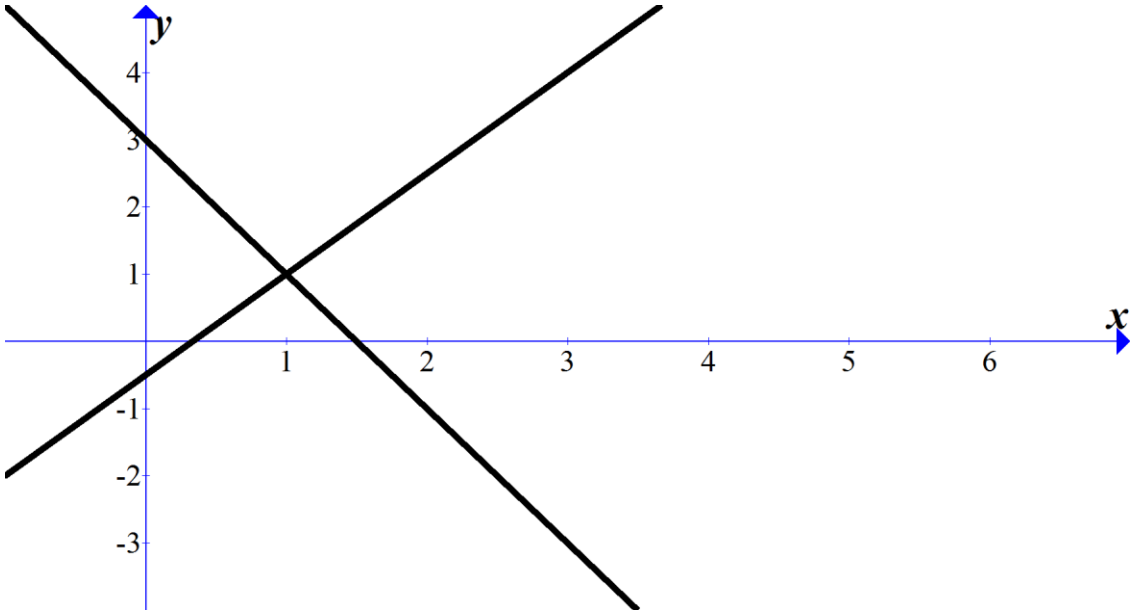


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b)

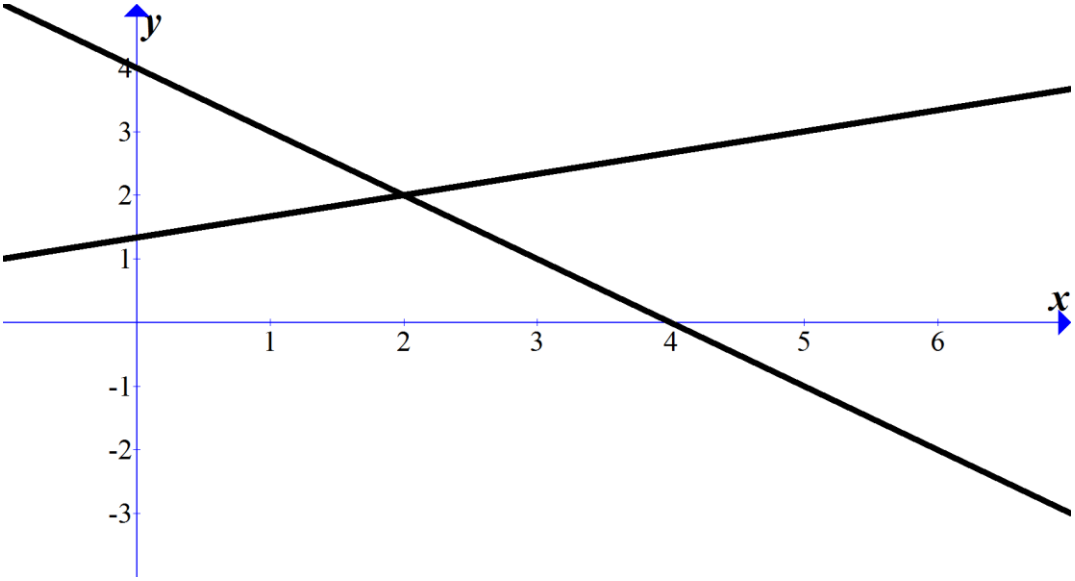


c)

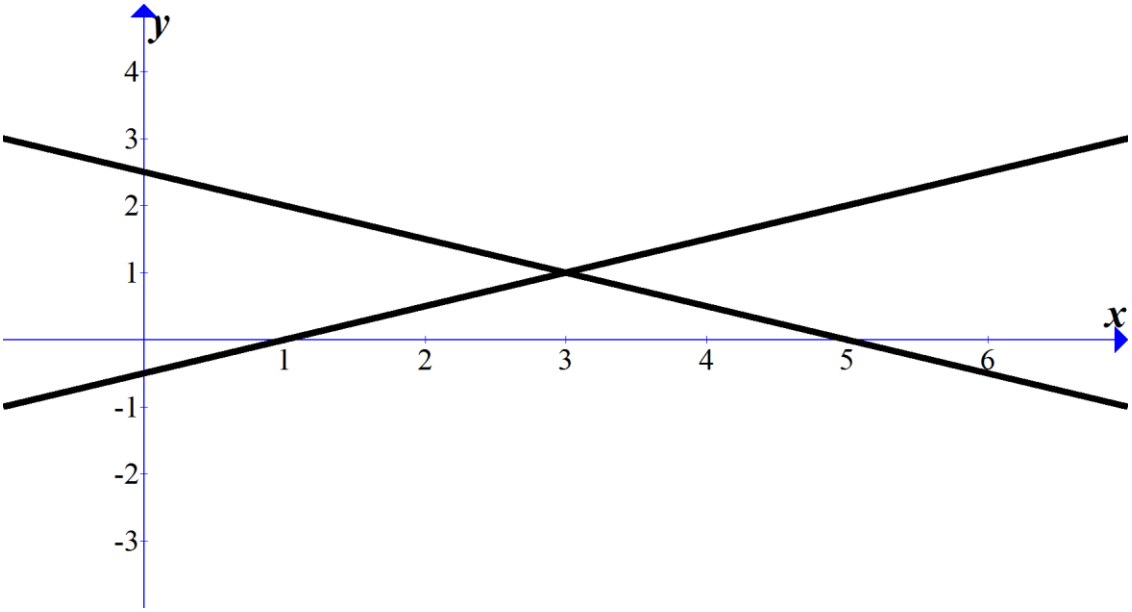


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d)

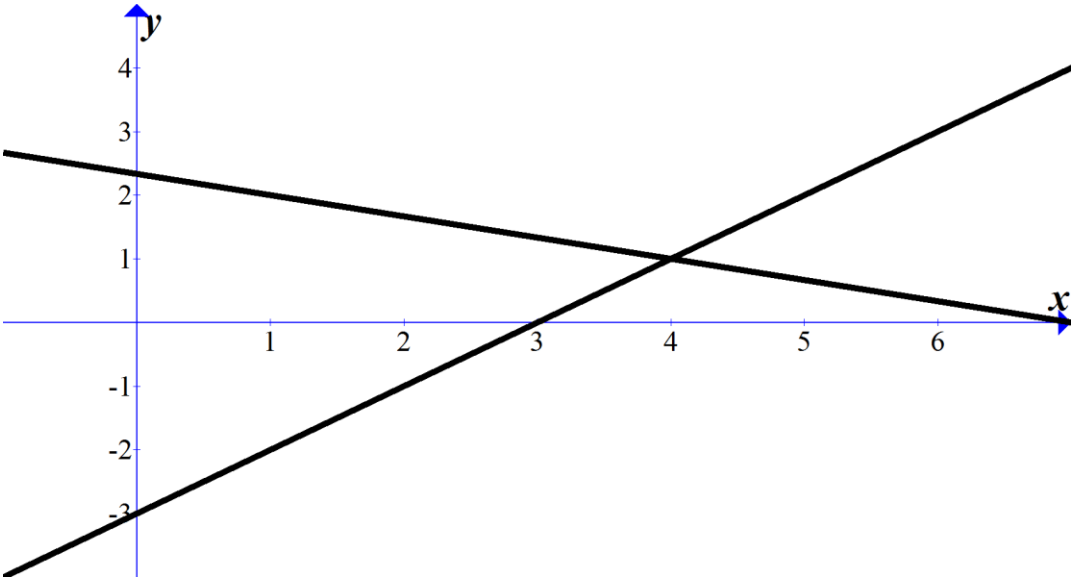


e)

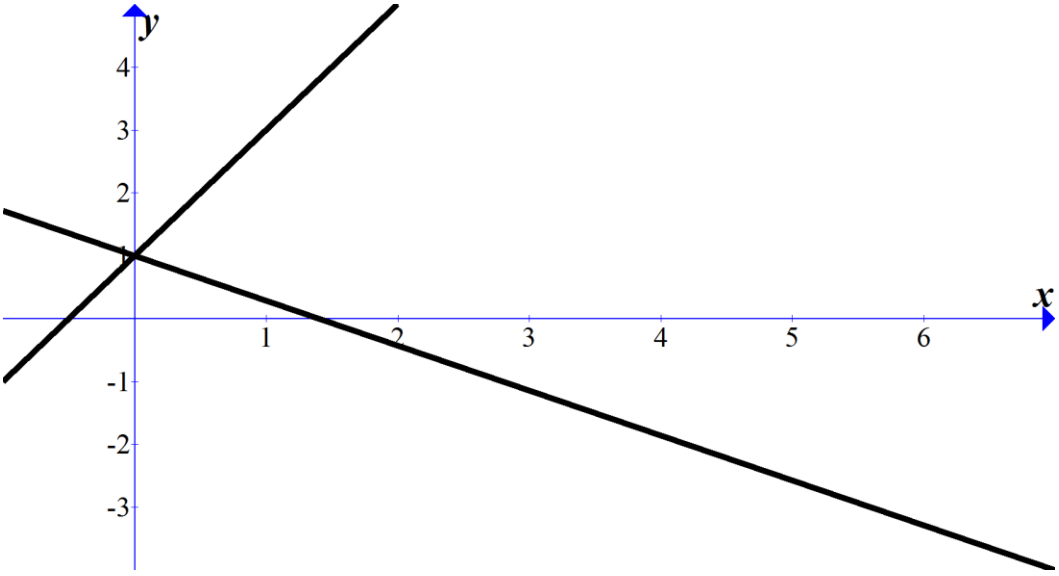


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f)



g)



4) Methods should verify solutions from q1

5)

a) 6 and 2

b) $2 and $2

c) 2 and 2

d) 12 and 6

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e) 45 degrees and 135 degrees

f) 14 by 7

6) 700 adults' and 300 children's

7) 10,000 @ 5% and 20,000 @ 8%

8) 125 of each dish

9) 30 x 2 point questions and 10 x 4 point questions

10) 30 people

11) 540

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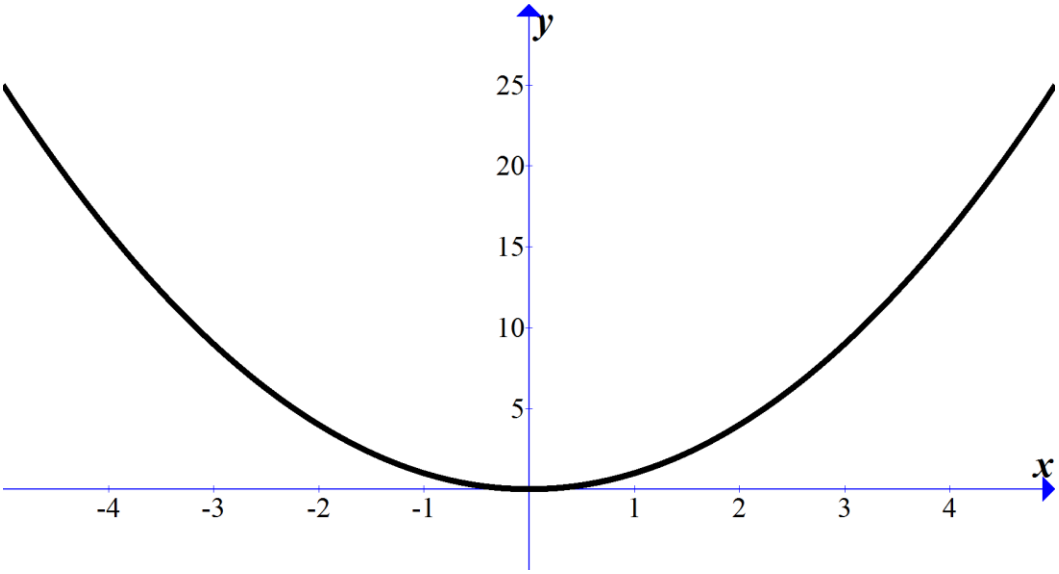
Further Mathematics: Solutions

Chapter 4: Graphs & Relations

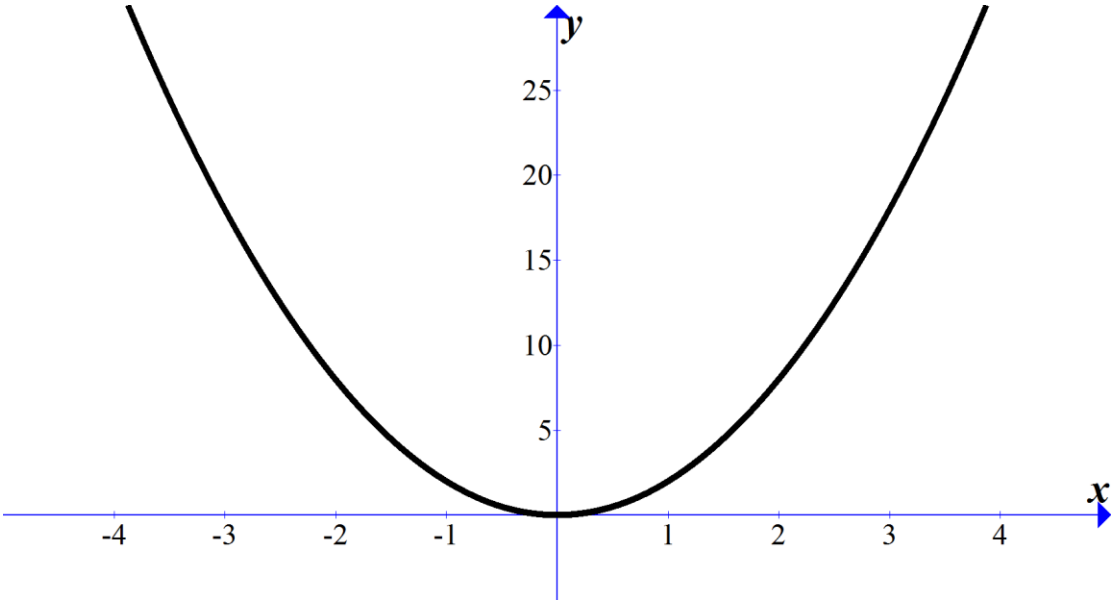
Exercise 3: Non-linear Relationships

1)

a)

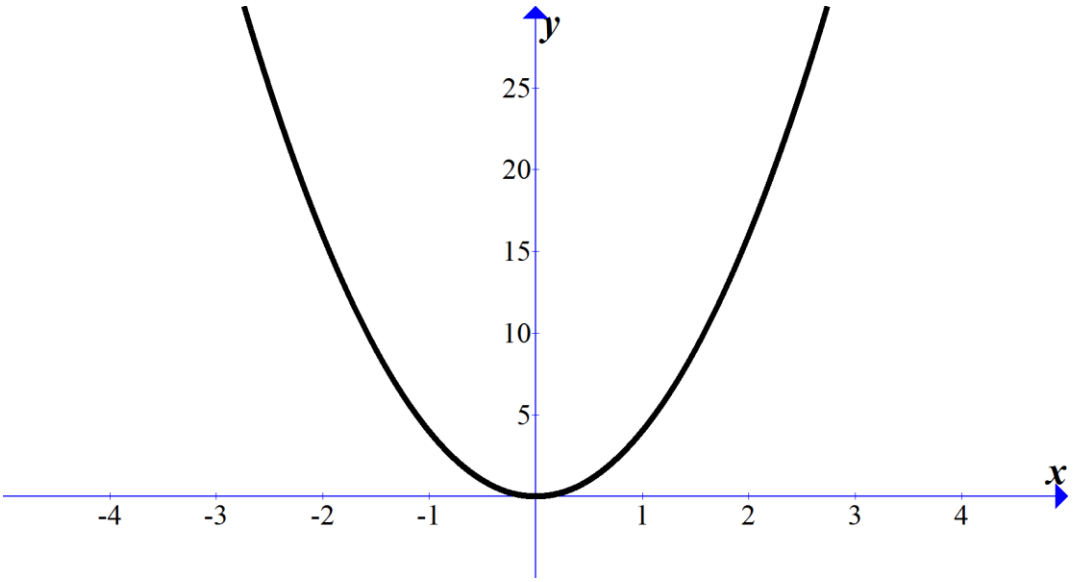


b)

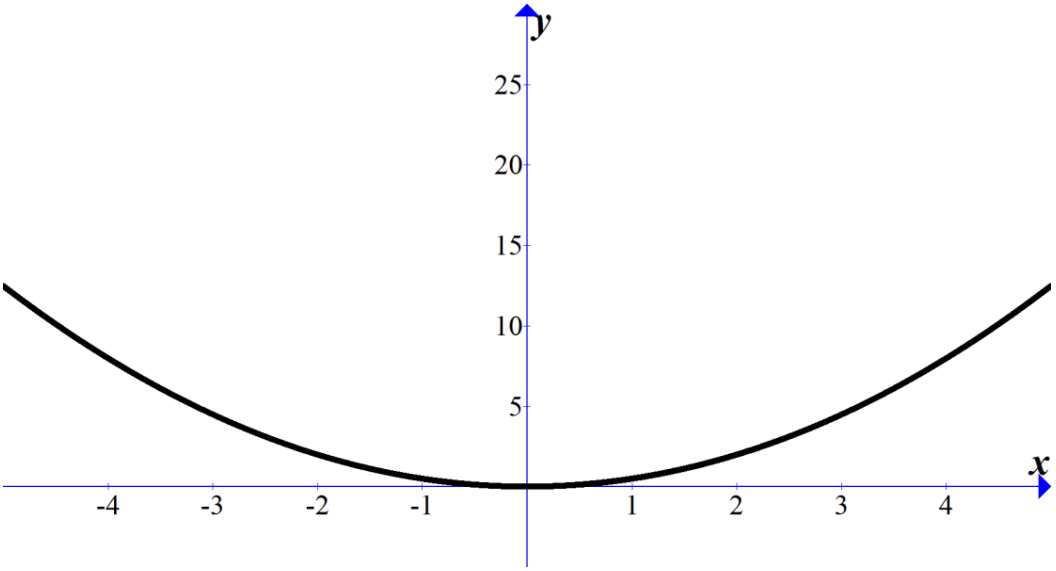


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c)

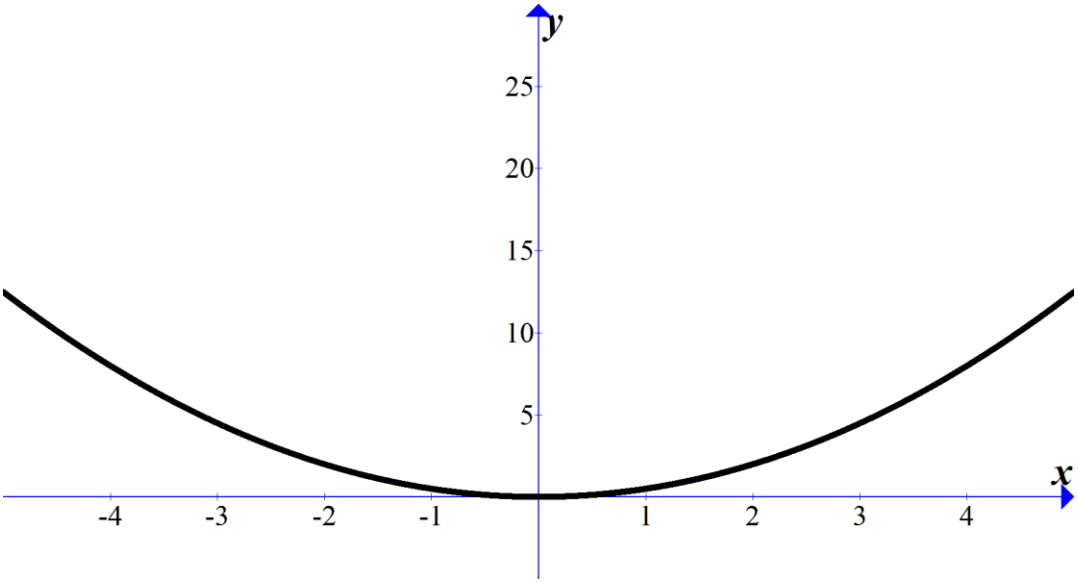


d)

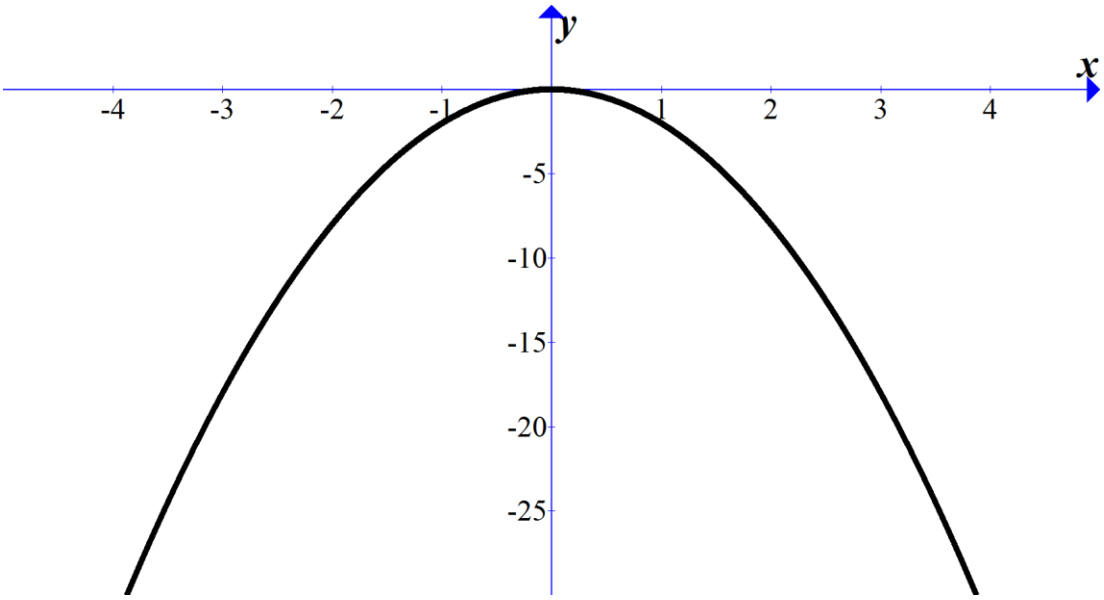


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e)



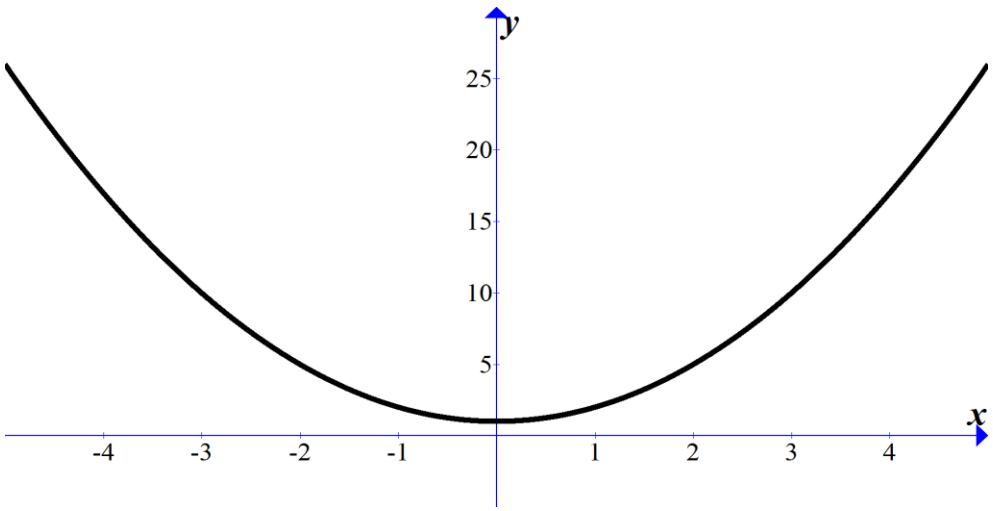
f)



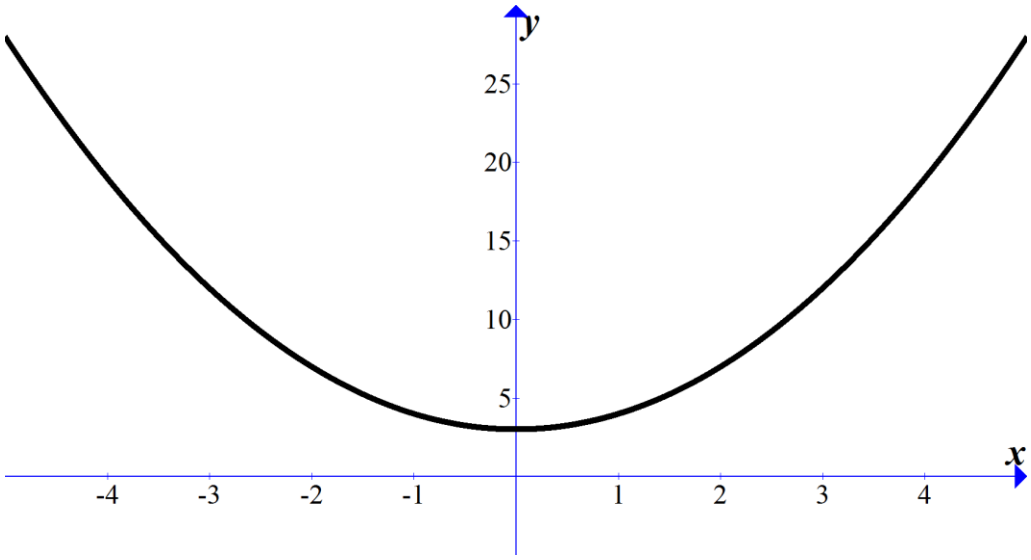
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2)

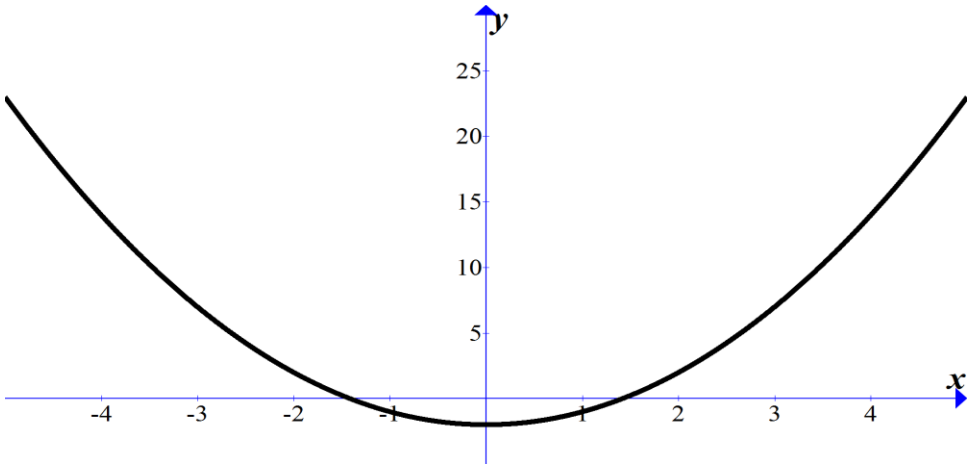
a)



b)

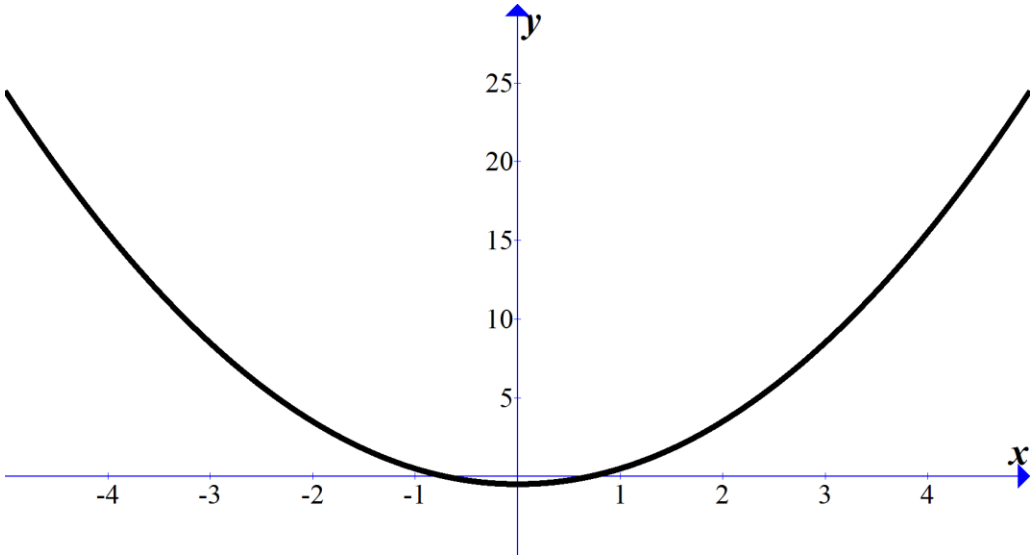


c)

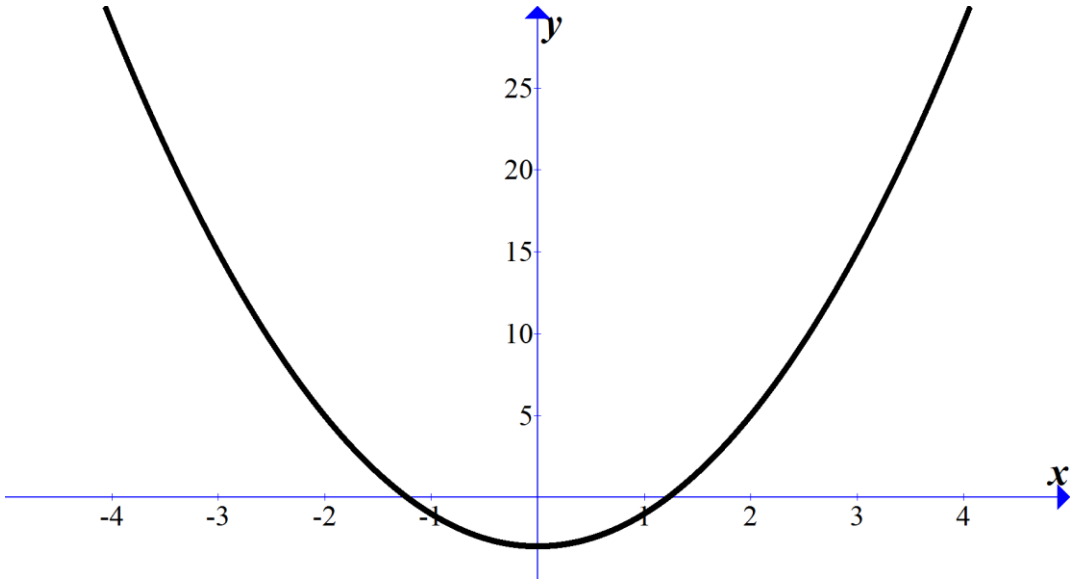


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d)



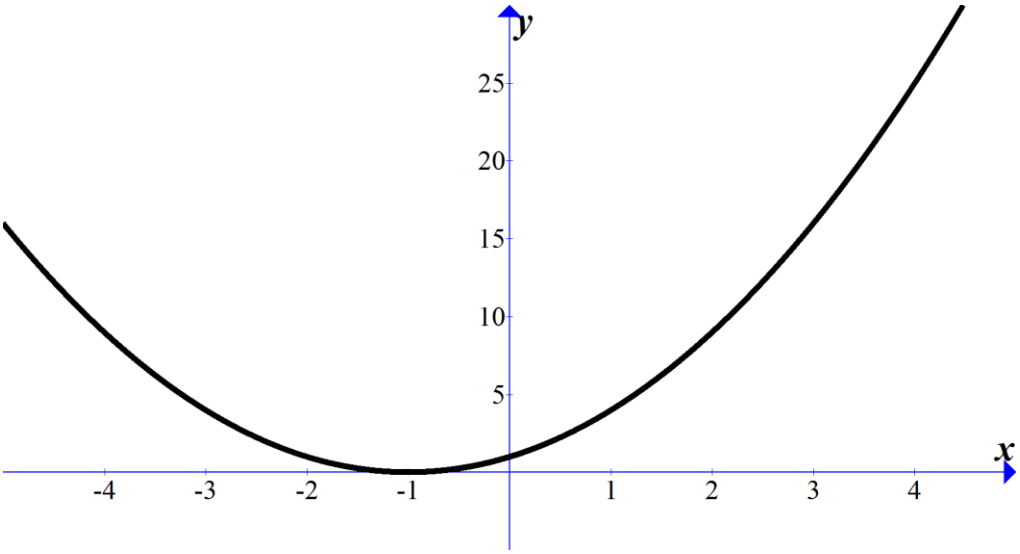
e)



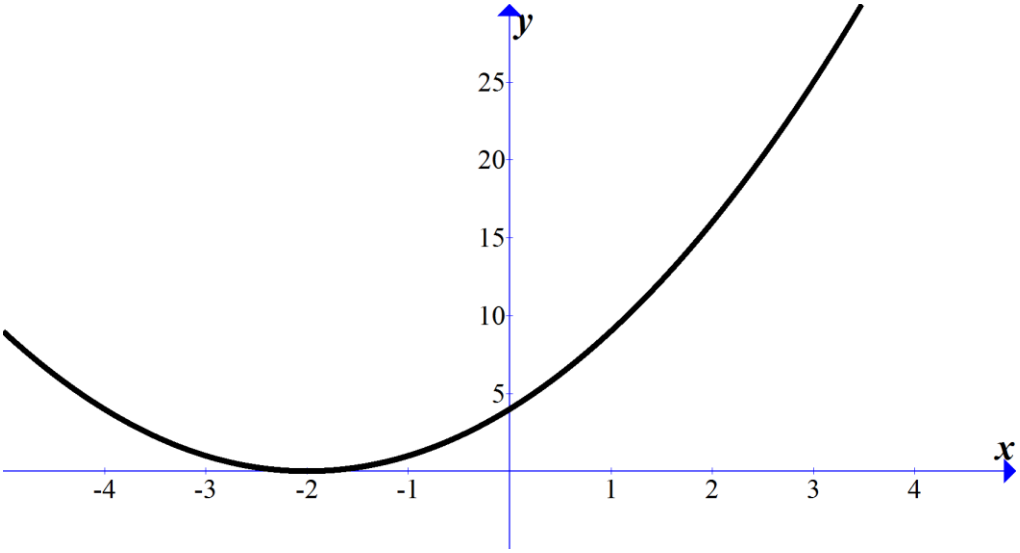
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3)

a)

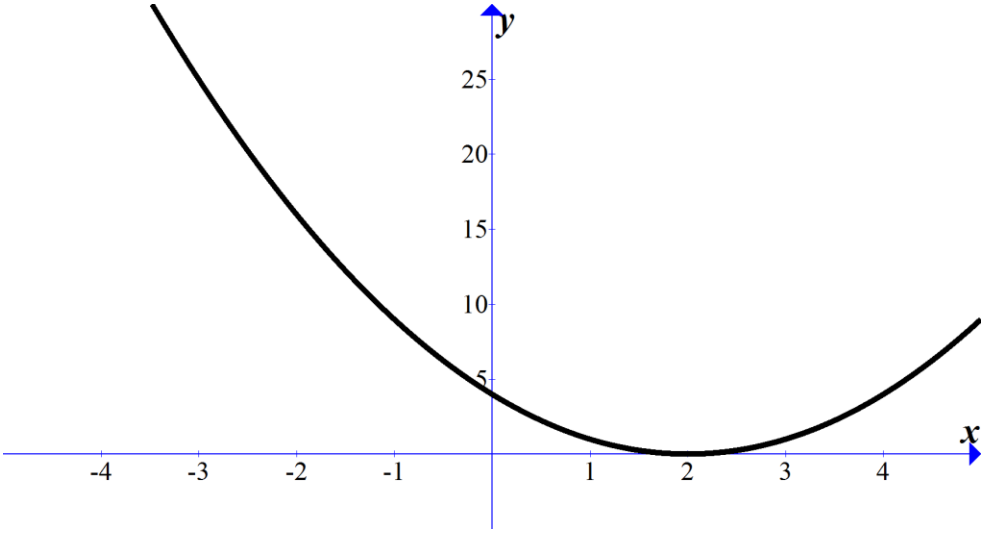


b)

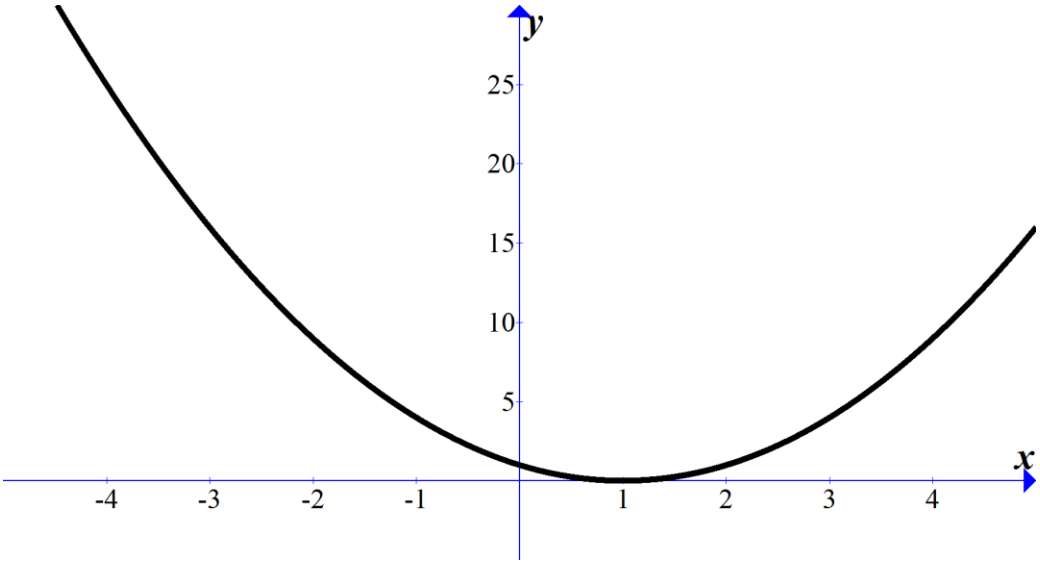


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c)



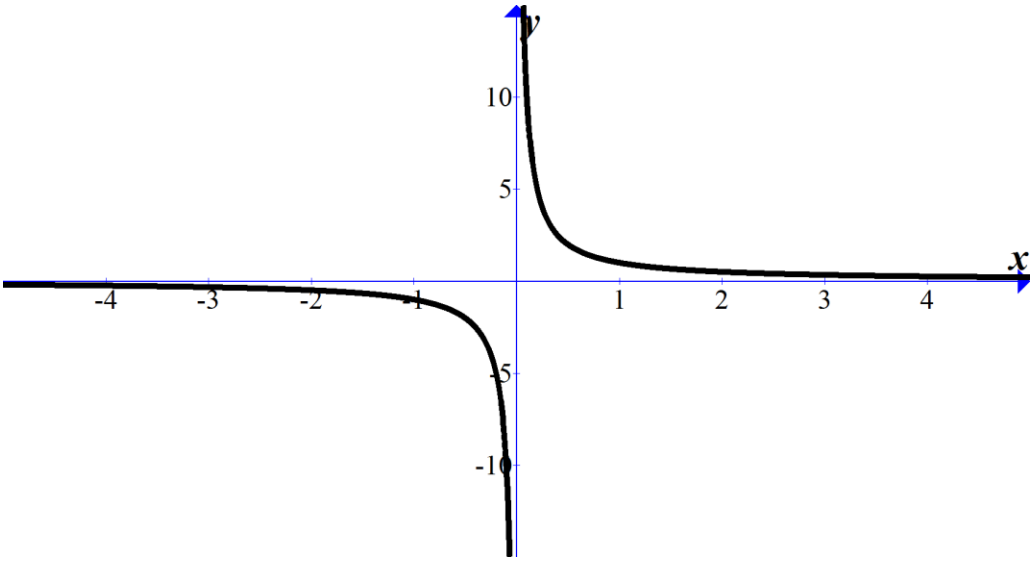
d)



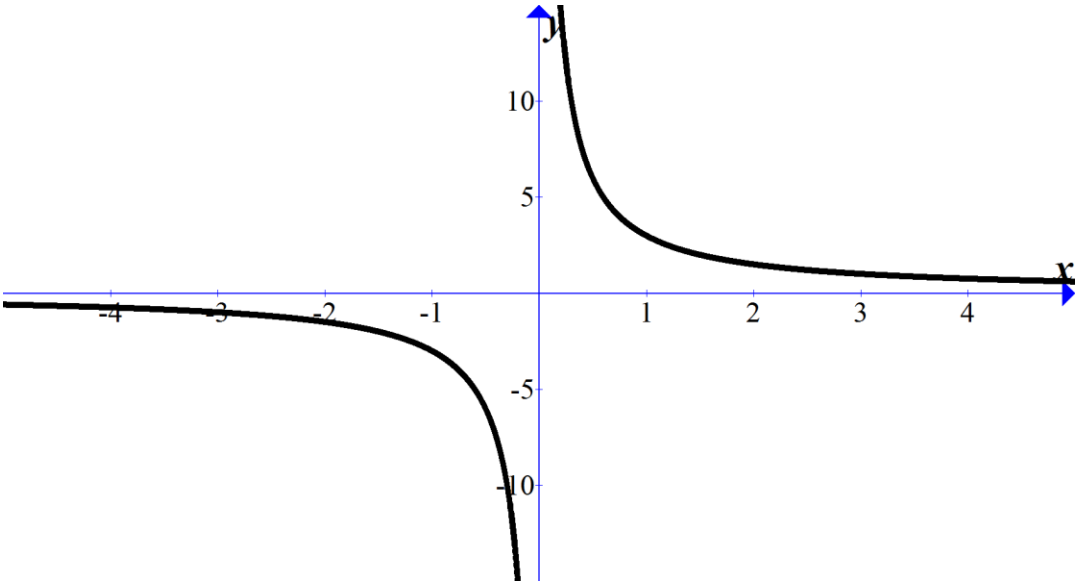
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4)

a)

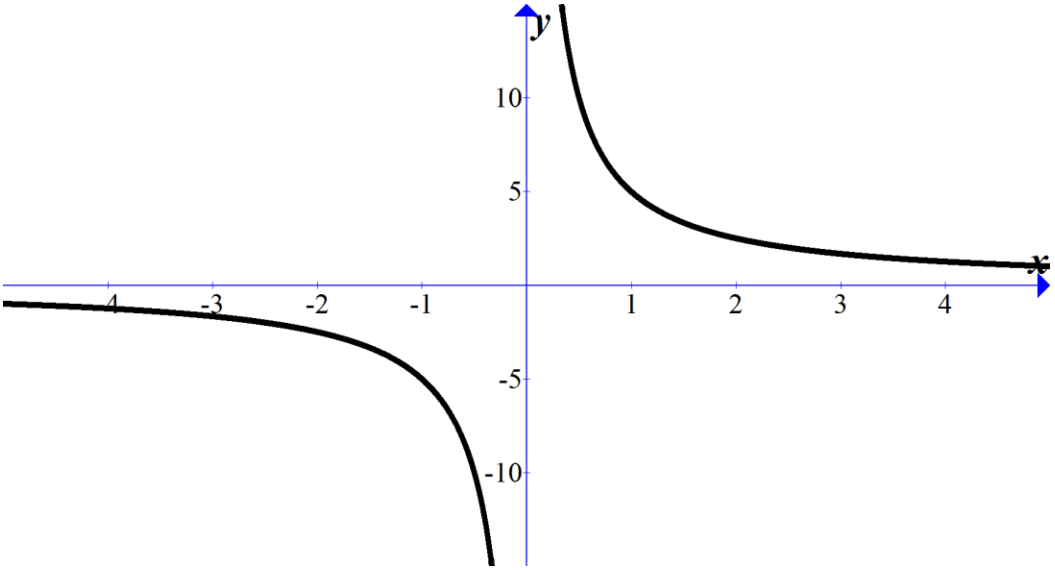


b)

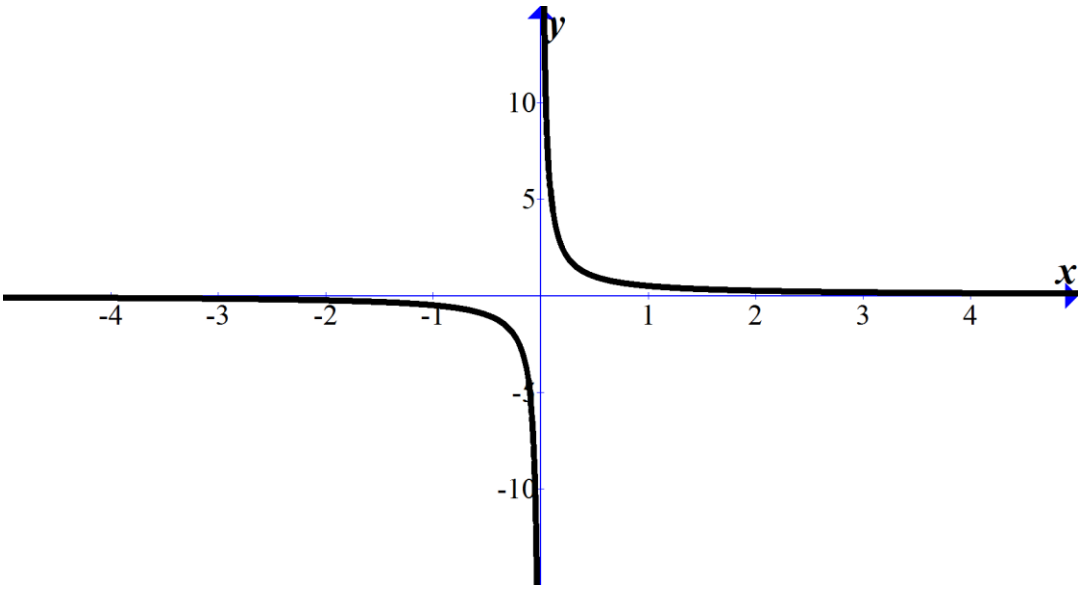


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c)

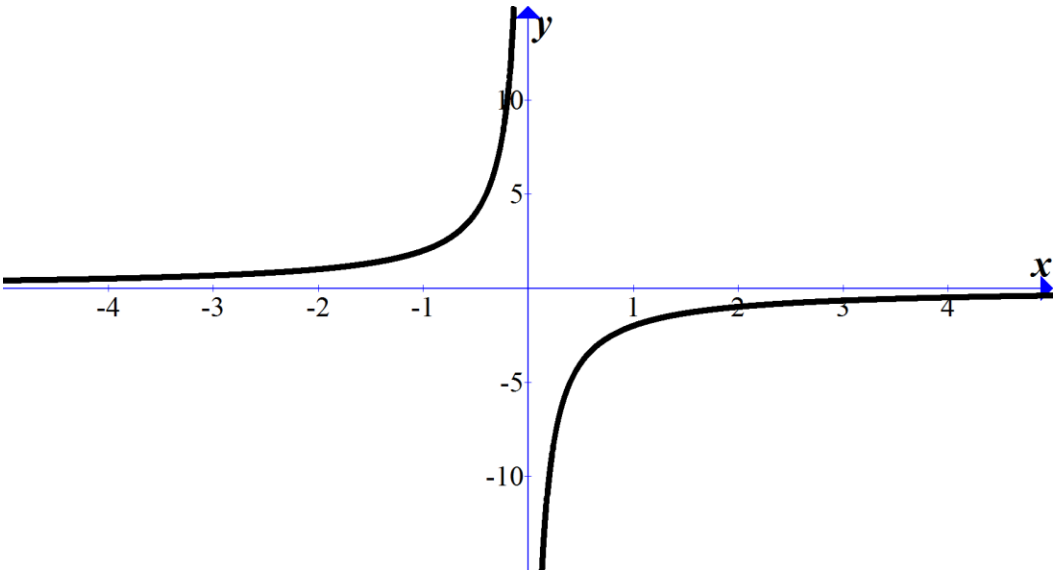


d)



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e)



5)

a) Parabola

b) Line

c) Parabola

d) Hyperbola

e) Line

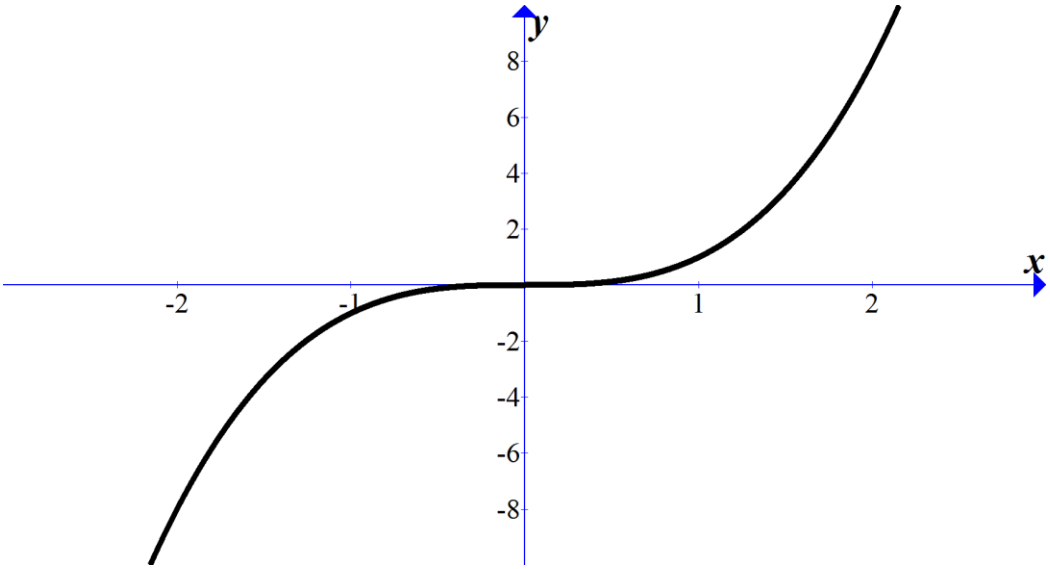
f) Parabola

g) Hyperbola

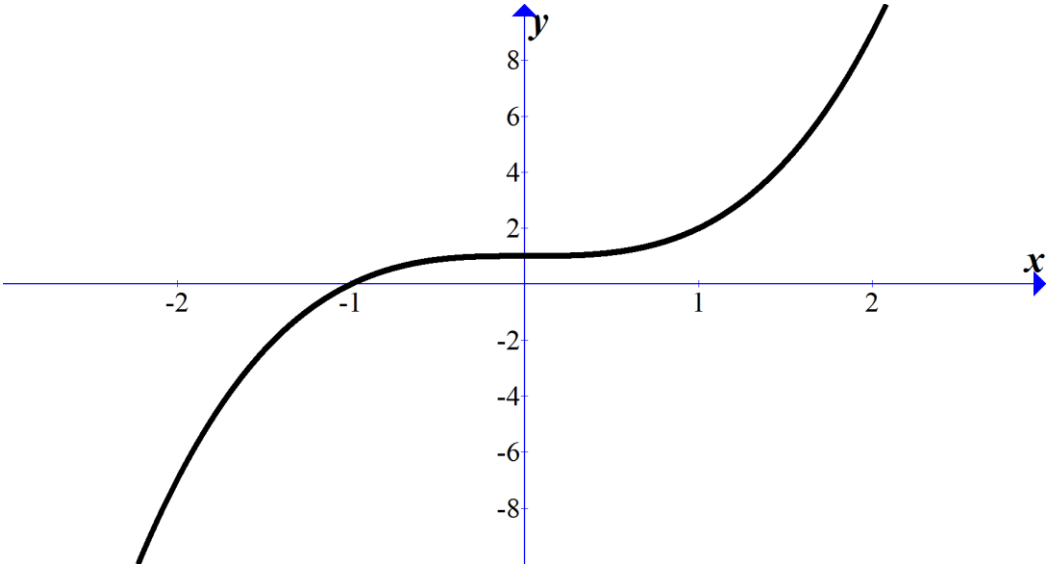
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6)

a)

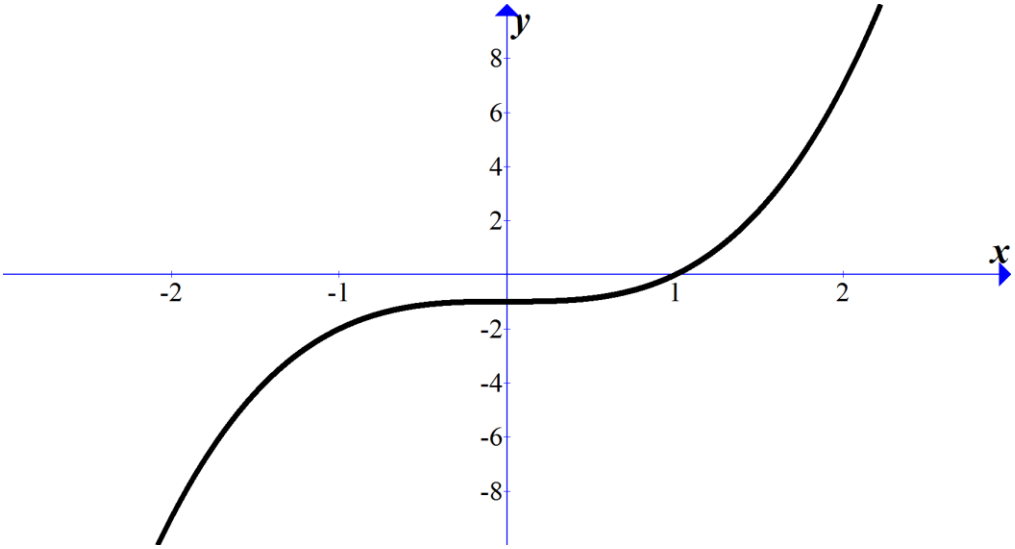


b)

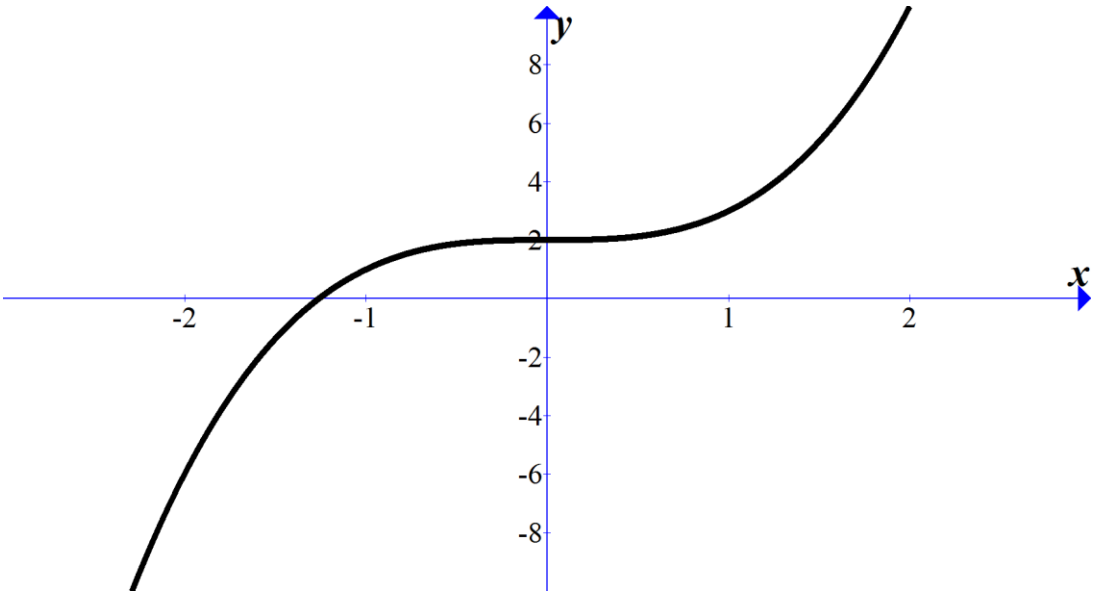


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c)

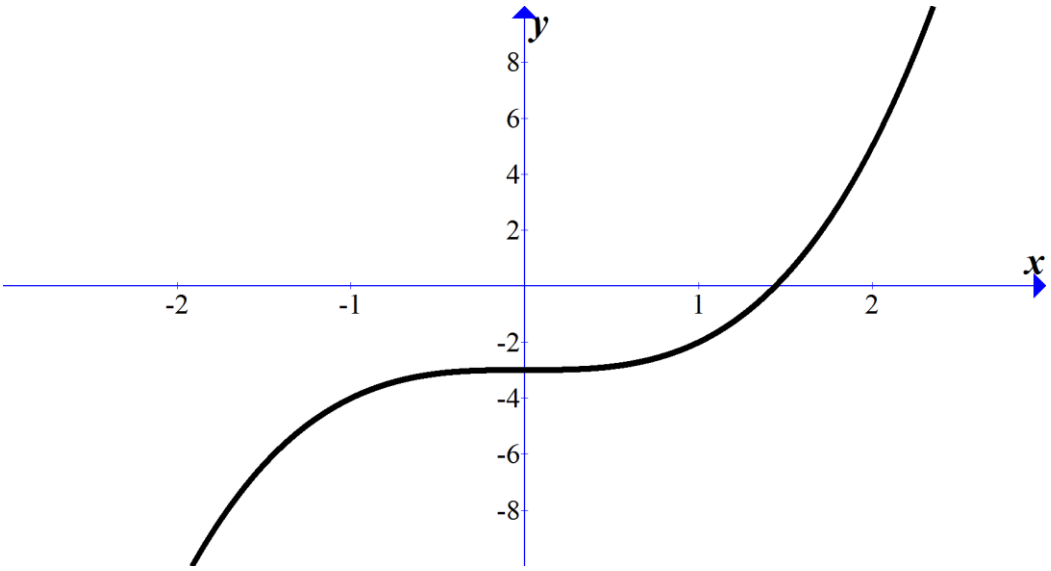


d)



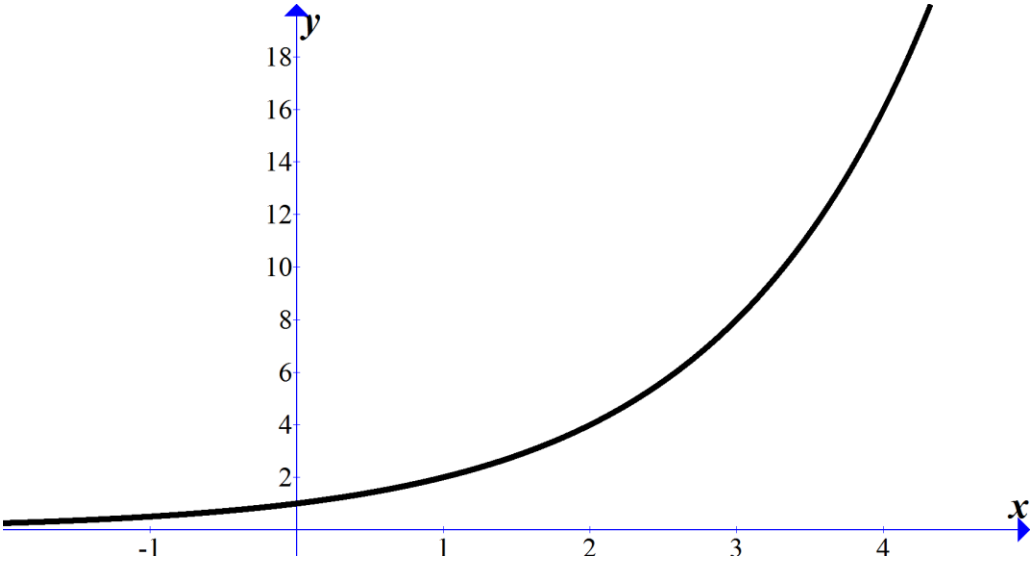
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e)



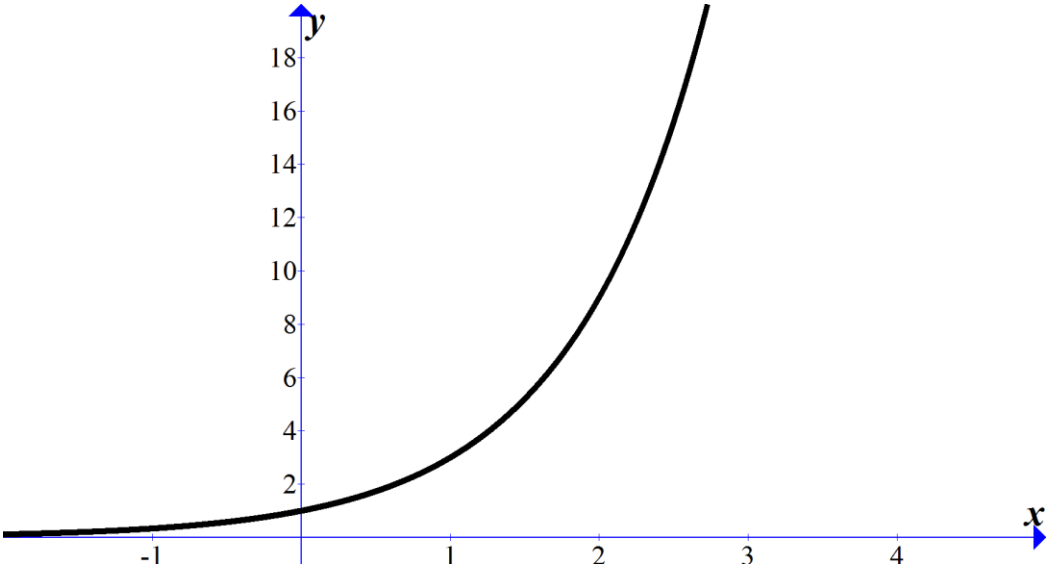
7)

a)

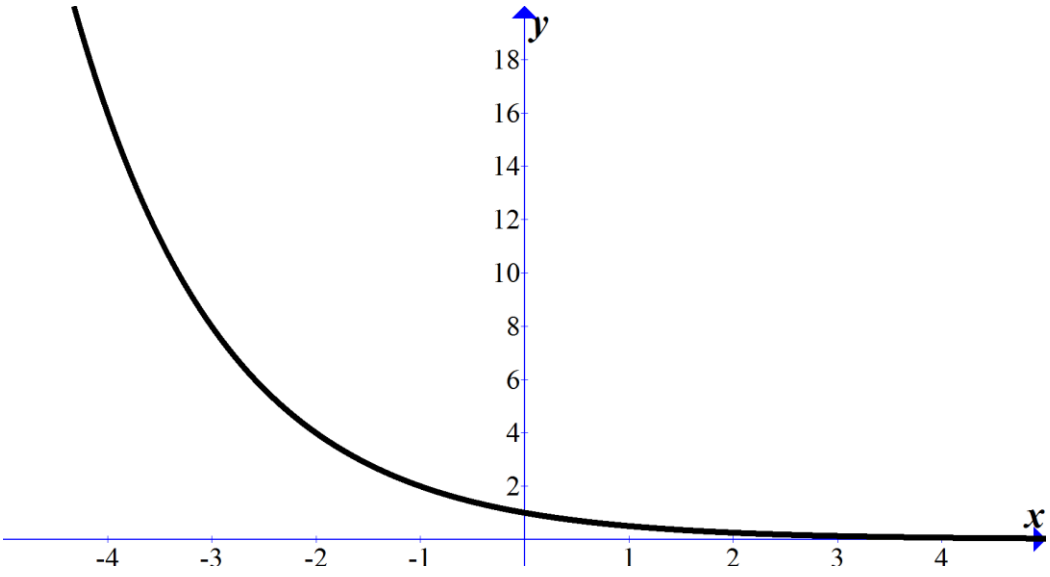


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b)

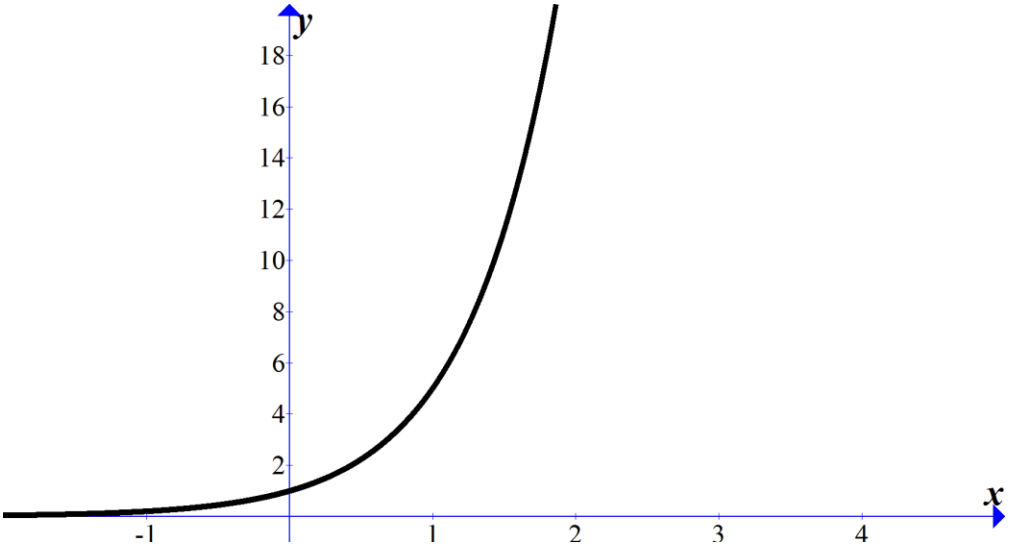


c)



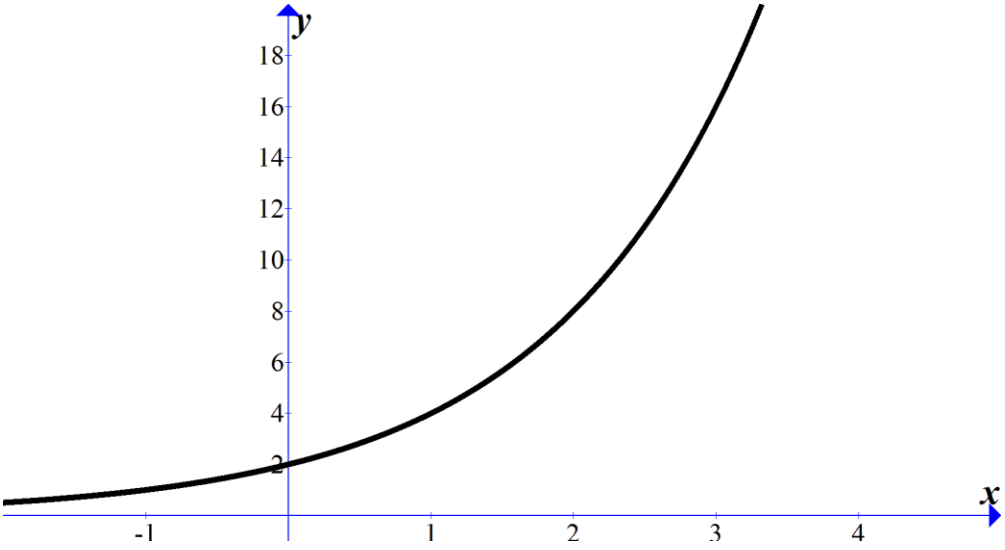
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d)



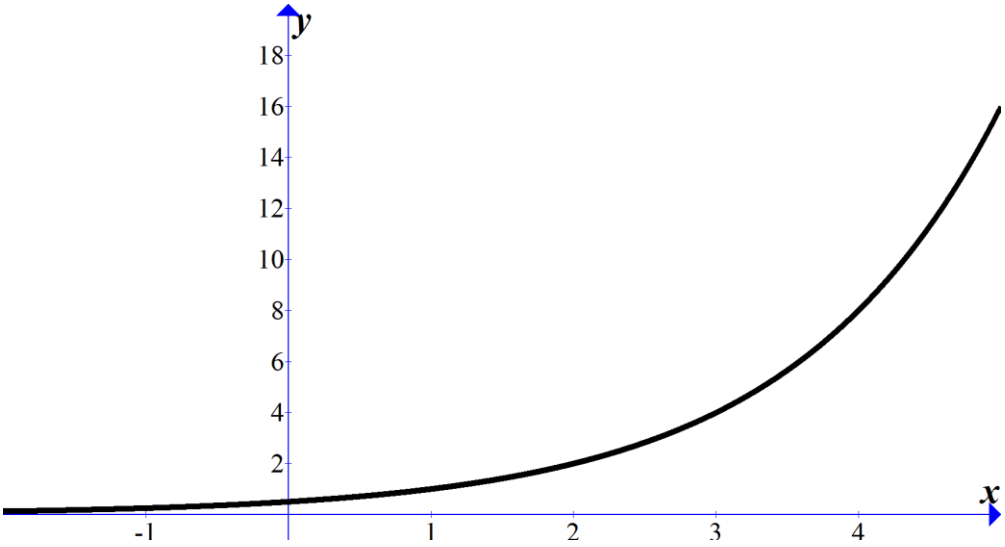
8)

a)

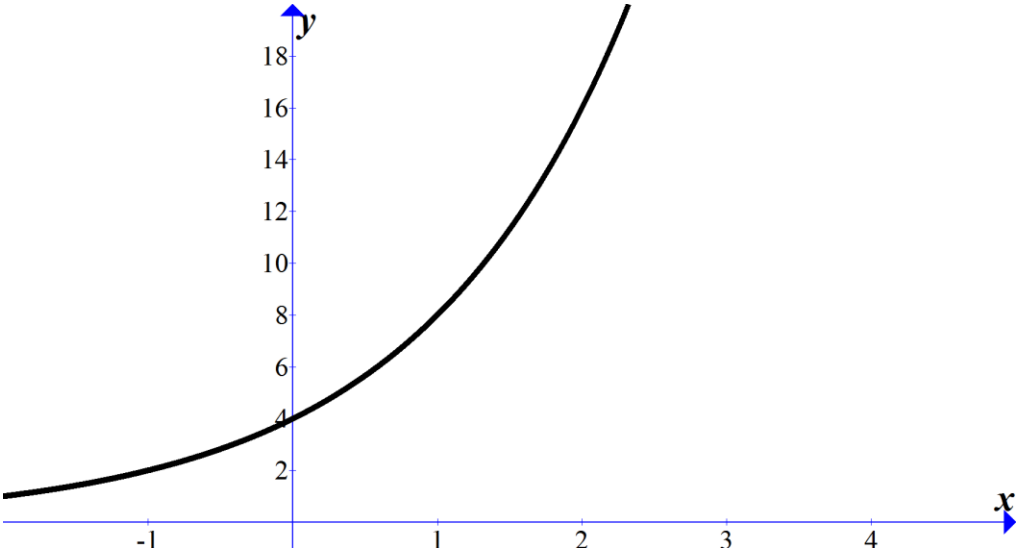


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b)

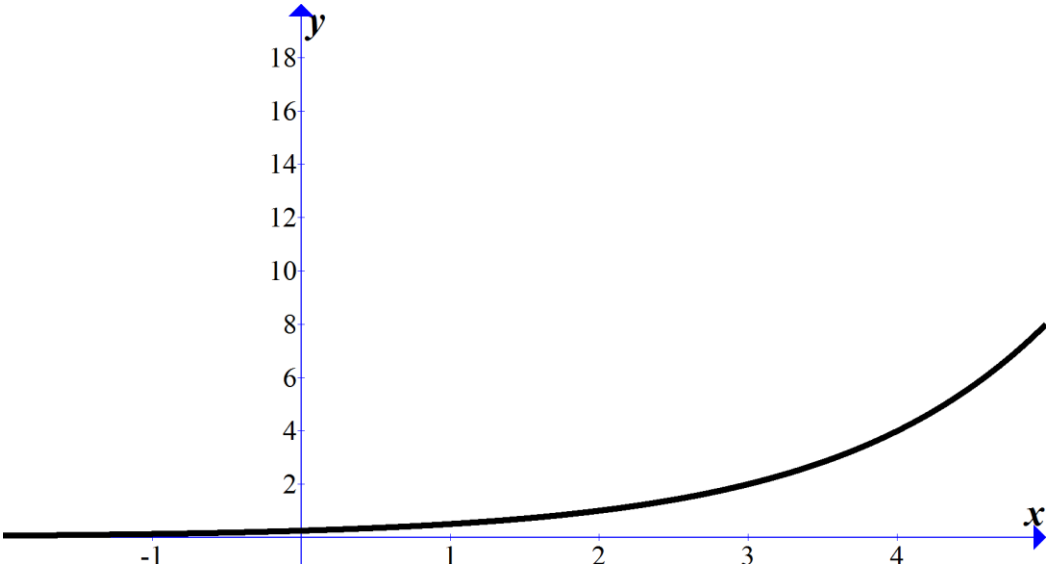


c)



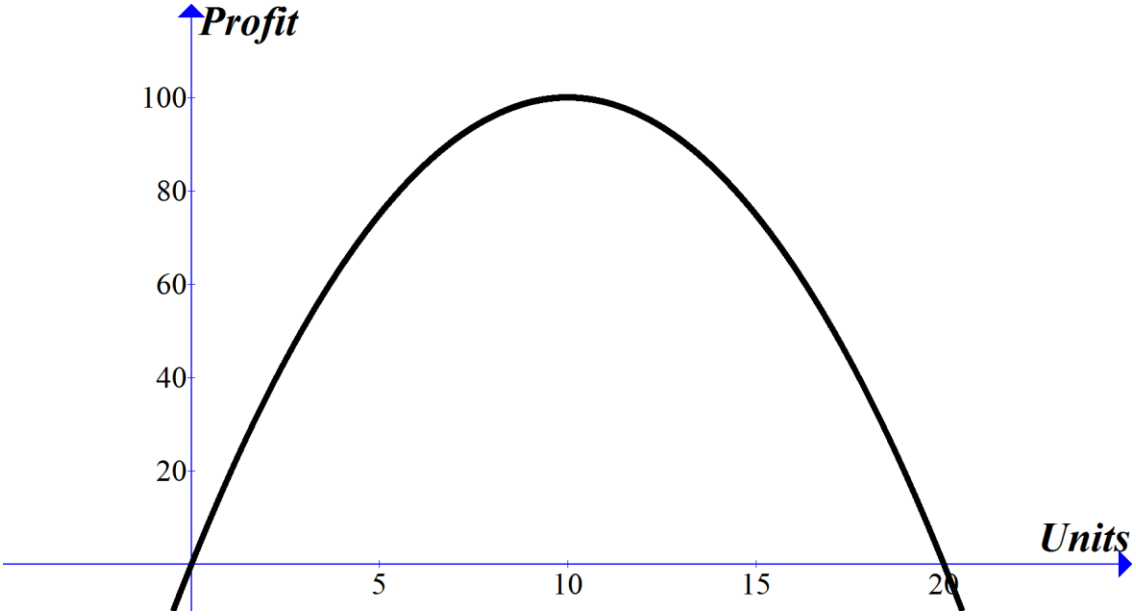
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d)



9)

a)



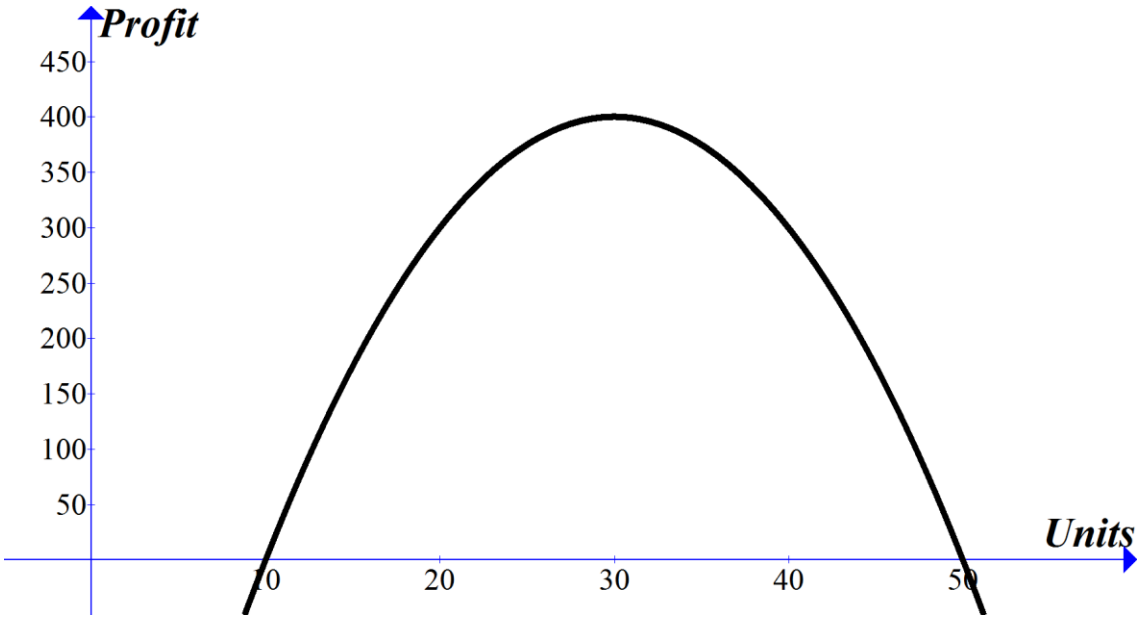
b) 0 and 20

c) 10 for a profit of $100

d) 5 or 15

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10)



Break even @ 10 units or 50 units

Maximum profit of $400 @ 30 units

11) 8 hours for a maximum population 128

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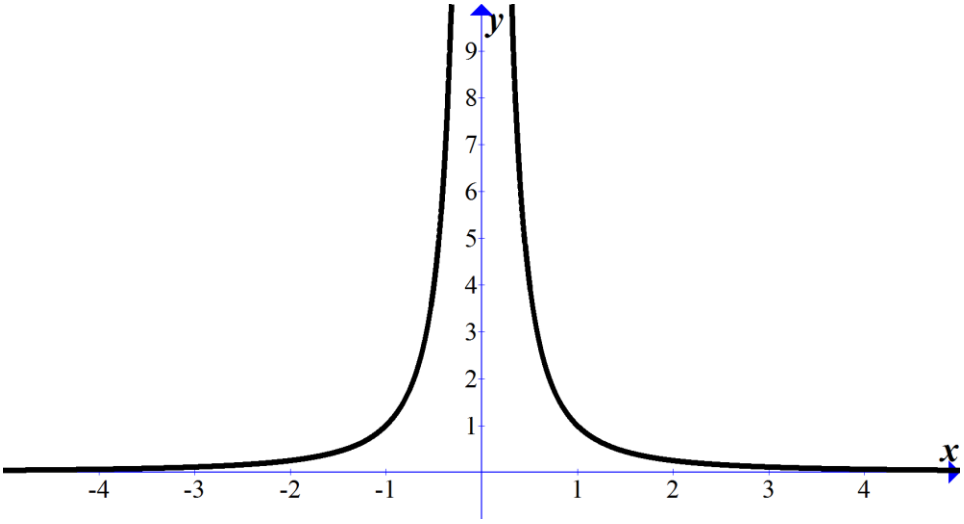
Further Mathematics: Solutions

Chapter 4: Graphs & Relations

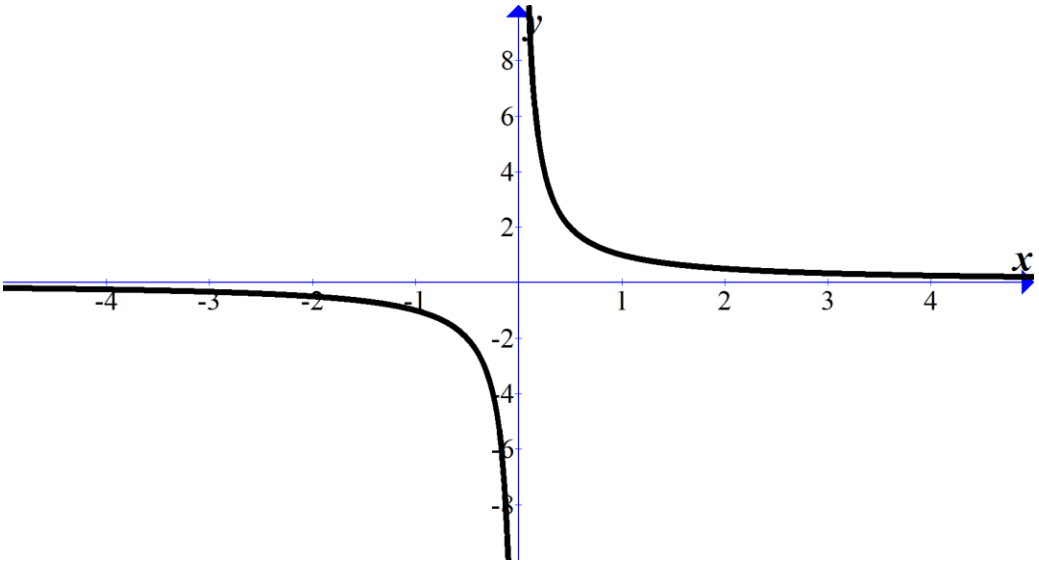
Exercise 4: Proportional Relationships

1)

a)

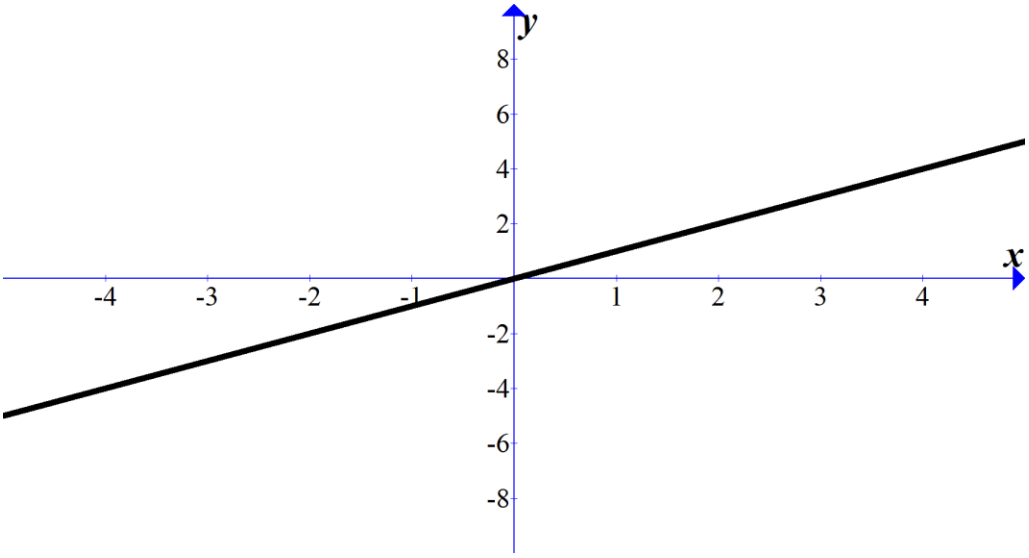


b)

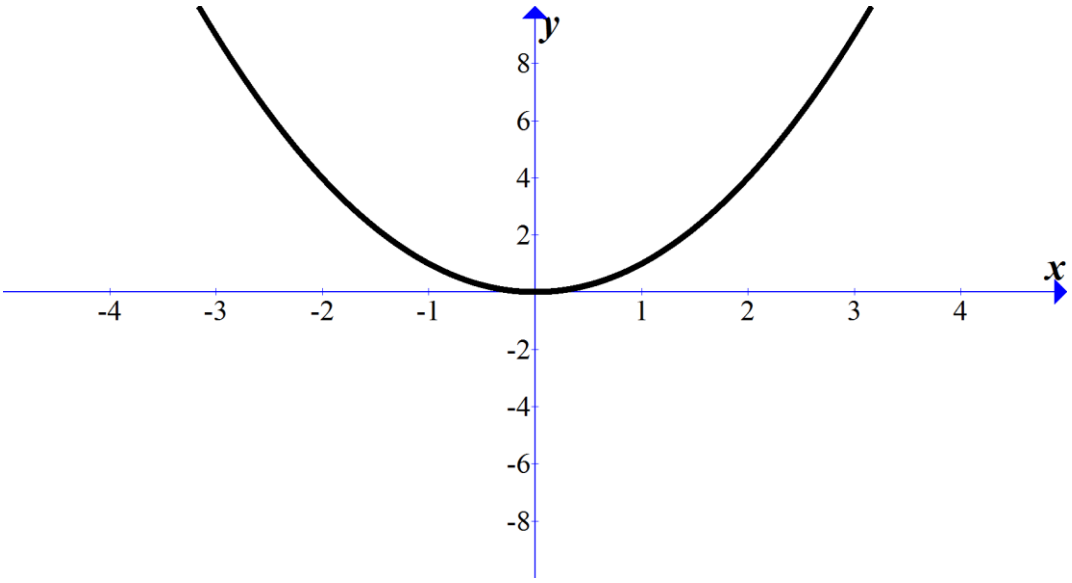


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c)

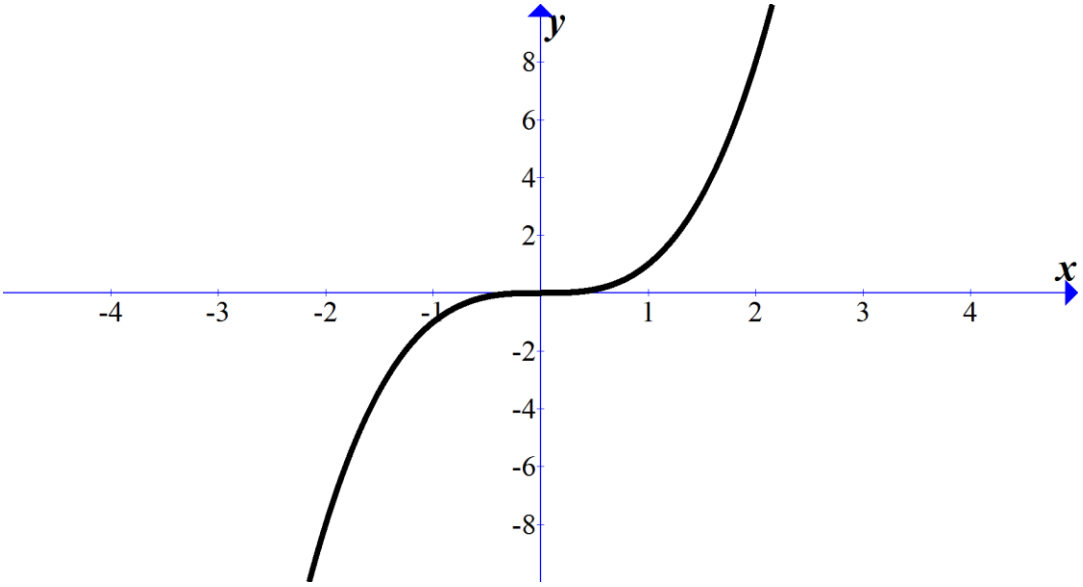


d) a



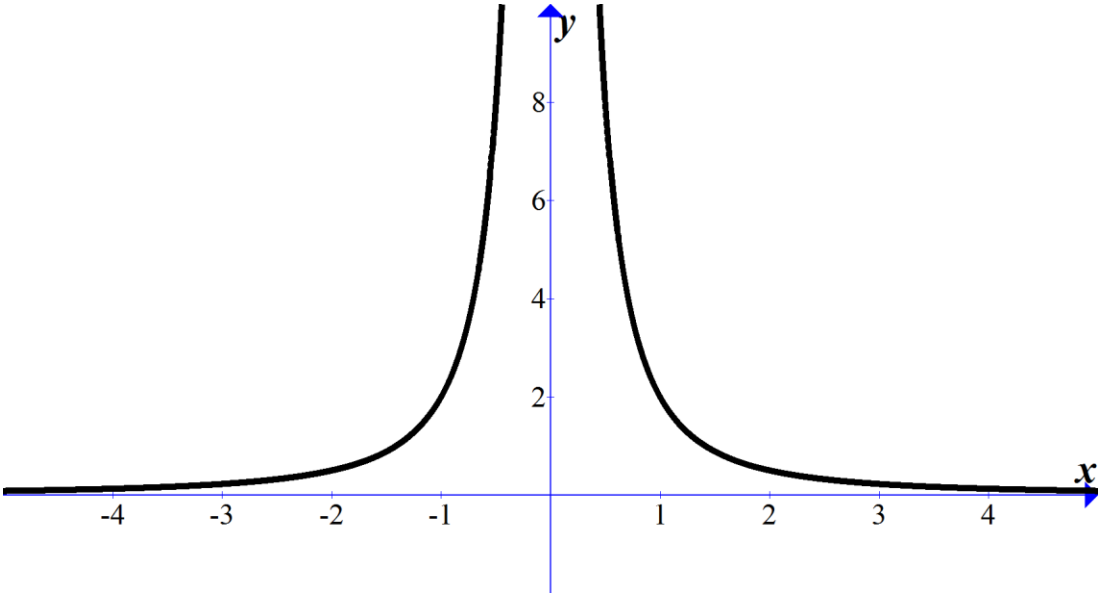
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e)



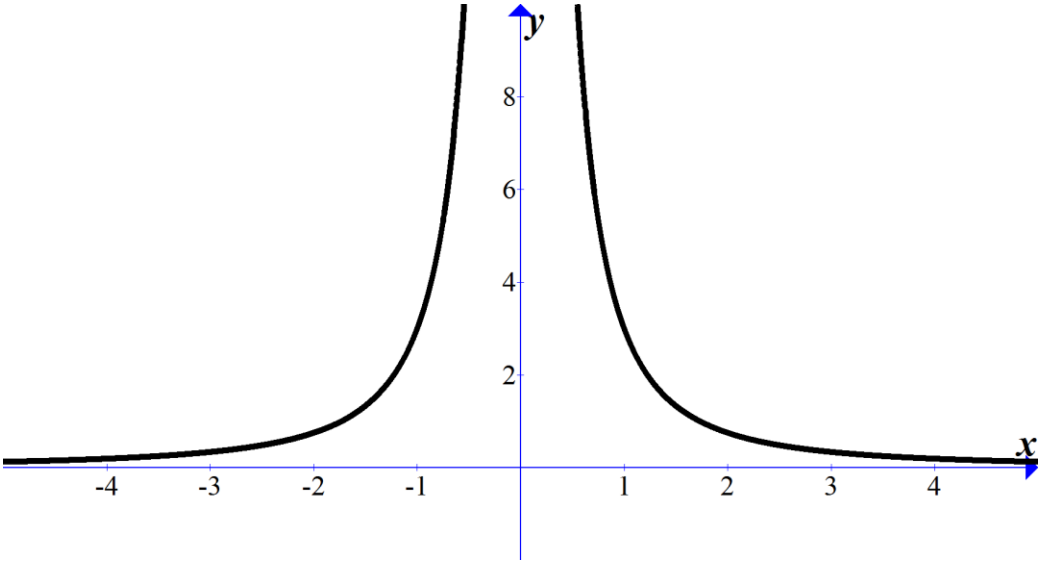
2)

a)

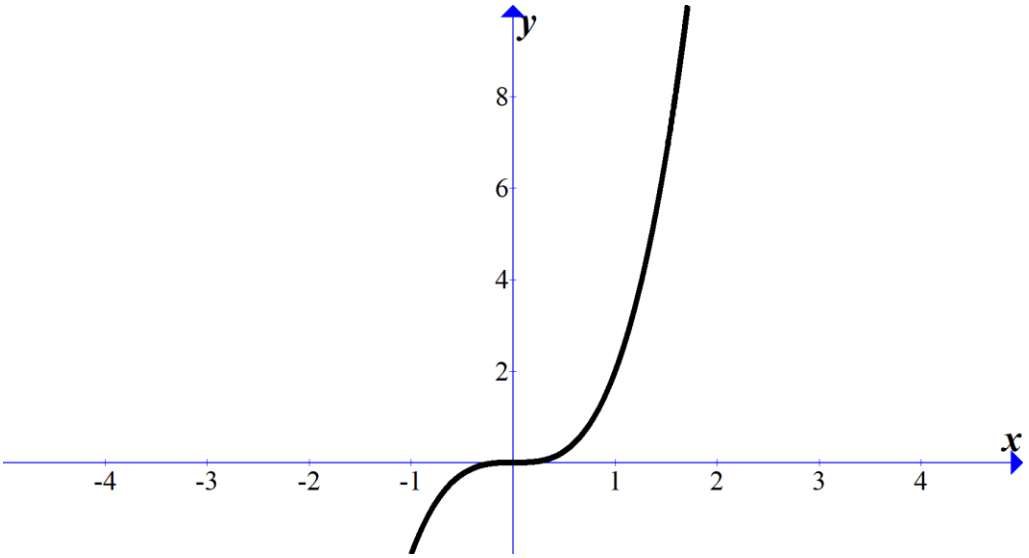


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b)

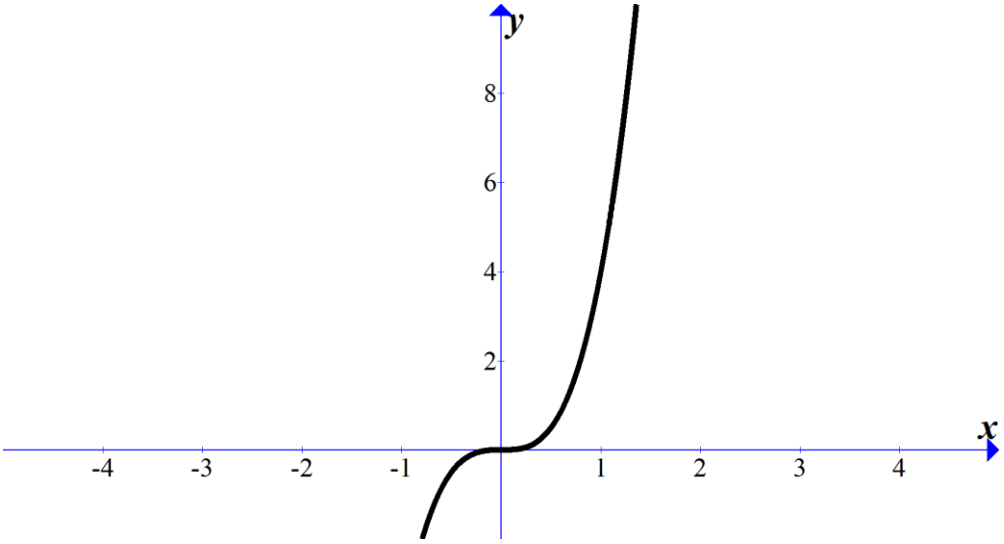


c)



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d)

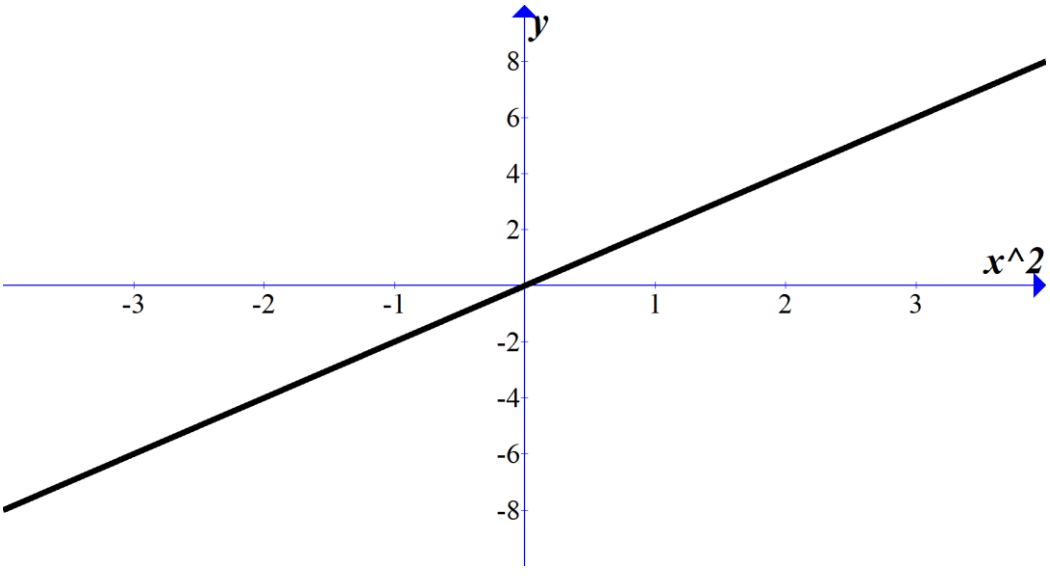


3)

- Negative values of n make some parts of the graph undefined
- For n=-1, part of the graph is inverted
- For n>1, the number of turning points increases
- As the value of k increases the graph becomes steeper more quickly

4)

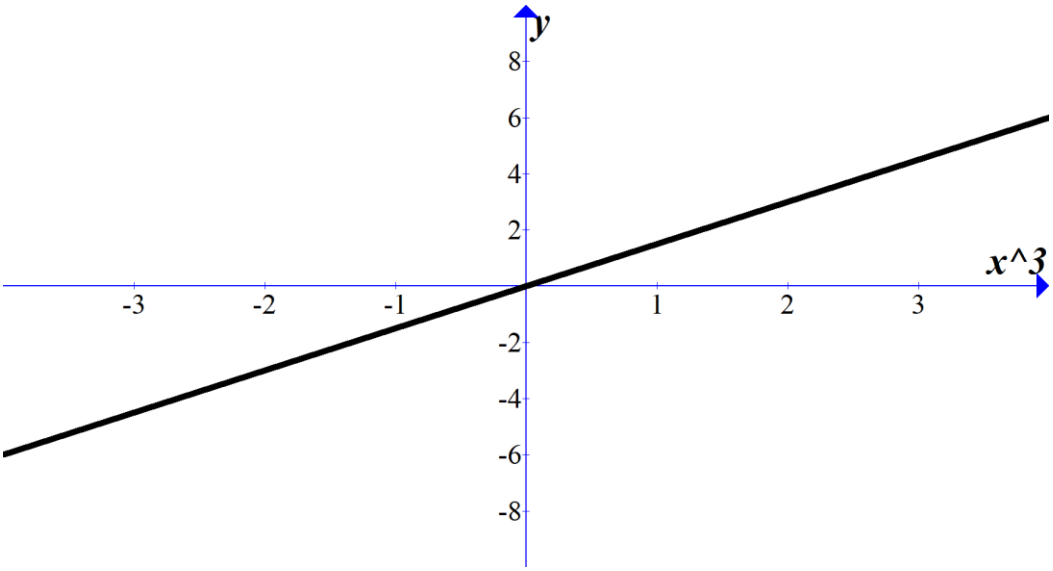
a)



b) The gradient of the line is 2, therefore the value of k is, and the equation is

5)

a)

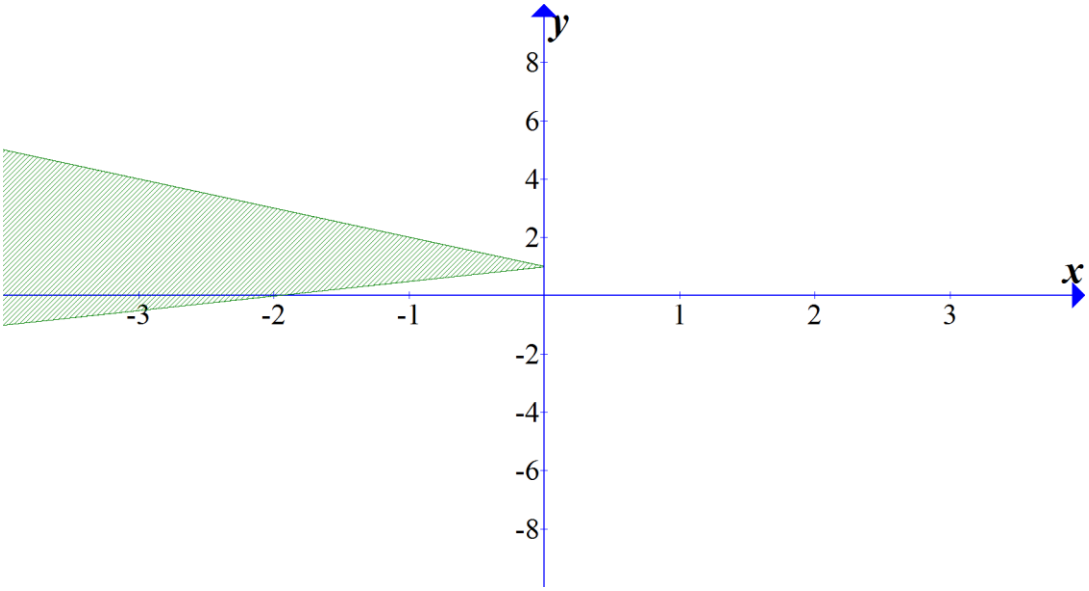


b) The gradient of the line is 1.5, therefore the value of k is 1.5, and the equation is

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1)

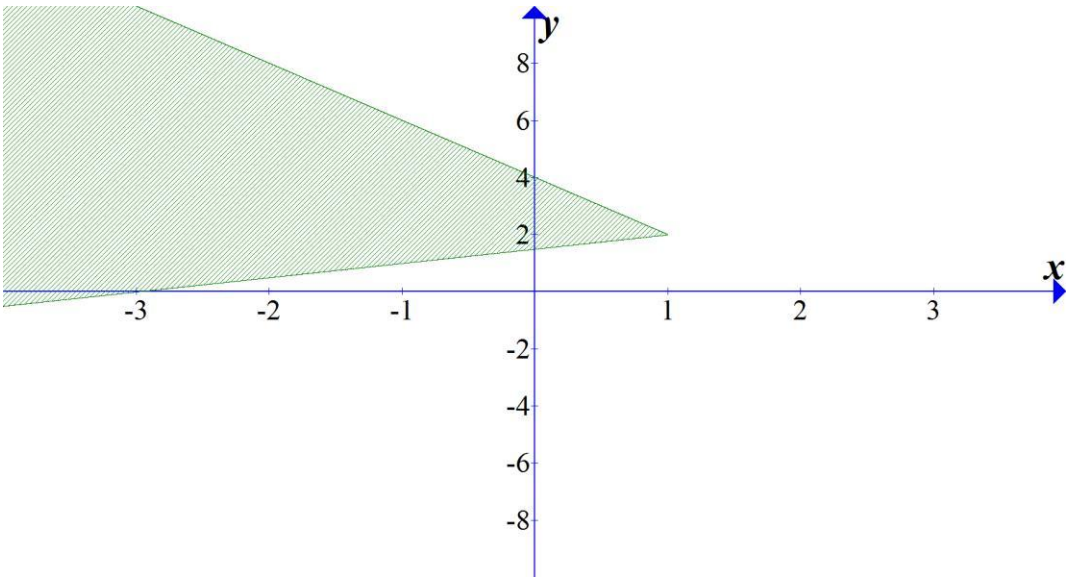
a)



b)

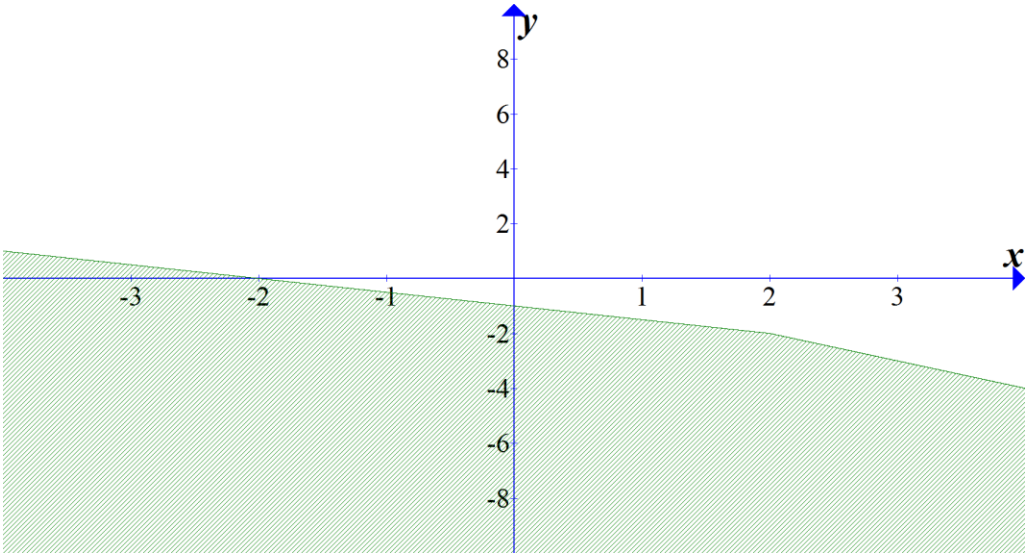
No feasible region

c)

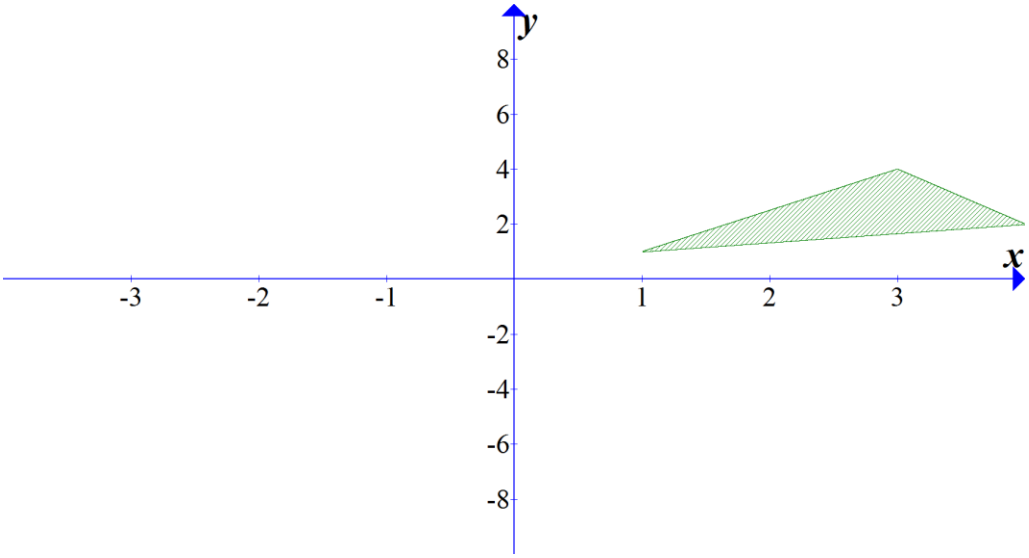


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d)



e)



2)

a) ≤ 20

b) $c = 3p$

c) $C = 400 + 25d$

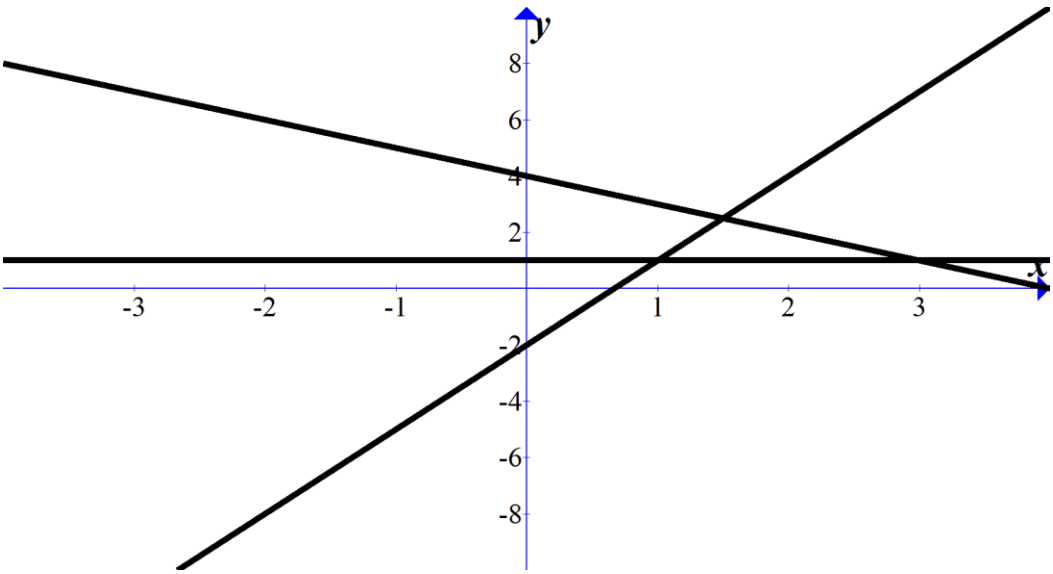
d) $R = 8x + 10y$

e) $200t + 125a \leq 4000$

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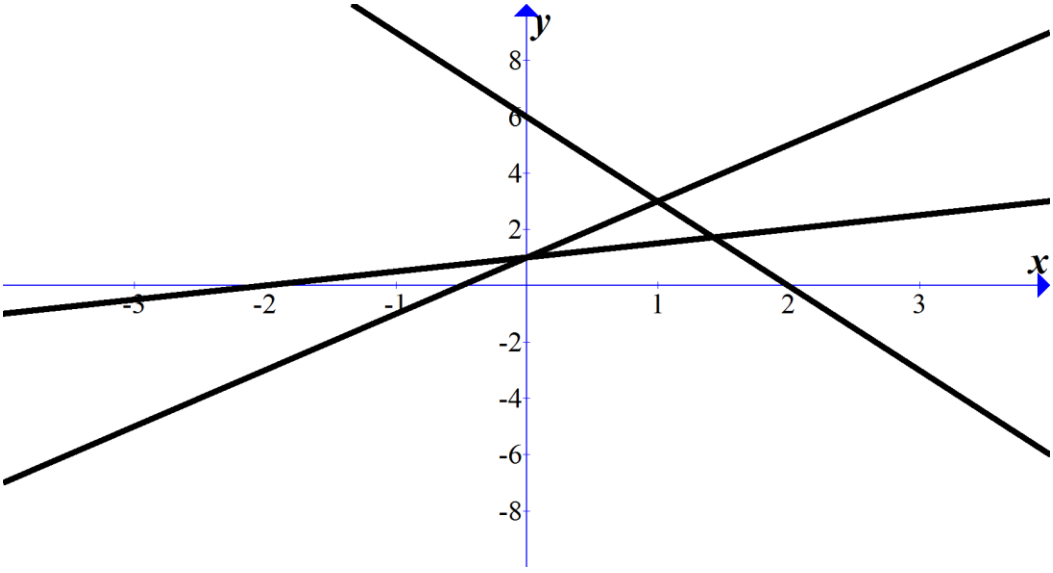
3)

a)



(1, 1) (3, 1) (1.5, 2.5)

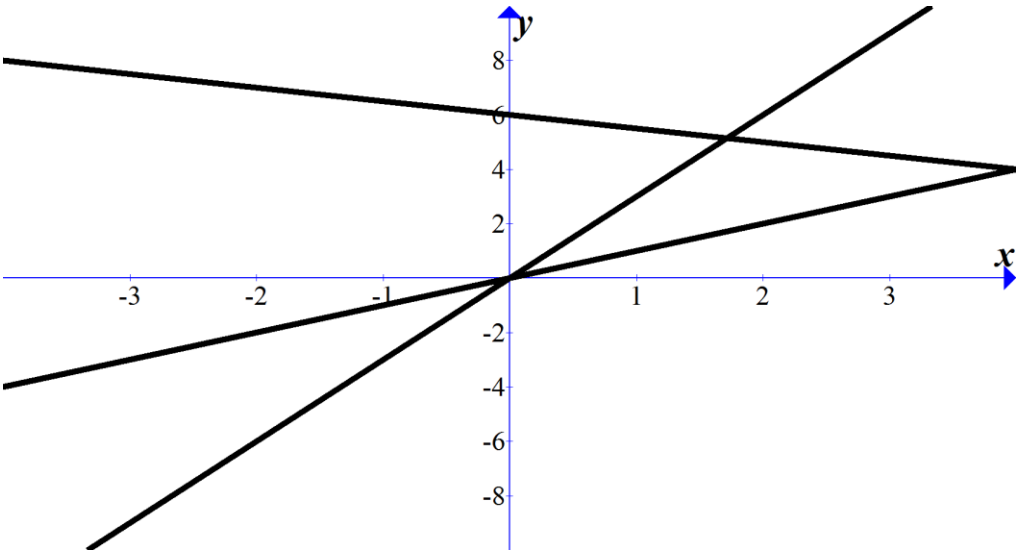
b)



(0, 1) (1, 3) (10/7, 12/7)

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c)



(0, 0) (12/7, 36/7) (4, 4)

4) To us e all the paint they must make 200 cans of B for a profit of $1000. Any other combination would not use all the paint

5)

a) 13 plastic and 2 ceramic for a profit of $2100

b) The same production for a profit of $3000

6) 1 x 50 seats and 8 x 40 seats for a cost of $5600

7) 600 bar and 200 air for a profit of $13800

8) 5 A class and 25 B class for a profit of $1325

9) 1000 adults and 4000 children for a profit of $23000

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Further Mathematics

Networks

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Further Mathematics: Solutions

Chapter 5: Networks

Exercise 1 Representation of Networks

1) Each vertex is named clockwise from top left

a) A odd, B odd

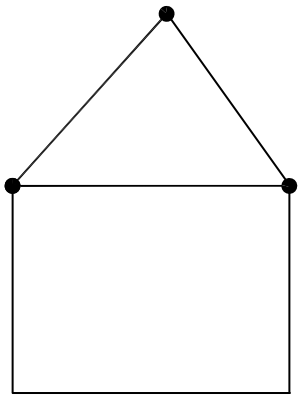
b) A even, B even, C even

c) A odd, B odd, C odd, D odd

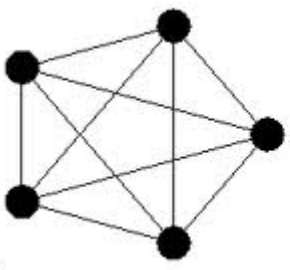
d) A even, B even, C odd, D odd, E even

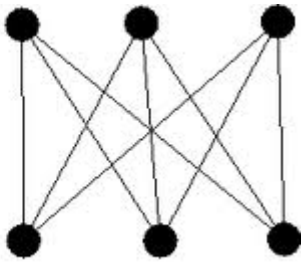
2) Every graph has an even number of vertices with odd degree

3) Planar



Non planar



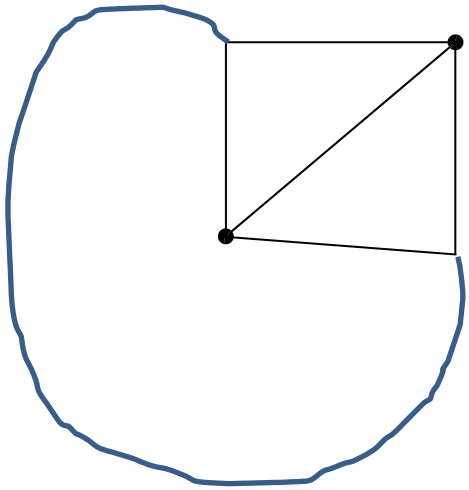


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4) F –E +V =2

For the planar graph in Q3, there are 3 faces (including outside the graph), 6 edges and 5 vertices, which satisfies the formula

5)



6) The graph does not obey Euler's rule so is not planar

7)

a) Yes, has 0 odd nodes

b) Yes, has 2 odd nodes

c) Yes, has 2 odd nodes

d) No, has 4 odd nodes

e) Yes, has 2 odd nodes

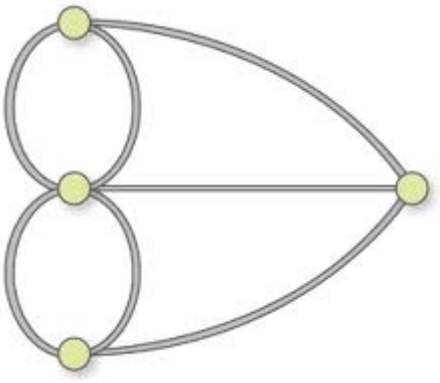
f) No, has 3 odd nodes

g) Yes, has no odd nodes

8) Only a can be traversed by visiting each vertex only once

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9)



The vertices are either degree 3 or degree 5; there are 4 nodes of odd degree, therefore the network is not traversable

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Further Mathematics: Solutions

Chapter 5: Networks

Exercise 2 Trees

1)

a) An undirected graph in which any two vertices are connected by a simple path

b) Any tree that includes all vertices of the original tree; a tree can have more than one spanning tree

c) If a tree has all edges weighted, a minimum spanning tree is the spanning tree from all possibilities that gives the minimum total sum of all paths

2) If for example a spanning tree shows costs of connecting two vertices, the minimum spanning tree is found that will produce a spanning tree from all possibilities at the least cost

3) Weighted paths between vertices, to allow calculation of spanning paths

4) Shortest transport routes, minimum cost of cabling etc., network connections in computing

5) Connections are AB, BC, BD for a total of 13 units

6) Lettering clockwise starting From node on far left as A, connections are AF (4), BE (8), BC (6), CD (4), and EF (3) for a total of 25 units

7) Minimum spanning tree is ED, EB, EC, AB for a total of 656

8) Minimum spanning tree is AE, BD, BC, AD, FG, CH, FH for a total of 28 units

9) Tree is Main to B, AB, BD, AE, CE for a total of 470 Km and a cost of $23500

10) Tree is AD, AE, CD, BE, EF, AG for a cost of 449,000

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Further Mathematics: Solutions

Chapter 5: Networks

Exercise 3 Paths & Flow

1) A directed graph allows flow in one direction only between vertices, this is indicated by arrows going into or away from them

2)

Vertex	Indegree	Outdegree
1	0	3
2	0	3
3	2	1
4	3	3
5	1	2
6	3	0
7	2	1

3)

$$R_1$$

	1	2	3	4	5	6	7
1	0	1	1	1	0	0	0
2	0	0	0	1	1	0	0
3	0	0	0	0	0	1	0
4	0	0	1	0	0	1	1
5	0	0	0	1	0	0	1
6	0	0	0	0	0	0	0
7	0	0	0	0	0	1	0

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R_2

	1	2	3	4	5	6	7
1	0	0	1	1	1	2	1
2	0	0	1	1	0	1	2
3	0	0	0	0	0	0	0
4	0	0	0	0	0	1	0
5	0	0	1	0	0	2	1
6	0	0	0	0	0	0	0
7	0	0	0	0	0	0	0

R_3

	1	2	3	4	5	6	7
1	0	0	1	1	0	3	2
2	0	0	1	0	0	4	1
3	0	0	0	0	0	0	0
4	0	0	0	0	0	0	0
5	0	0	0	0	0	2	0
6	0	0	0	0	0	0	0
7	0	0	0	0	0	0	0

R_T

	1	2	3	4	5	6	7
1	0	1	3	3	1	5	3
2	0	0	2	2	1	5	3
3	0	0	0	0	0	1	0
4	0	0	1	0	0	2	1
5	0	0	1	1	0	4	2
6	0	0	0	0	0	0	0
7	0	0	0	0	0	1	0

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4)

	A	B	C	D	E
A	0	1	1	1	
B	0	0	1	0	0
C	0	0	0	0	0
D	0	1	1	0	0
E	0	1	1	1	0

Rankings are A, E, D, B, C

5)

a) 150 litres per hour

b) 800 litres per hour

6) Should return same answer if cut through AB and AD

7) The flows from left to right are 8, 7, 13 and 8, (the maximum flow is 7)

8) Minimum cut is V1V3, V3V2, V3V4, V4 Sink which gives 23. Note the value of V3V2 is zero since the flow is opposite the flow through the other parts of the cut

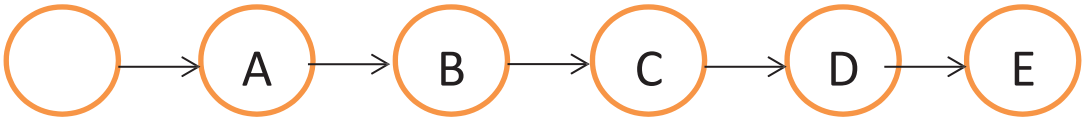
Upon inspection the maximum flow is also 23

9) 18

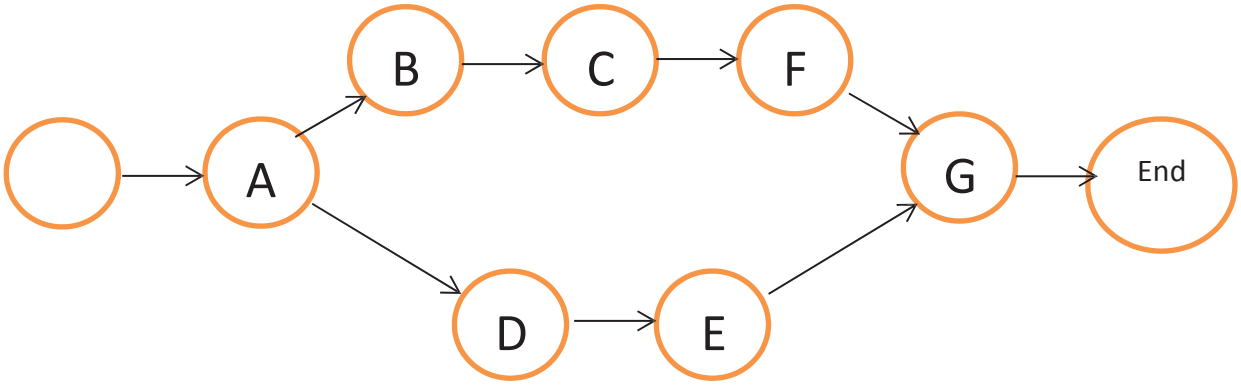
10) The new road must be able to carry more than 1400 vehicles per hour

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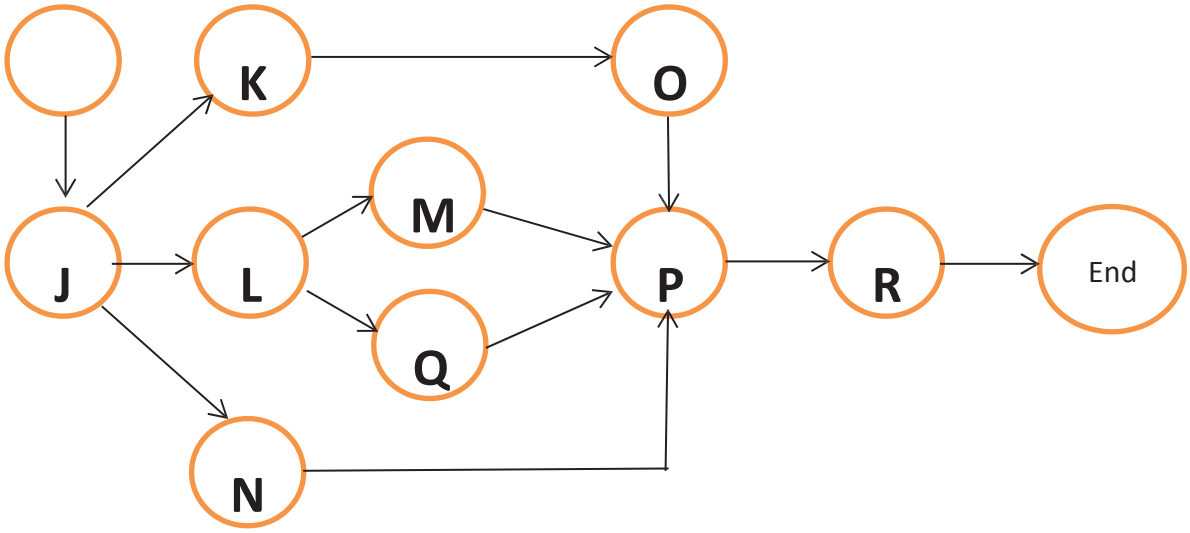
1)



2)



3)



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4) Critical path is ACEFGH, production time is 16 days

5) Task B has a float time up to 4 days, as does task D

6) Critical path is ABECF (16) and task D has a float time of 5 units

7) Sydney to Darwin
Melbourne to Brisbane
Adelaide to Perth

Total cost $1335

8) Truck 1 to pit 2
Truck 2 to pit 4
Truck 3 to pit 3
Truck 4 to pit 1

Total distance 275 km

9) Alan PHP
Jill Java
Peter C++
Boris Html

Total cost $285

10) Bob Amenities
Carol Desks
Glen Mop
Jason Windows
Rachel Rubbish

Total efficiency 75

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Further Mathematics

Matrices

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Further Mathematics: Solutions

Chapter 6: Matrices

Exercise 1 Representation & Operations

1)

a) 2 x 2

b) 1 x 1

c) 3 x 5

d) 1 x 2

e) Cannot be multiplied

2)

a) $\begin{bmatrix} 5 & 5 \\ 5 & 5 \end{bmatrix}$

b) $\begin{bmatrix} 0 & -1 \\ 0 & 2 \end{bmatrix}$

c) $\begin{bmatrix} 4 & 5 \\ 1 & 4 \end{bmatrix}$

d) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

e) $\begin{bmatrix} 84 & 763 \\ 1234 & -499 \end{bmatrix}$

3)

a) $\begin{bmatrix} 2 & -1 \\ 2 & -1 \end{bmatrix}$

b) $[10]$

c) $\begin{bmatrix} -2 & 2 & -1 \\ 2 & -1 & 1 \end{bmatrix}$

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d) Cannot be multiplied

4)

a) $\begin{bmatrix} 8 & 16 & 32 \end{bmatrix}$

b) $\begin{bmatrix} -6 \\ -1 \\ -8 \end{bmatrix}$

c) Cannot multiply a 3 x 1 matrix by a 3 x 3

d) Cannot multiply a 3 x 3 matrix by a 1 x 3

e) $\begin{bmatrix} -6 & 96 & 128 \end{bmatrix}$

f) $\begin{bmatrix} 34 & 6 & 26 \end{bmatrix}$

g) $\begin{bmatrix} -1 & -8 & 9 \\ 8 & -4 & 4 \\ 3 & -10 & 5 \end{bmatrix}$

h) $\begin{bmatrix} 2 & -7 & 11 \\ 12 & -1 & 7 \\ 4 & 1 & -1 \end{bmatrix}$

i) $\begin{bmatrix} 80 \end{bmatrix}$

j) Cannot multiply D (3 x 1) by B (3 x 3)

5)

A=

	Roll	Fruit	Drink
Red	10	8	7
Blue	12	6	9
Green	11	10	5

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6)

B=

	Roll	Fruit	Drink
Red	12	6	8
Blue	9	9	7
Green	11	11	8

7)

C=2B=2

	Roll	Fruit	Drink
Red	12	6	8
Blue	9	9	7
Green	11	11	8

=

	Roll	Fruit	Drink
Red	24	12	16
Blue	18	18	14
Green	22	22	16

8)

A+B+C

=

	Roll	Fruit	Drink
Red	46	26	31
Blue	39	33	30
Green	44	43	29

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9)

a)

D=

	Cost
Roll	$3.50
Fruit	0.75
Drink	$2.45

b) $19.50

c) $71.05

d) (A + B + C) x D =

(A+B+C) x D=

	Cost
Red	$256.45
Blue	$234.75
Green	$257.30

10)

a)

$$\begin{bmatrix} 0.6 \\ 0.15 \\ 0.25 \end{bmatrix}$$

b) AR since A is a 3 x 3 matrix and R is a 3 x 1

c) Peter 77.55

Brett 80.85

Amy 74.95

Karen 66.5

Sue 84.9

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11)

C=

	A	B	C
Per day	66	0	30
C per km	0	0.48	0.25

P =

Days	KM
4	560

PC=

	A	B	C
Total Cost	$264	$268.80	$260

Company C is cheapest

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Further Mathematics: Solutions

Chapter 6: Matrices

Exercise 2 Simultaneous Equations

1)

a) $\frac{1}{8}\begin{bmatrix} 1 & 1 \\ -4 & 4 \end{bmatrix}$

b) $\frac{1}{3}\begin{bmatrix} 3 & -6 \\ -0.5 & 2 \end{bmatrix}$

c) $-\frac{1}{9}\begin{bmatrix} 6 & -3 \\ -5 & 1 \end{bmatrix}$

d) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

e) Inverse does not exist (determinant is zero)

2)

a) $\begin{bmatrix} 2 & 1 & 4 \\ 1 & -1 & 2 \end{bmatrix}$

b) $\begin{bmatrix} 3 & 2 & 7 \\ 1 & 3 & 1 \end{bmatrix}$

c) $\begin{bmatrix} 2 & -1 & -3 \\ 2 & 1 & -1 \end{bmatrix}$

d) $\begin{bmatrix} 1 & -1 & 0 \\ 2 & -1 & -4 \end{bmatrix}$

e) $\begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 4 \end{bmatrix}$

f) $\begin{bmatrix} 1 & -1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$

3)

a) $\begin{bmatrix} 1 & 2 \end{bmatrix}$

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b) $[1 \quad 3]$

c) $[1 \quad 2]$

d) $[0 \quad 2]$

e) No solution

4) 5 and 1

5) 15 x 5 cm

6) 7 and (-5)

7) Books are $1.60 each, and pens are 50 cents each

8) $\frac{2}{5}$

9) 34 and 12

10) The faster runner travels 4 km per hour, and the other 3 km per hour

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Further Mathematics: Solutions

Chapter 6: Matrices

Exercise 3 Transition Matrices

1)

a) $\begin{bmatrix} 0.7 & 0.68 \\ 0.3 & 0.32 \end{bmatrix}$

b) $\begin{bmatrix} 0.69 & 0.688 \\ 0.31 & 0.312 \end{bmatrix}$

c) $\begin{bmatrix} 0.689 & 0.688 \\ 0.311 & 0.3112 \end{bmatrix}$

d) $\begin{bmatrix} 0.68889 & 0.68888 \\ 0.31111 & 0.31112 \end{bmatrix}$

e) Matrix A

2)

a) 0.45

b) John 0.428745, Ken 0.571255

3)

a) $\begin{bmatrix} 0.96 & 0.04 \\ 0.08 & 0.92 \end{bmatrix}$

b) 3.04 million and 860000

c) 2,899,848 and 1,000152

4) Cools 28,874 and Woolless 31126

5) During the fifth year; 30,000 customers each

6) 52 in the city and 48 at the airport

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7) 30 (or 31) in the city, 17 at the airport, and 13 (or 12) in the suburbs

8)

a) 23810

b) 23810, 33333, 19047, 23810

c) 12116, 21356, 28836, 37692

d) 8809, 19253, 30638, 41300

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